

The Compass



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Community Mathematics Centre

Schools of the Krishnamurti Foundation India
(KFI)

Statement of intent

The magazine is published by the Community Mathematics Centre. It is a collaborative initiative of the schools of the Krishnamurti Foundation India (KFI).

Its intention is:

- To share materials for mathematics and science teachers who teach at the high school or middle school levels, and for students who study at these levels.
- To serve as a platform where teachers and students can share their findings and discoveries with others who share their love for the subject.

In addition to mathematical content, in each issue we shall share a few ‘snippets’ from the world of science — short extracts abstracted from articles in online magazines, selected on the basis of intrinsic interest.

The Compass is published as an online magazine free of cost. Articles may be downloaded and used freely by teachers, but with proper attribution.

Edited by Shailesh Shirali, Padmapriya Shirali, Shashidhar Jagadeeshan, Sai Jagan Mohan, Chris John & Swati Sircar on behalf of schools of the KFI

Contact id ... compass.math.2024@gmail.com

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Colour coding

The magazine has articles addressed to teachers who teach at different levels: Lower Primary level; Upper Primary and High School levels; and Senior Secondary and Junior College levels. There is also a Problem Corner. In addition we have the Snippets.

For the convenience of the reader, we use the following colour coding to distinguish between the various levels. Of course, such a division can never be exact. So, our apologies in advance for errors of judgement!

- Primary & Upper Primary (Classes 4, 5, 6, 7)
- Upper Primary & High School (Classes 6, 7, 8, 9, 10)
- Senior Secondary & Junior College (Classes 10, 11, 12)
- Problem Corner
- Snippets

Part I

Letters to the Editor

Letter from Prof. Jyotirmoy Sarkar

Dear Editors,

I am delighted to see two proofs of the same result, each authored by a different person. See [1, 2]. I admire the editor for placing the two papers one after another. Hopefully, this will counter the prevailing misconception that there is only one way to arrive at the truth.

In that spirit, may I present a third proof? This one requires only a picture, whose caption says it all.

A Third Proof

Result: Let $T_n = 1 + 2 + 3 + \dots + n$.

Then $1^3 + 2^3 + 3^3 + \dots + n^3 = T_n^2$.

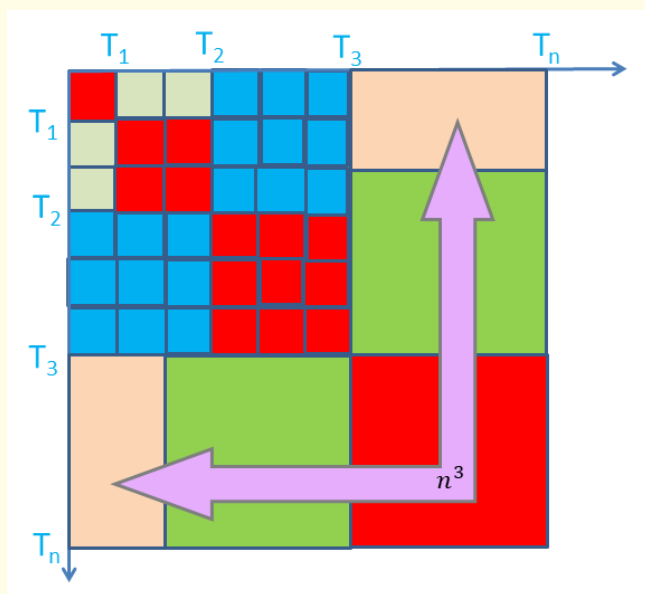


Figure 1: Break each n^3 into n copies of n^2 . If n is even, also break one n^2 into two $n \times n/2$ rectangles. Rearrange these pieces (in an elbow pattern about the previous pieces) to make a $T_n \times T_n$ square.

Further reading

- [1] Saibal Gupta. Summing the Cubes of the Natural Numbers – I. *The Compass*, 3-2 77–80, (2026).
- [2] S. C. Dutta Roy. Summing the Cubes of the Natural Numbers – II. *The Compass*, 3-2 81–82, (2026).

About the author

JYOTIRMOY SARKAR is a professor at the School of Science (Dept... of Statistics & Biostatistics), Indiana University Indianapolis (USA). His main research areas include enumeration, probability, statistics, and reliability theory. He enjoys reading, 'riting, 'rithmetic and R-coding. He may be contacted at jsarkar@iu.edu.

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Note from the editor: Later in this issue we present *yet one more* of this identity! Please see page 39.

Part II

Math & Science Education

The Role of Discussions in Math Teaching

Article contributed by Padmapriya Shirali

The role of discussions in math classes is gaining in importance. Building communication and reasoning skills in students, developing self-reliance in testing the correctness of their reasoning through peer interaction, increasing collaboration, and deepening students' appreciation of concepts and processes are seen as the obvious benefits.

The discussions can be organised in different ways.

Student pairs can work together, discuss their ideas, reason them out, and jot down their ideas.

A “fishbowl” arrangement can be created where half the class sits in an inner circle and the other half sits behind them. The inner circle can discuss the question while the outer circle listens and takes down points. The two sets of students can then exchange places for different questions.

The teacher can give a few prompts as questions to promote larger participation and guide the discussion. The teacher can carefully develop, among students, the skill of summarising another person's reasoning before presenting a different approach or asking clarifying questions.

The teacher can decide how to present the discussions between pairs, or in groups of four, to the rest of the class. It can take the form of short presentations, if time permits, or of putting up the jotted-down points on a display board. It is also a time to find multiple ways of solving problems and to address various types of errors.

I share here a few sample discussions.

Discussion 1

Grade level: 6. Students work in groups of four.

Problem posed: Can you use the information on the cards to draw the shape being described? (Please check the figure on the next page.)

Group A

S1: Let's take the first statement and look for shapes that have two parallel lines.

S2: There are two shapes, a rectangle and a square.

T: Could there be more?

S3: Yes. A parallelogram and a diamond shape.

S4: What you are calling a diamond shape is a rhombus.

S1: Now, what other statement should we take?

The shape has two pairs of parallel sides.	The area of the shape is 24cm^2 .
The shape has four right angles.	The shape's perimeter is numerically larger than its area.
The length of each side is an even number.	The shape is irregular.
The shape is a quadrilateral.	The shape has two lines of symmetry.

Figure 2

S2: The shape has two lines of symmetry.

S3: Hmm. The rectangle and square have lines of symmetry. But what about the other two shapes?

S4: They don't have lines of symmetry.

T: Are you sure that they do not have lines of symmetry? How will you justify that?

S4: Let me make a cut-out of a parallelogram. See, if you make a fold at the centre, the edges don't match.

S2: Let's try with the rhombus. But the rhombus does have lines of symmetry.

S3: So we have the square, rectangle, and rhombus.

T: Are you sure that all of them have two lines of symmetry?

S4: The rectangle has two, one vertical and one horizontal.

S1: The square also has vertical and horizontal.

S2: Doesn't the square have diagonal symmetry too?

S1: Oh yes, we cannot consider the square. So only the rectangle and rhombus.

T: Now how will you proceed?

S3: The shape is irregular.

S1: The rhombus is regular. So it has to be a rectangle. We got it!

T: But what about its size?

S: The length of each side is an even number.

S2: But there will be so many such rectangles.

T: Do you need more information? What will you use?

S3: The area of the shape is 24 square cm. That means it can be 3×8 .

S4: But the lengths have to be even. So it will be 4×6 or 2×12 . So we have two solutions.

T: Is there any other card that you have not used that can help you decide the dimensions?

S3: The shape's perimeter is larger than its area. That may help.

S2: Let's try. If the dimensions are 4×6 , the perimeter will be 20 cm. That doesn't work.

S4: If the dimensions are 2×12 , the perimeter will be 28. That fits.

S1: We got it!

T: Did you use all the pieces of information?

Students: No. We didn't use the information about its being a quadrilateral and having right angles.

Students presented their solution to the rest of the class.

T (to Group B): What clues did you start with? How did you proceed?

Group B: We first looked at the card that said the shape's perimeter was larger. That wasn't useful. We knew two parallel lines meant square, rectangle, parallelogram, or rhombus. Then we used the card that said four right angles; it had to be a square or rectangle. Then we used the fact that the shape was irregular, so it had to be a rectangle. Finally we used the area, and the fact that the perimeter was larger than the area.

T (to Group C): What clues did you start with? How did you proceed?

Group C: We first looked at the card that said the shape has four right angles. That obviously meant either a square or a rectangle. Then we saw that it was irregular, so it had to be a rectangle. Then we used the area, and the fact that the perimeter is larger than the area, to get the dimensions.

T: Did you need all the information that was given? Why or why not?

Group C: We didn't need all the information, as we could infer that it was a quadrilateral. We didn't need the information about symmetry, since we used the fact that it was irregular.

The class enjoyed the activity and the opportunity to discuss their assumptions and ideas. The teacher intervened when necessary and posed questions to help students clarify their learning.

Discussion 2

Grade level: 8. Students work in groups of four.

Problem posed: Eight of my pets aren't dogs, five aren't rabbits, and seven aren't cats. How many pets do I have?

Group A

S1: This seems tough to crack!

S2: There is no positive information.

T: What do you mean?

S3: All three pieces of information are given negatively.

T: In what way can we interpret the statement 'Eight of my pets aren't dogs'? What does that imply?

S4: It means that the other pets are eight in number.

T: Proceed.

S1: The other pets are rabbits and cats.

S2: Their total is eight.

S3: Ah! We can write that information as an equation:

$R + C = 8$ (where R = the number of rabbits and C = the number of cats)

T: Go on. I will see what the other group is doing.

S4: Similarly, the statement ‘five aren’t rabbits’ implies

$$D + C = 5$$

S3: Let’s write the third equation too:

$$D + R = 7$$

S1: Now we have three equations and three variables. We have not learnt how to solve such equations!

S4: Yes. We know how to solve equations with two variables.

S2: Let’s see. Don’t give up so fast.

All of them stare at the three equations noted down on paper.

$$R + C = 8 \quad (\text{equation 1})$$

$$D + C = 5 \quad (\text{equation 2})$$

$$D + R = 7 \quad (\text{equation 3})$$

A few minutes pass. S3 sees a connection between equations 2 and 3.

S3: Hey! Look! If $D + C = 5$ and $D + R = 7$, then the difference of R and C must be 2.

S1: I didn’t understand.

S2: I understood.

S2: Look. (Begins to explain to S1.) D is common to both equations. Right?

S1: Yes.

S2: We see that one equation totals 5 and the other totals 7. Why is there a difference in the sums?

S1: Because the other letters are different — they are C and R .

S3: That means their difference is 2.

S1: Oh! I get it. $C - R = 2$.

S3: The other way round. The equation with R is greater. So $R - C = 2$.

S1: What difference does it make?

S3: You are thinking that since $R + C$ is the same as $C + R$, it will be the same in subtraction. But $5 - 2$ is not the same as $2 - 5$.

S1: I understand.

S4: Right, now what do we have? $R + C = 8$ and $R - C = 2$.

S1: I know how to solve these two equations. We can add them.

S2: Let's do it quickly.

Solving $R + C = 8$ and $R - C = 2$, they get $2R = 10$, hence $R = 5$.

S3: Now we can substitute the value of R in equation 1 and get $C = 3$.

S1: Now we can get D easily. Since $D + 3 = 5$, $D = 2$.

S4: Let's verify and see if our answer is correct. (He draws a diagram to check.)

S2: That matches the equations.

S1: But what if there were other pets, like fish?

S2: You are thinking of fish because you have an aquarium at home.

S3: I wonder what it would be like if other pets were allowed.

S4: The teacher has finished going round to the other groups. Do you think they used the same method as us?

Group B

Problem posed: Eight of my pets aren't dogs, five aren't rabbits, and seven aren't cats. How many pets do I have?

S4: What kind of problem is this?

S3: We have never done such problems before!

S1: That is why we are supposed to discuss and figure it out.

S2: I don't like domestic animals. I like wild ones, like lions and bears.

S4: OK. We can change these animals to wild ones!

S1: Let's try to solve the problem.

S4: Can I guess and solve?

S3: You can guess, but we have to see if it fits the given information.

S4: Let us say that the total number of animals is eight.

S1: How can it be only eight? The statement says, 'Eight of my pets aren't dogs.' That means, if you include the dogs, the total number has to be more than eight.

S1: OK, let us say 12. Then, since eight are not dogs, the remaining dogs have to be four.

S2: All right. Among the eight, there are rabbits and cats. Five are not rabbits, so those five have to be cats. And seven are not cats, so seven will be rabbits.

S4: But four dogs, five cats, and seven rabbits do not make 12. That is 16 pets.

S3: So we should try with a smaller number, between 8 and 12. Let's try 10.

S1: (Draws 10 marks on the paper.) These eight are not dogs. The remaining two are dogs.

S2: OK. Out of these eight, there are five that are not rabbits, so the remaining three are rabbits.

S3: The five are cats.

S4: That fits: two dogs, three rabbits, and five cats. We got it!

Group C

Problem posed: Eight of my pets aren't dogs, five aren't rabbits, and seven aren't cats. How many pets do I have?

S1: In algebra, we always start with the unknown. Right?

S4: OK. How many pets?

S1: Right. Let us call the total number of pets P .

S4: I can make an equation for D . $D = P - 8$.

S3: Yes. And the equation for R is $R = P - 5$.

S2: Obviously, $C = P - 7$.

S4: Now what do we do?

S1: Hmm.

S2: I know. We can add the three numbers.

S3: They are not numbers. They are expressions.

S2: I meant that. $P - 8$, $P - 5$, and $P - 7$.

S4: Well, that becomes $3P - 20$.

S1: How does that help?

S3: But that should be the same value as P .

S4: Oh yes! $3P - 20 = P$.

S1: Yes. That makes $2P = 20$, so $P = 10$.

S3: Teacher, we're ready to present.

The teacher notices that some of the groups have used different approaches. Realising the potential for peer learning, she encouraged the students to present their solutions and reason it out. Students from other groups asked questions and sought clarifications.

T: How did you begin to think about this problem?

S2: In the beginning we were confused by the wording. Then we looked at only one statement to understand what it means. If eight are not dogs, it implied those eight were the other animals.

T: How did you organise your information?

Students: We showed the information in a graphic way.

Teacher: What methods did you use?

S1:

T: What did not work?

T: Have you tried making a guess?

T: What else have you tried?

Teacher: Can you restate the method used by Group 1 in your own words?

S1: I didn't understand the method the third group used.

T: What part of what they said do you understand?

T: What would happen if other pets are allowed?

S3: I asked the same question.

T: Is there any other way you could solve this problem?

Students: Not sure, we will think about it.

About the author

PADMAPRIYA SHIRALI has been a math educator for the past four decades, teaching at various levels from Class 1 to Class 10. She has a strong interest in math pedagogy and conducting explorations in maths. She is a member of the NCERT textbook committee. She has worked closely with Shri P.K. Srinivasan, an eminent math educationist, and was part of the Rishi Valley Rural Education team which developed the multi-grade learning material known as 'school in a box.' She may be contacted at padmapriya.shirali@gmail.com

Is Geometry Uniquely Human?

Adapted from: “New Research Challenges the Idea That Geometry Is Uniquely Human.” <https://scitechdaily.com/new-research-challenges-the-idea-that-geometry-is-uniquely-human/>. Author: New York University

The article says that geometry may not be a skill that is uniquely human after all. Instead, it argues that the roots of geometric thinking may come from basic navigation abilities that humans share with animals, and that language helps humans turn those spatial skills into more abstract math.

In simple terms, the article is challenging an old idea: that people have a special built-in mental system just for geometry. Some researchers have thought that humans are born with a kind of “math module” that gives us geometric understanding. But the new review, by psychologist Moira Dillon, suggests a different explanation. It says that the mental tools we

use for geometry may have first evolved for moving through space, finding our way, and remembering directions.

The article points out that many animals can do surprisingly good navigation. Rats, chickens, fish, and even babies can handle distance, direction, and shape in ways that resemble basic geometry. These abilities are not full school-style geometry, but they do show that the foundations of geometric thinking are not limited to humans.

A major point in the article is the idea that human language is what takes these basic spatial abilities to a higher level. According to Dillon, language lets people think about geometry without actually moving around. That means we can imagine shapes, compare angles, and solve problems in our heads. In this view, language does not create geometry from nothing; it helps us use older, shared mental abilities in a more powerful way.

The article notes that this question has been around for a long time. Philosophers such as Plato, Descartes, and Kant all thought about where geometry comes from. Scientists began studying the issue experimentally much later. This new review is part of that ongoing debate, but it leans strongly toward the idea that geometry grew out of everyday navigation rather than a uniquely human inner math system.

Overall, the article is saying that geometry may be less about a special human gift and more about a shared ability to move through the world, with language giving humans an extra edge.

Acknowledgment: This article has been prepared from the published article whose source is given above. The summary was made using the *Gemini 3.1 Pro/Perplexity* LLM.

Part III

Miscellaneous Articles – 1

Three Squares in a Triangle

Article contributed by Swati Sircar & Ganesh Chakali

Problem

Study the following diagram.

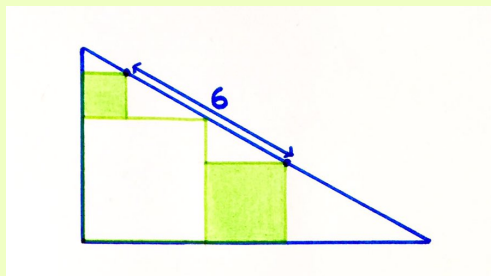
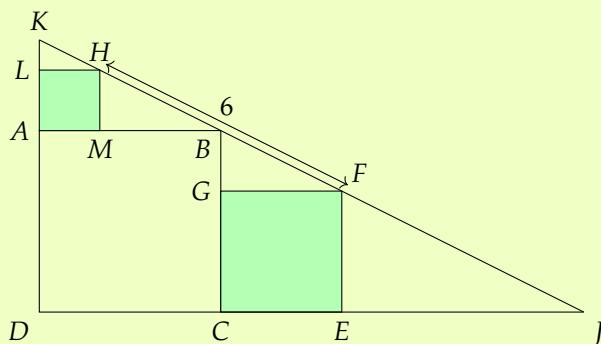


Figure 3

There are three squares inside the triangle, placed as shown. One length is marked. What is the sum of the areas of the two shaded squares?

Before proceeding with the solution, let us name all the relevant points in the diagram.



Solution

Can this problem be solved using only concepts known at the middle school level?

The natural approach to solve this problem would make use of similar triangles. But it is only after class 8 that similarity of triangles is introduced in the curriculum. So, is it possible to solve this problem without using similarity and using only simpler concepts?

This solution reveals one such way to solve this problem.

Before solving the main problem, let us prepare the following simple, yet powerful, ingredient. We shall use it to solve the problem.

Consider a rectangle $ABCD$ and any point E inside the rectangle. Line segments GI and FH are drawn parallel to AB and CB respectively, through the point E .

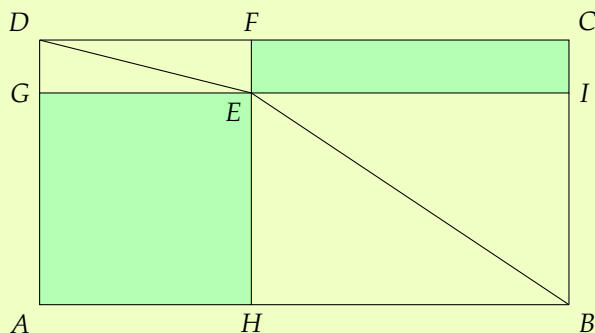


Figure 4

Proposition 1. *Points D, E and B are collinear if and only if the shaded areas are equal.*

To prove proposition 1 we must prove that:

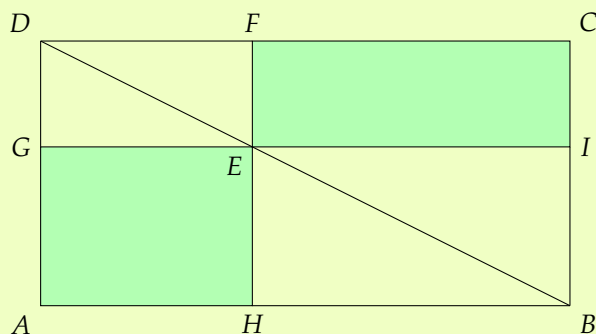
- (i) If D, E and B are collinear, then the shaded areas are equal.
- (ii) If the shaded areas are equal, then the points D, E and B are collinear.

Proof of (i): Suppose D, E and B are collinear. Then it is obvious that E is a point on the diagonal DB .

Since BD is a diagonal of the rectangle $ABCD$, the areas of triangles ABD and BDC are equal. It is trivial to observe that $\triangle DGE$ and $\triangle DFE$ are congruent. Hence they have equal area.

Similarly, $\triangle BEH$ and $\triangle BEI$ have equal area.

As a result, the shaded areas are equal.



Proof of (ii): Let us use proof by contradiction to prove that if the shaded areas are equal then points D, E and B are collinear.

Suppose the shaded areas are equal and the points D, E and B are non-collinear.

As the shaded areas are equal, we have:

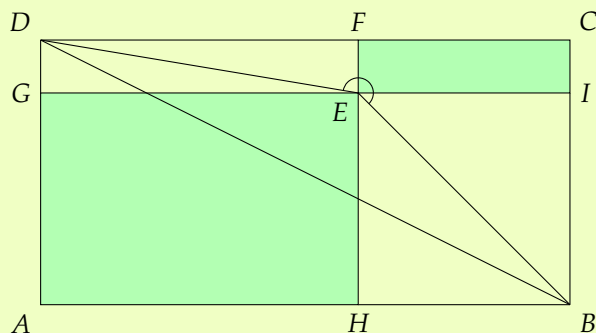
$$\text{Area (GAHE)} = \text{Area (FEIC)} \quad (1)$$

We can see that $\triangle DGE$ and $\triangle DFE$ are congruent, since DE is a diagonal of the rectangle $DGEF$. Hence their areas are equal.

Similarly, $\triangle BEH$ and $\triangle BEI$ have the same areas.

Since points D, E and B are non-collinear, E must either lie strictly within $\triangle ABD$ or strictly within $\triangle CBD$.

Without loss of generality, we may suppose that E lies strictly within $\triangle CBD$, as shown in the figure.



Then quadrilateral $DEBC$ is concave and quadrilateral $DABE$ is convex. Therefore we have

$$\text{Area}(DEBC) < \text{Area}(\triangle CBD) = \text{Area}(\triangle ABD) < \text{Area}(DABE),$$

and so,

$$\text{Area}(DEBC) < \text{Area}(DABE).$$

Now, observe that:

$$\text{Area}(DEBC) = \text{Area}(\triangle DFE) + \text{Area}(FEIC) + \text{Area}(\triangle BEI),$$

$$\text{Area}(DABE) = \text{Area}(\triangle DGE) + \text{Area}(GAHE) + \text{Area}(\triangle BEH).$$

We also have:

$$\text{Area}(\triangle DFE) = \text{Area}(\triangle DGE),$$

$$\text{Area}(\triangle BEI) = \text{Area}(\triangle BEH).$$

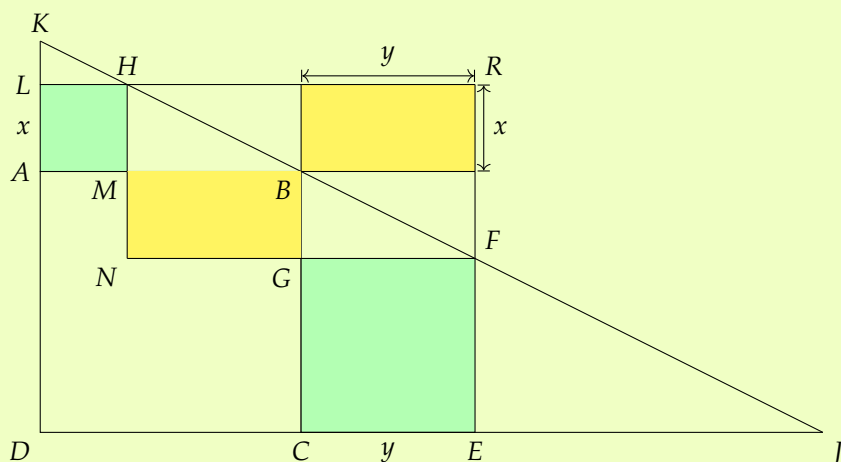
Hence if $\text{Area}(DEBC) < \text{Area}(DABE)$ then the following must be true:

$$\text{Area}(FEIC) < \text{Area}(GAHE).$$

But this contradicts the stated information that the areas of the shaded rectangles are equal. We therefore conclude that our initial assumption must be incorrect. Therefore, if the shaded areas are equal, then points D, E and B are collinear. ■

Hence Proposition 1 is proved.

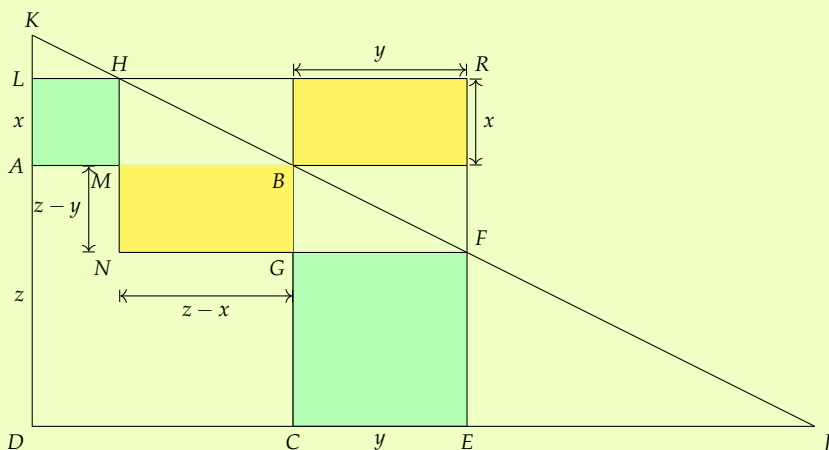
Coming back to the given problem, let the side lengths of the shaded squares be x and y , as shown. Construct additional rectangles as shown:



Since points H, B and F lie on the hypotenuse, they are collinear. Hence from Proposition 1, the yellow shaded areas are equal.

Hence Area $(MNBG) = xy$.

Suppose the length of AD is z , then we have the following:



The sides of rectangle $MNGB$ are $z - x$ and $z - y$, hence

$$(z - x)(z - y) = xy,$$

$$\therefore z^2 - (x + y)z + xy = xy,$$

$$\therefore z^2 = (x + y)z,$$

$$\therefore z = x + y.$$

Since $z = x + y$, we get $NH = 2x$ and $NF = 2y$. Considering $\triangle HNF$, by Pythagoras's theorem, we obtain:

$$(2x)^2 + (2y)^2 = HF^2 = 6^2, \quad \therefore x^2 + y^2 = 9.$$

It follows that the sum of the shaded square areas is 9.

About the authors

SWATI SIRCAR is a consultant with Azim Premji University. Mathematics is the second love of her life (the first being drawing). She has a B.Stat-M.Stat degree from the Indian Statistical Institute and an MS in math from University of Washington, Seattle. She has been doing mathematics with children and teachers for more than 15 years and is deeply interested in anything hands on, origami in particular. She initiated *Math Space*, a mathematics laboratory at the Bengaluru campus of Azim Premji University. It caters to schools, teachers, parents, children, NGOs working in school education and teacher educators. Swati may be contacted at swati.sircar@apu.edu.in.

GANESH CHAKALI is a math educator, Intern for math education and communications at the Raising A Mathematician Foundation, and a budding independent educational researcher. He holds an Integrated B.Sc., B.Ed. degree in Mathematics from Azim Premji University, Bengaluru. His journey in Math education has been significantly shaped by the mentorship of and collaboration with Swati Sircar. Ganesh may be contacted at ganesh.chakali.studies@gmail.com.

Sums of Cubes – 2

We continue our study of expressing the integers as sums of cubes and describe a few miscellaneous results in this topic. Each is of interest in itself.

Remark: When we say “cubes” we include the cubes of negative integers.

Any integer can be written as the sum of five cubes

We start with a simple and easily verified identity:

$$6n = (n + 1)^3 + (n - 1)^3 + (-n)^3 + (-n)^3. \quad (2)$$

This shows that every multiple of 6 is a sum of four cubes. Therefore, to deal with non-multiples of 6, we only have to focus attention on the gap between the given integer and the nearest multiple of 6.

Any integer M that is not a multiple of 6 can be written in the form $6n + r$ where $r \in \{\pm 1, \pm 2, 3\}$. So let us focus on each of these possibilities.

- If $M = 6n \pm 1$, then since $\pm 1 = (\pm 1)^3$, we obtain a representation of M as a sum of five cubes.
- If $M = 6n \pm 2$, then we write

$$6n + 2 = 6(n - 1) + 8 = 6(n - 1) + 2^3,$$

$$6n - 2 = 6(n + 1) - 8 = 6(n + 1) + (-2)^3,$$

and thereby obtain the required representation.

- Finally, if $M = 6n + 3$, we write

$$6n + 3 = 6(n - 4) + 27 = 6(n - 4) + 3^3,$$

and this yields the required representation.

This concludes the proof of the stated result. ■

Reference: <https://mathsinteresting.quora.com/Prove-that-any-integer-is-a-sum-of-five-cubes>

Is every integer the sum of 4 cubes?

The short answer to this is: we don't know! It is a much-researched question, but concrete results are difficult to obtain. Extensive computational work has been carried out and every integer tested (within a huge range) has been found to be expressible as the sum of four cubes. Positive results have been obtained for integers in particular congruence classes. But a general result that covers all integers has proved elusive till now.

Specifically, a positive result has been obtained for the following congruence classes.

- Integers of the form $0 \pmod{6}$. This follows from the identity

$$6n = (n+1)^3 + (n-1)^3 + (-n)^3 + (-n)^3.$$

- Integers of the form $3 \pmod{6}$. This follows from the identity

$$6n+3 = n^3 + (-n+4)^3 + (2n-5)^3 + (-2n+4)^3.$$

- Integers of the form $1 \pmod{18}$. This follows from the identity

$$18n+1 = (2n+14)^3 + (-2n-23)^3 + (-3n-26)^3 + (3n+30)^3.$$

- Integers of the form $7 \pmod{18}$. This follows from the identity

$$18n+7 = (n+2)^3 + (6n-1)^3 + (8n-2)^3 + (-9n+2)^3.$$

- Integers of the form $8 \pmod{18}$. This follows from the identity

$$18n+8 = (n-5)^3 + (-n+14)^3 + (-3n+29)^3 + (3n-30)^3.$$

By multiplying throughout by -1 , we obtain more such results. E.g., from the one immediately above, we get:

$$-18n-8 = (-n+5)^3 + (n-14)^3 + (3n-29)^3 + (-3n+30)^3,$$

$$\therefore 18n-8 = (n+5)^3 + (-n-14)^3 + (-3n-29)^3 + (3n+30)^3,$$

$$\therefore 18(n-1)+10 = (n+5)^3 + (-n-14)^3 + (-3n-29)^3 + (3n+30)^3,$$

showing that integers of the form $10 \pmod{18}$ are also expressible as the sum of four cubes.

Taken together, the above relations show that all integers of the following forms

$$0, \pm 1, \pm 3, \pm 6, \pm 7, \pm 8, 9 \pmod{18}$$

can be expressed as the sum of four cubes.

This leaves integers of the following congruence classes:

$$\pm 2, \pm 4, \pm 5 \pmod{18}.$$

Further research has eliminated the first of these classes, by showing that all integers of the form $\pm 2 \pmod{18}$ can be expressed as the sum of four cubes. Now we are left only with integers of the following congruence classes:

$$\pm 4, \pm 5 \pmod{18}.$$

Or, more simply put, integers of the form $\pm 4 \pmod{9}$.

The general problem remains open, though particular integers in this class have been dealt with. Interested readers should consult [3].

What about sums of 3 cubes?

Having discussed at length the problem of expressing integers as sums of 4 cubes, we turn to the question about sums of 3 cubes.

Let T represent the set of all natural numbers which are sums of three or fewer cubes:

$$T = \left\{ n : n = a^3 + b^3 + c^3, a \in \mathbb{Z}, b \in \mathbb{Z}, c \in \mathbb{Z} \right\}.$$

Since every cube is of one of the forms $9k, 9k \pm 1$, it follows that a sum of three or fewer cubes must be of one of the following forms: $9k, 9k \pm 1, 9k \pm 2, 9k \pm 3$. We therefore have the following result:

Theorem. *If a number is of the form $9k \pm 4$, then it does not belong to T .*

It is easy to generate elements of T , as many as we may want, simply by giving all possible integer values to a, b, c in some specified range (and with $|a| \leq |b| \leq |c|$ to avoid duplication of elements) and then computing the value of $a^3 + b^3 + c^3$.

But verifying whether a given number belongs to T (or not) is far more difficult.

To get a glimpse of the difficulty involved, consider the following. The expressions for 99, 98 and 97 are easily found, as they involve relatively small numbers:

$$99 = 2^3 + 3^3 + 4^3,$$

$$98 = 0^3 + (-3)^3 + 5^3,$$

$$97 = (-1)^3 + (-3)^3 + 5^3.$$

Similarly we have these expressions for 91, 92 and 93:

$$91 = 0^3 + 3^3 + 4^3,$$

$$92 = 1^3 + 3^3 + 4^3,$$

$$93 = (-5)^3 + (-5)^3 + 7^3.$$

There are, of course, no such expressions for 94 and 95, as these are of the forbidden forms $9k \pm 4$.

But for 96, we have all of a sudden:

$$96 = 10853^3 + 13139^3 + (-15250)^3.$$

And for 75, we have the following:

$$75 = 435203083^3 + (-435203231)^3 + 4381159^3.$$

To discover such relations, one clearly needs powerful computational facilities. The complexities are huge.

The following question now poses itself:

*If a number n is **not** of the form $9k \pm 4$, then does n belong to T ?*

Offhand, we see no reason for supposing that this is true (or that it is false). But computational evidence suggests otherwise!

The evidence overwhelmingly suggests that every number not of the form $9k \pm 4$ belongs to T .

It is interesting to note the evolution of this observation. Over the decades, mathematicians tried to express various numbers as sums of three cubes (of positive or negative integers), making use of powerful computational resources. By 1960, the only numbers less than 100 which had not yet been expressed in the required form were

30, 33, 39, 42, 52, 74, 75, 80, 84, 87, 91, 96.

Then in the 1960s the following relations were discovered:

$$87 = 4271^3 + (-4126)^3 + (-1972)^3,$$

$$96 = 13139^3 + (-15250)^3 + 10853^3,$$

$$91 = 83538^3 + (-67134)^3 + (-65453)^3,$$

$$80 = 103532^3 + (-112969)^3 + 69241^3.$$

In the 1990s, the following relations were discovered:

$$39 = 134476^3 + (-159380)^3 + 117367^3$$

$$75 = 435203083^3 + (-435203231)^3 + 4381159^3$$

$$84 = 41639611^3 + (-41531726)^3 + (-8241191)^3$$

In the first decade of the 2000s, the following were discovered (the numbers kept getting bigger and bigger!):

$$30 = 2220422932^3 + (-2218888517)^3 + (-283059965)^3,$$

$$52 = 23961292454^3 + (-61922712865)^3 + 60702901317^3,$$

$$74 = 66229832190556^3 + (-284650292555885)^3 + 283450105697727^3.$$

By 2019, the only numbers below 100 which were not of the form $9k \pm 4$ and had not yet been expressed in the required form were 33 and 42. Then in mid-2019, the following mind-boggling relations were discovered:

$$33 = 8866128975287528^3 + (-8778405442862239)^3 + (-2736111468807040)^3,$$

$$42 = 80435758145817515^3 + (-80538738812075974)^3 + 12602123297335631^3.$$

If you are like me and find it difficult to imagine how such relations could ever have been found, here is an account of the story, excerpted from [7].

Mathematicians Solve '42' Problem With Planetary Supercomputer

MICHELLE STARR, 9 SEP 2019

Mathematicians have finally figured out the three cubed numbers that add up to 42. This has settled a problem that has been pondered for 65 years ... The problem,

set in 1954, is ...: $x^3 + y^3 + z^3 = k$. Here k is each of the numbers from 1 to 100; the question is, what are x , y and z ?

Over the decades, solutions were found for the easier numbers. In 2000, mathematician Noam Elkies of Harvard University published an algorithm to help find the harder ones.

This year, just the two most difficult ones remained: 33 and 42.

After watching a YouTube video [8] about the problem with 33 on the popular maths channel Numberphile, mathematician Andrew Booker from the University of Bristol in the UK was inspired to write a new algorithm. He ran this through a powerful supercomputer at the university's Advanced Computing Research Centre, and got the solution for 33 after just three weeks.

So, we were left with the hardest one of them all: 42. This proved a much more obstinate problem, so Booker enlisted the aid of fellow MIT mathematician Andrew Sutherland, an expert in massively parallel computation.

As you already know from the headline of this article, they figured it out. They also did a fun reveal of their success: according to The Aperiodical, both mathematicians quietly changed their personal websites to the solution, and named the pages "Life, the Universe, and Everything," a fitting nod to Douglas Adams.

Of course, it wasn't simple. The pair had to go large, so they enlisted the aid of the **Charity Engine**, an initiative that spans the globe, harnessing unused computing power from over 500,000 home PCs to act as a sort of 'planetary supercomputer.' It took over a million hours of computing time, but the two mathematicians found their solution. ...

"I feel relieved," Booker said. "In this game, it's impossible to be sure that you'll find something. It's a bit like trying to predict earthquakes ... So, we might find

what we're looking for with a few months of searching, or it might be that the solution isn't found for another century."

Is that it, then? Well ...no. That's just 1 to 100 covered. Go up an order of magnitude to 1000, and there are still plenty of numbers to solve: 114, 165, 390, 579, 627, 633, 732, 906, 921 and 975 are all awaiting a solution to the sum of three cubes.

Got any ideas?

We'll leave that question for the reader ...

Further reading

- [1] Wikipedia, "Taxicab number,"

https://en.wikipedia.org/wiki/Taxicab_number

- [2] Kevin A. Broughan, "Characterizing the Sum of Two Cubes", Journal of Integer Sequences, Vol. 6 (2003), <https://cs.uwaterloo.ca/journals/JIS/VOL6/Broughan/broughan25.pdf>

- [3] Wikipedia, "Sums of four cubes," https://en.wikipedia.org/wiki/Sum_of_four_cubes_problem

https://en.wikipedia.org/wiki/Sum_of_four_cubes_problem

- [4] Wikipedia, "Sums of three cubes,"

https://en.wikipedia.org/wiki/Sums_of_three_cubes

- [5] Hisanori Mishima, "How to search the solutions of $n = x^3 + y^3 + z^3$,"

<http://www.asahi-net.or.jp/~KC2H-MSM/mathland/math04/cube01.htm>

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<http://www.asahi-net.or.jp/~KC2H-MSM/mathland/math04/matb0100.htm>
- [7] ScienceAlert, "Mathematicians Solve '42' Problem With Planetary Supercomputer," <https://www.sciencealert.com/the-sum-of-three-cubes-problem-has-been-solved-for-42>
- [8] Numberphile, "The Uncracked Problem with 33,"
<https://www.youtube.com/watch?v=wymmCdLdPvM>

Article contributed by Shailesh Shirali

Summing the Cubes of the Natural Numbers – IV

In the April 2026 issue of the magazine, we presented two different proofs ([1] and [2]) of the identity

$$1^3 + 2^3 + 3^3 + \cdots + n^3 = (1 + 2 + 3 + \cdots + n)^2,$$

and in this issue we presented Prof Sarkar's visual proof of the same result (page 2). Now we describe a *fourth proof* of this identity! It may well be the simplest proof of all. Let

$$S_n = 1^3 + 2^3 + 3^3 + \cdots + n^3,$$

$$T_n = 1 + 2 + 3 + \cdots + n.$$

Then it is obvious that

$$T_n - T_{n-1} = n. \tag{3}$$

We are also familiar with this identity:

$$T_n + T_{n-1} = n^2. \tag{4}$$

Multiplying the corresponding sides of (3) and (4), we get:

$$T_n^2 - T_{n-1}^2 = n^3. \quad (5)$$

Now it is a simple matter of telescoping:

$$\begin{aligned} 1^3 + 2^3 + \cdots + n^3 &= T_1^2 + (T_2^2 - T_1^2) + \cdots + (T_n^2 - T_{n-1}^2) \\ &= T_n^2. \end{aligned}$$

So $S_n = T_n^2$, and we are through. 

Article contributed by Shailesh Shirali

Further reading

- [1] Saibal Gupta. Summing the Cubes of the Natural Numbers – I. *The Compass*, **3-2** 77–80, (2026).
- [2] S. C. Dutta Roy. Summing the Cubes of the Natural Numbers – II. *The Compass*, **3-2** 81–82, (2026).

Part IV

Contributed Articles – 1

Differences of Squares in the Powers of 5

Article contributed by Lakshmi Prasad Karnati

Mathematical patterns often emerge from the application of basic algebraic identities to specific numeric sequences. An example is found when we study the differences of squares of powers of 5. Thus we have,

$$(5^{n+1})^2 - (5^n)^2 = 600 \times 25^{n-1} = 30 \times 20 \times 25^{n-1}$$

Derivation

Applying $a^2 - b^2 = (a + b)(a - b)$ with $a = 5^{n+1}$ and $b = 5^n$, we get,

$$\begin{aligned}(5^{n+1})^2 - (5^n)^2 &= (5^{n+1} + 5^n)(5^{n+1} - 5^n) \\ &= 5^n(5 + 1) \cdot 5^n(5 - 1) \\ &= 6 \cdot 4 \cdot 5^{2n} = 24 \cdot 25^n = 600 \cdot 25^{n-1}\end{aligned}$$

Geometric Progression

It follows that the sequence of differences of the squares of consecutive powers of 5 is a geometric progression with common ratio 25.

Numerical Verification

n	Equation	Factored Form	Result
1	$25^2 - 5^2$	30×20	600
2	$125^2 - 25^2$	$30 \times 20 \times 25$	15,000
3	$625^2 - 125^2$	$30 \times 20 \times 25^2$	375,000
4	$3125^2 - 625^2$	$30 \times 20 \times 25^3$	9,375,000

About the author

LAKSHMI PRASAD KARNATI may be contacted at klprasad@gmail.com.

On the Sum of Circular Exterior Angles in a Polygon

Article contributed by Advik Rajaani

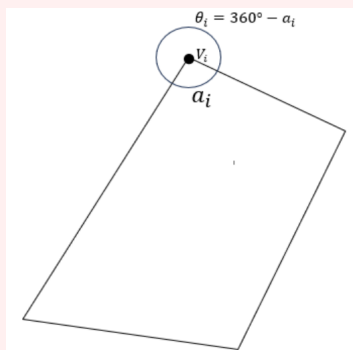
Classical exterior angles of polygons, defined by extending sides, yield a sum of 360° for convex polygons but become ambiguous at vertices with reflex angles. We study *circular exterior angles*, defined at each vertex as the angle outside the polygon that completes a full revolution with the interior angle. This definition is well-defined for all simple polygons.

We prove that the sum of circular exterior angles for an n -sided polygon is $180^\circ(n + 2)$, forming an arithmetic progression as n increases.

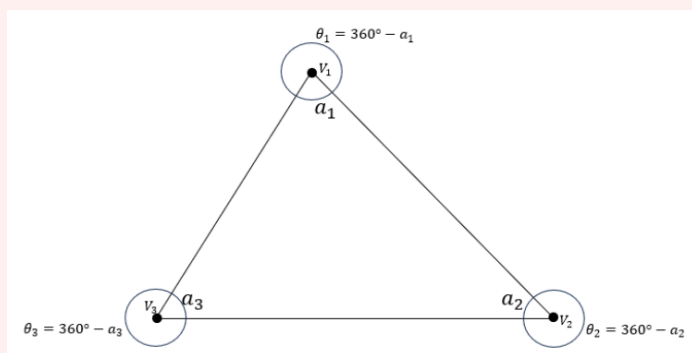
Preliminaries

Let P be a simple, counter-clockwise-oriented polygon with $n \geq 3$ sides and vertices $V_1, V_2, \dots, V_n$. Let a_i denote the interior angle at V_i . If $a_i \leq 180^\circ$, it is convex; if $a_i > 180^\circ$, it is reflex.

The **circular exterior angle** at V_i is defined as $\theta_i = 360^\circ - a_i$. The total sum of circular exterior angles is $S(P) = \sum_{i=1}^n \theta_i$.



(a)



(b)

Figure 5: (a) The circular exterior angle θ_i at vertex V_i is the angle outside the polygon that completes a full revolution with the interior angle a_i . (b) A triangle $P = (V_1, V_2, V_3)$ with interior angles a_1, a_2, a_3 and corresponding circular angles $\theta_1, \theta_2, \theta_3$

Main Result

Theorem. Let P be a simple polygon with $n \geq 3$ sides. The total sum of its circular exterior angles is:

$$S(P) = \sum_{i=1}^n \theta_i = 180^\circ(n + 2)$$

Proof

By definition, $\theta_i = 360^\circ - a_i$. Summing over all vertices:

$$S(P) = \sum_{i=1}^n (360^\circ - a_i) = 360^\circ n - \sum_{i=1}^n a_i$$

Using the standard formula for the sum of interior angles, $\sum_{i=1}^n a_i = 180^\circ(n - 2)$, we obtain:

$$S(P) = 360^\circ n - 180^\circ(n - 2) = 180^\circ(n + 2). \quad \blacksquare$$

Remark 1: Unlike the standard exterior angle sum (360°), the circular exterior angle sum grows linearly with n .

Remark 2: Arithmetic progression: For $n \geq 3$, let S_n be the sum of circular exterior angles for an n -sided polygon. Then the sequence $\{T_k\}_{k=1}^\infty$ defined by $T_k = S_{k+2}$ forms an arithmetic progression with a first term of 900° and a common difference of 180° .

This is true as $T_k = S_{k+2} = 180^\circ((k+2) + 2) = 180^\circ(k+4)$. The first term is $T_1 = 180^\circ(5) = 900^\circ$. The difference between successive terms is:

$$T_{k+1} - T_k = 180^\circ((k+1) + 4) - 180^\circ(k+4) = 180^\circ$$

Thus, $\{T_k\}$ is an arithmetic progression. $\blacksquare$

Remark 3: (Non-convex Polygons): For vertices with reflex angles (i.e., $a_j > 180^\circ$), extending sides yields an ambiguous standard exterior angle. However, the circular exterior angle $\theta_j = 360^\circ - a_j$ remains well-defined, positive, and strictly less than 180° . Thus, the theorem holds uniformly for both convex and non-convex polygons.

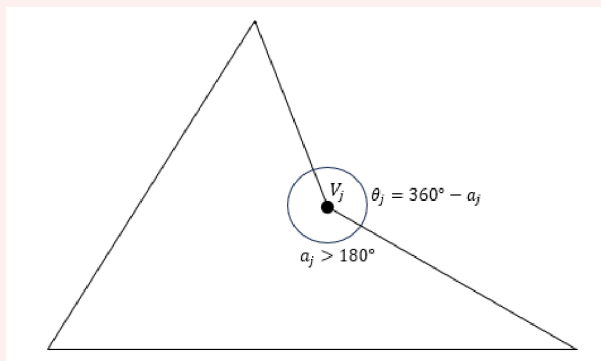


Figure 6

Discussion: The above result does not contradict the 360° exterior angle theorem for convex polygons; rather, it captures different geometric information (remaining rotation about a point instead of directional turning).

Degenerate Cases: If consecutive vertices become collinear, the interior angle is 180° , but the result continues to hold.

Further reading

[1] Euclid, *Elements*, Book I.

[2] S. Lozanovski, *A Beautiful Journey Through Olympiad Geometry*.

[3] H. S. M. Coxeter, *Introduction to Geometry*.

Generative AI Disclosure

ChatGPT (GPT-5.1, OpenAI) was used for language editing and improving clarity of exposition. The mathematical content, ideas, and proofs are entirely the author's own.

About the author

ADVIK RAJAANI is a student at Mother's International School, New Delhi. He has actively participated in Mathematical Olympiads since Grade 7 and is deeply interested in pure mathematics. He has interests in Algebra, Number Theory, Combinatorics, and Geometry, with a particular appreciation for elegant and insightful problem-solving. He may be contacted at advikrajaani@gmail.com.

Part V

News From Around the World

AI Just Solved an 80-Year-Old Problem

Adapted from: “AI cracks 80-year-old mathematics challenge — researchers are astonished.” <https://www.nature.com/articles/d41586-026-01651-0>. Author: Davide Castelvecchi

The article reports that OpenAI’s AI system has apparently solved an 80-year-old geometry problem posed by Paul Erdős, surprising mathematicians with an autonomous result that was independently verified. It frames the finding as a notable sign that AI is beginning to contribute original work in mathematics, not just assist humans.

Main idea

The article centres on Erdős’s unit-distance problem, which asks for the best way to place points on a plane so that as many pairs as possible are the same distance apart. Erdős believed no arrangement could beat the

growth rate he had found in his own construction, but OpenAI says its model found a counterexample. The model used algebraic number theory to choose point coordinates that satisfied particular equations, which led to a better arrangement than Erdős had conjectured was possible.

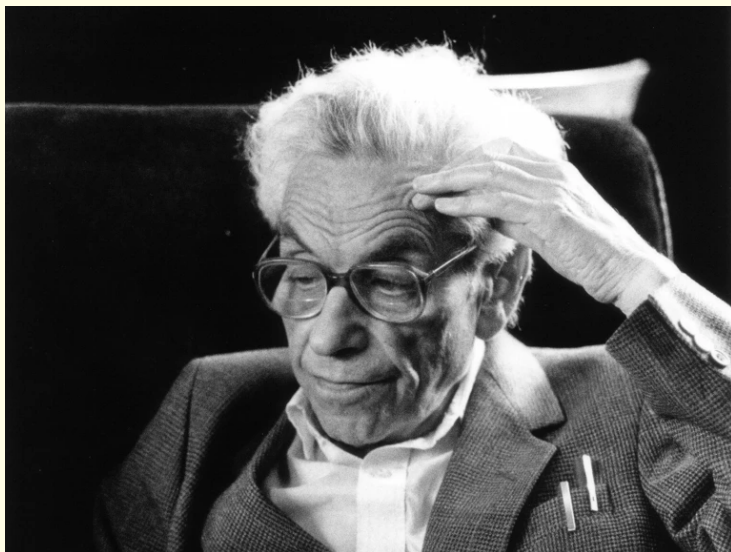


Figure 7: Paul Erdős, who published more than 1,500 papers during his lifetime, also left a legacy of more than 1,000 open research questions, some of which are now being solved with AI. Credit: George Csicsery

Why it matters

The result is presented as a milestone because it was not just a model helping with calculations, but a system producing a meaningful mathematical advance on its own. OpenAI's Sebastien Bubeck says this may be the first

time AI has autonomously produced an important result in any field, while other mathematicians quoted in the piece describe the finding as genuinely impressive.

The article also emphasizes the symbolic weight of the achievement. Erdős left behind many open problems, and the piece suggests that AI may now be able to chip away at some of the hardest ones in ways that were not previously practical for humans alone.

How it worked

According to the article, the AI solved the problem after a single prompt from mathematicians at OpenAI. The prompt was framed as an open-ended question about whether Erdős's conjecture could be true or false, and the system reportedly produced a long chain of reasoning that led to the proof.

A key detail is that the model used tools from algebraic number theory, a highly specialized area of math. The article highlights this as evidence that AI systems are starting to move beyond narrow "silos" of knowledge and combine ideas from across mathematical literature in ways that could be difficult for a person to do comprehensively.

What experts said

Several mathematicians quoted in the article responded with strong surprise or admiration. Tom Trotter says Erdős would have been delighted, Tony Feng calls the result incredible, and Daniel Litt says it is the first AI-produced result he finds interesting in itself.

At the same time, the article notes that OpenAI has not fully disclosed the system's identity or the complete 125-page reasoning document. That leaves some details opaque, even though the main result has been published and independently checked.

Overall message

The article's broader message is that AI may be entering a new phase in scientific work. Earlier systems often needed humans to guide them through repeated trial and error, but this example is presented as more direct and autonomous.

In short, the page argues that this is not just another flashy demo. It is being treated as a serious mathematical result that could change expectations about what AI can contribute to research in the near future.

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Adapted from: "An OpenAI model has disproved a central conjecture in discrete geometry." <https://openai.com/index/model-disproves-discrete-geometry-conjecture/>. Author: OpenAI

This webpage is an OpenAI announcement about a major result in discrete geometry: an internal model produced a proof that disproves a long-standing conjecture about the maximum number of unit-distance pairs among n points in the plane. The article presents the result both as a mathematical breakthrough and as evidence that advanced AI systems can contribute original ideas in research.

Main claim

The central problem is the planar unit distance problem, first posed by Paul Erdős in 1946, which asks how many pairs of points can be exactly 1 unit apart if you place n points in the plane. For decades, mathematicians believed the best constructions were essentially based on rescaled square grids, and that the growth rate of the maximum number of unit-distance pairs should be close to linear, written in the article as $n^{1+o(1)}$.

The new result says that this belief is false. The model found an infinite family of point configurations with at least $n^{1+\delta}$ unit-distance pairs for some fixed $\delta > 0$, which is a genuine polynomial improvement over the previously expected behaviour. The article notes that a later refinement by Princeton mathematician Will Sawin shows δ can be taken as 0.014.

Why the problem mattered

The article explains that the unit distance problem is famous because it is easy to state but very hard to solve. Erdős even offered a prize for it, and the problem became one of the best-known questions in combinatorial geometry. The piece also situates the new result against the background of earlier bounds: a lower-bound construction from the 1940s had remained essentially unchanged, while the best general upper bound has been $O(n^{4/3})$ since work by Spencer, Szemerédi, and Trotter in 1984.

The article emphasizes that the result is surprising not just because it settles a famous conjecture, but because it overturns a widely accepted geometric intuition. The old square-grid constructions were thought to be near-optimal, so the discovery that much denser examples exist changes how mathematicians may think about the problem.

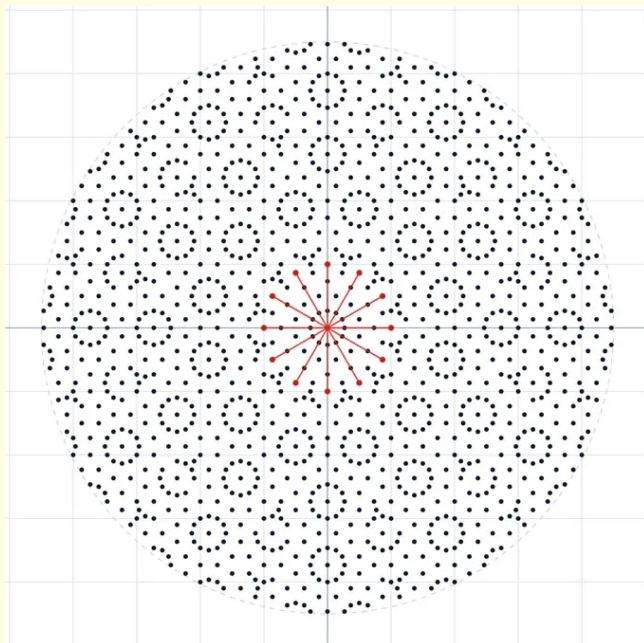


Figure 8: The AI produced a counterexample to Erdős's grid arrangement, which isn't easy to visualize. It involved building a complicated grid in a high-dimensional space then projecting it onto a flat plane. But this picture gets at the gist of the idea. Credit: Kai Williams/ChatGPT, based on an idea by Will Sawin

How the proof works

One of the most interesting parts of the article is that the proof uses ideas from algebraic number theory rather than purely geometric methods. The familiar starting point is Erdős's original idea, which can be interpreted

using the Gaussian integers, but the new construction replaces them with more complicated number fields that have richer symmetries and can generate more unit-length differences.

The article says the proof relies on deep tools such as infinite class field towers and Golod-Shafarevich theory to show that the required number fields exist. That is part of why the result is notable: a problem in elementary-looking Euclidean geometry is resolved using machinery from a very different area of mathematics. The article presents this as a striking example of unexpected connections between fields.

Role of the AI model

The piece makes a strong claim about the method of discovery: the proof came from a general-purpose reasoning model, not one specially trained or scaffolded for mathematics or for the unit distance problem in particular. According to the article, the model was tested on a broader set of Erdős problems as part of an effort to see whether advanced models can contribute to frontier research, and in this case it produced a proof that resolved the open problem.

The article frames this as an important milestone for both mathematics and AI because it is described as the first time a prominent open problem at the centre of a subfield has been solved autonomously by AI. It argues that mathematics is a particularly good test case for AI reasoning because proofs can be checked rigorously, and a long argument only works if every step is coherent.

Reactions from mathematicians

The article includes supportive remarks from prominent mathematicians. Tim Gowers calls the result “a milestone in AI mathematics,” while Arul Shankar says current AI models can do more than assist human mathematicians and are capable of “original ingenious ideas” carried through to completion.

It also quotes Thomas Bloom in a companion note, where he says the result teaches mathematicians something new about the problem and suggests that number-theoretic constructions may have more to say about discrete geometry than previously suspected. Bloom’s comments also point toward future research, suggesting that other open problems may be approachable through similarly deep algebraic ideas.

Broader significance

The article argues that the importance of the result goes beyond the specific conjecture it resolves. In the authors’ view, stronger mathematical reasoning in AI could make models better research partners by helping them maintain complicated arguments, connect distant ideas, and propose promising directions that humans might not prioritize immediately. They suggest this could matter not only in mathematics, but also in areas like biology, physics, materials science, engineering, and medicine.

The final message is cautiously optimistic: AI may become a powerful creative tool in research, but human judgment still matters. The article ends by stressing that people will still choose the problems, interpret the findings, and decide what to pursue next.

Overall takeaway

In plain terms, the page says an OpenAI model helped solve a famous and long-standing geometry problem by finding a surprising proof that uses deep number theory. The article presents this as both a mathematical surprise and an early sign that AI may soon play a much more active role in original research.

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Adapted from: “AI cracked an Erdős math problem. Now experts want guardrails.” <https://www.sciencenews.org/article/ai-guardrails-erdos-math-problem>. Author: Kathryn Hulick

OpenAI used a general-purpose large language model to produce a mathematical proof that disproved Paul Erdős’s long-standing unit-distance conjecture, and outside experts judged the result mathematically interesting and verifiable. The achievement has sparked excitement about AI’s usefulness in mathematics but also prompted a declaration calling for strict guardrails because of concerns about reliability, attribution, and access.

Context and the result

The unit-distance problem asks, for any number of points on a plane, what is the maximum number of pairs separated by exactly one distance; Erdős proposed a conjectured answer in 1946 that stood unproven for decades. OpenAI researchers fed the conjecture to an internal reasoning-trained large language model and found the model produced a counterexample and a formal proof, which OpenAI published on May 20; outside experts

reviewed and commented on the solution as part of accompanying material.

Why mathematicians find it interesting

Experts say the AI's counterexample is notable because it links tools from algebra and number theory to a geometric question, showing cross-field techniques can be unexpectedly powerful; several reviewers called the mathematics "beautiful" and helpful as inspiration for new approaches. The construction behind the counterexample uses a high-dimensional grid projected to the plane (a technical but conceptually describable approach), and interactive visualizations were created to illustrate the idea.

Concerns about AI's role

Many mathematicians caution that the result does not necessarily show a creative leap by AI — rather, the model's strength may be perseverance across many tries rather than insight—and some argued publicly that similar proofs could be reproduced with currently available models. A group of experts published a declaration urging tight guardrails for AI in mathematical research; the declaration had gathered many signatures within days, reflecting broad community concern about how such tools are used and governed.

Verification, transparency, and attribution issues

A central worry is how to verify AI-produced mathematics reliably: the model's reasoning can be brittle, AI-generated proofs can be long and opaque, and some users may produce pages of math they cannot them-

selves vet, making it hard to tell correct work from nonsense. OpenAI reported the prompt produced the correct solution in roughly half of repeated trials and claimed improvements in the model's ability to say "I cannot solve it," but it has not released full trial data or details about runtime, limiting external assessment. Experts also raise ethical questions about attribution because large language models draw on a broad corpus of prior work without a clear way to credit the human sources that inspired the ideas.

Access and the future of the field

There is concern that if the most capable systems remain private and costly, mathematical research could become less open and equitable, altering incentives for learning and practice; nonetheless, leading mathematicians remain cautiously optimistic that AI will become an important tool in research if used responsibly.

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Acknowledgment: This article has been prepared from the published article whose source is given above. The summary was made using the *Gemini 3.1 Pro/Perplexity* LLM.

Is AI Making Our Brains Weaker?

Adapted from: “Is AI Making Our Brains Weaker?” TIME magazine.
<https://time.com/article/2026/05/19/is-ai-making-our-brains-weaker/>. Author: Markham Heid

This is a TIME article by Markham Heid titled “Is AI Making Our Brains Weaker?” that argues AI may be changing how people think, learn, and persist on difficult tasks. Its main message is cautious: AI can be useful, but heavy reliance on it may weaken some cognitive skills if people use it as a substitute for independent thinking rather than a support.

Main argument

The article opens with a simple but important question: when does a technological aid become a crutch or a handicap? It uses recent research on AI chatbots and large language models to suggest that, at least in

some short-term experiments, leaning on AI can reduce people's own performance on tasks such as math, reading comprehension, and essay writing.

One study cited in the article found that people who used AI for about 10 minutes on math or reading tasks did worse later when solving similar problems without AI, and they gave up on hard problems more quickly. The authors interpret this as evidence that AI assistance can lower persistence as well as unaided performance.

Evidence from studies

The article also discusses a related MIT-led study in which people used ChatGPT to help write essays. According to the piece, those participants underperformed on neural, linguistic, and behavioural measures compared with people who wrote without AI, and they struggled to recall what they had written shortly afterward.

The article does not present these studies as final proof that AI permanently damages intelligence. Instead, it treats them as warning signs that dependence on AI may have immediate cognitive costs, especially if people repeatedly outsource tasks that normally build memory, reasoning, and writing skills.

Cognitive offloading

A major concept in the article is *cognitive offloading*, meaning the use of external tools to help with remembering, reasoning, and other mental work. The article says some researchers believe excessive offloading can cause short-term decline in the abilities that those tools replace.

The article emphasizes that this concern may be especially serious for children, because their cognitive abilities are still developing. It suggests that if AI keeps kids from practising core skills, the effects could be more harmful than they are for adults.

Different expert views

The article is not one-sided. It includes the view of researchers who are less worried, arguing that using AI may not simply take away mental effort but may shift it elsewhere. For example, if someone uses AI to draft an email, they may then have time and energy for more important work.

One expert quoted in the article compares mental skills to muscles: if you never use a specific ability, it may weaken. But the article also notes the possibility that people could gain new skills at the same time, such as learning how to judge AI output critically.

Why AI feels different

A key point in the article is that generative AI may be more concerning than older tools like calculators or search engines because it does more of the work for the user. Search engines still require people to read, compare, and synthesize information, while AI can produce a polished answer instantly, which may make users less critical over time.

The article introduces the idea of an “AI-trust spiral,” in which frequent use leads to more trust, which leads to more offloading, which then reduces critical thinking even further. This is presented as one of the biggest risks of widespread AI adoption.

Best-use debate

The article also explores how AI might be used in healthier ways. One expert argues that AI can improve thinking if people use it at the end of their own reasoning process, for example by asking it what they missed or requesting opposing views.

However, the article says many people use AI in the opposite way: they ask the system first and then try to think after the answer is already in front of them. The concern is that once a persuasive answer is generated, it becomes hard to keep thinking independently.

Overall takeaway

The article's overall conclusion is not that AI should be avoided, but that it should be used carefully, especially in education and other settings where people are still building foundational skills. It argues that society has not yet fully learned the best way to integrate AI without weakening the very abilities it is meant to support.

In short, the piece warns that AI may be helpful as a tool, but risky as a replacement for thinking. Its central message is that the more we let AI do our mental work, the more we may lose practice in the skills that make independent thought possible.

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Adapted from: "Is AI ruining our skills? Early results are in — and they're not good" <https://www.nature.com/articles/d41586-026-01947-1>. Author: Mariana Lenharo

The article argues that AI can cause real skill erosion, not just change how work gets done, and it backs that claim with early evidence from medicine and software engineering. It says the main warning is not that AI makes people useless, but that heavy reliance on AI can weaken the human ability to perform and learn tasks independently.

In medicine, the article focuses on a study of experienced Polish endoscopists who used an AI system that flagged adenomas during colonoscopies. When the tool was later unavailable, their adenoma-detection performance fell, from 28.4% before the tool's introduction to 22.4% afterward in colonoscopies done without AI help. The piece presents this as evidence that even highly skilled professionals can become less effective when they grow used to AI assistance.

The article also highlights concern among health workers more broadly, citing a survey in which 70% of nurses and 77% of physicians said they worried about losing skills because of over-reliance on AI systems. It quotes researchers who argue that awareness of this risk should prompt people to think carefully about which abilities they want to preserve and which they are comfortable outsourcing.

A second example comes from software development. Researchers at Anthropic ran a randomized controlled trial with 52 software engineers doing a coding task, and half were encouraged to use an AI assistant. Afterward, those who used AI scored worse on a quiz about what they had learned, averaging 50% versus 67% for the group without AI, especially on questions that required debugging and understanding the code.

The article uses that result to make a broader point: AI may help people produce good outputs while reducing the depth of their understanding. One quoted researcher describes this as a disconnect between performance

and learning, where people borrow capability from the AI rather than building it themselves. That concern is especially strong for students and younger workers who may be using these tools while still developing foundational skills.

It then widens the lens to other fields and technologies. The article notes that older tools like GPS already changed human skills, but argues generative AI is different because it automates parts of thinking and interpretation that were once considered distinctly human. It also cites earlier research on accountants using automated systems, where people forgot routine tasks once the tool was removed.

The article ends with a practical warning rather than a rejection of AI. The suggested response is mindfulness: understand how much work you are offloading, know the limits of the system, and avoid trusting AI outputs without questioning them. In short, the piece says AI is useful, but it may weaken expertise unless people deliberately keep practising the underlying skills themselves

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AI Can Translate Instantly. Why Learn Another Language?

Adapted from: “If AI can translate instantly, why learn another language?” <https://theconversation.com/if-ai-can-translate-instantly-why-learn-another-language-280310>.

Author: Olivia Maurice

The article argues that even though AI can translate quickly and accurately, learning another language still matters because the effort itself builds deeper thinking, memory, attention, and cultural understanding.

It says translation tools are useful for convenience, but they cannot replace the cognitive and human value of actually becoming multilingual.

Olivia Maurice begins by noting how real-time translation is now built into everyday tools, from video calls to social apps, which raises a fair question:

if machines can do the job, why spend years learning a language? She answers that humans have always used tools to reduce mental workload, but language learning is different from simple convenience because it involves active engagement rather than passive substitution. In other words, AI can support communication, but it should not become a shortcut that eliminates the learning process entirely.

A major theme of the article is that effort matters in learning. Maurice points to the idea of “desirable difficulties” — meaning that tasks that feel difficult in the moment often create stronger long-term learning. When people struggle with grammar, vocabulary, and meaning across languages, they exercise memory, attention, and cognitive flexibility more deeply than they would through passive translation. She links sustained multilingual use with cognitive resilience, suggesting that regular mental effort across languages may help the brain maintain function as people age.

The article then examines research on multilingualism and explains that the popular idea of a broad “bilingual advantage” is too simplistic. Some studies do find benefits in areas like attention or working memory, but others do not, so the effects appear to be more selective than universal.

Maurice discusses a recent study of 94 adults, aged 18 to 83, which measured multilingual experience on a spectrum rather than treating people as simply bilingual or monolingual. That study found that multilingual and monolingual participants were similar on most tasks, but people with richer multilingual experience performed better in visuospatial working memory, especially older adults.

The article uses this to argue that multilingualism may help preserve specific mental abilities over time rather than boosting every type of cognition equally.

Another key point is that AI translation cannot reproduce the full experience of language. Maurice acknowledges that AI is excellent for speed and accessibility, but says it relies on pattern recognition instead of lived understanding. It can miss humour, emotional tone, cultural context, and subtle social meaning, especially in languages that are less represented in training data. She illustrates this with the idea that a translated romantic scene may preserve the literal words but lose the vulnerability and sincerity that make the original exchange meaningful. For her, translation is not the same as participation.

The article also emphasizes that language learning is tied to identity and culture, not just communication. Maurice argues that speaking a language is a way of understanding how others think, how values are expressed, and how meaning changes across context and history. She cites multilingual speakers who describe different languages as serving different roles in their lives, such as one language for emotion and another for business, to show that languages can shape different “selves”. The conclusion is that AI may transform how people learn languages, but it cannot replace the deeper cognitive and cultural work involved.

Overall, the article’s message is that AI translation is useful, but learning another language still offers benefits that technology cannot fully duplicate: mental exercise, cultural insight, and richer human expression.

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AI May Worsen Wealth Inequality

Adapted from: “A.I. May Worsen Wealth Inequality. We Are Witnessing the Rise of a New Aristocracy.”

The New York Times.

<https://www.nytimes.com/2026/04/08/opinion/ai-wealth-inequality-jobs-investment.html>.

Author: Jennifer M. Harris

This opinion piece argues that artificial intelligence is likely to **intensify wealth inequality** rather than reduce it. Jennifer M. Harris says AI will most likely automate lower- and middle-income jobs, strengthen the bargaining power of high earners and investors, and concentrate both wealth and political influence in the hands of a small elite.

Main argument

The article opens by framing current inequality as already extreme, noting that a small number of wealthy households have added enormous sums to their fortunes and that this concentration is historically unusual. Harris argues that AI enters this environment as an accelerant, not a remedy: it could remove many jobs that sit in the middle and lower parts of the wage scale while leaving top earners largely protected. In her view, that means more money flowing upward, more pressure on affordability, and a weaker public sector because tax revenues would shrink.

Jobs and wages

A major part of the piece focuses on labour-market effects. Harris rejects the optimistic claim that AI will broadly equalize work by hurting elite knowledge workers more than everyone else. Instead, she says the first and hardest impacts will fall on administrative staff, office support workers, sales workers, and lower-level programmers, especially roles with salaries in the roughly \$40,000 to \$100,000 range. She points to recent job losses and corporate restructuring as evidence that this process has already begun, while higher-skill workers with networks, judgment, and elite credentials are more insulated.

She also argues that AI can reduce wages even for workers who keep their jobs. If fewer people are employed, remaining middle-class workers have less bargaining power, and if AI makes tasks easier to perform, employers can hire from a larger pool of workers at lower pay. The result, she says, is not just displacement but a broader weakening of the middle class.

Wealth concentration

The article then shifts from wages to ownership. Harris argues that the AI boom is enriching investors far more than workers, especially because much of the capital is flowing through private markets rather than widely held public stock markets. She contrasts this with earlier tech booms in which ordinary investors and employees could benefit through public-market exposure and retirement accounts.

By comparison, she says, major AI firms have raised huge sums from venture capital, private equity, and sovereign wealth funds, which keeps the upside concentrated among a very small group.

She emphasizes that AI companies are also relatively small employers compared with older tech giants, so their wealth creation is not broadly shared through payrolls.

In her telling, the result is a more extreme version of the “winner-take-most” economy: a few owners capture the gains, while the rest of society sees little direct benefit.

Economic costs

Harris argues that high inequality is not only unfair but economically harmful. She says when the top share of income rises too much, growth can slow because lower- and middle-income households spend more of their earnings than wealthy households do. That means less demand circulating through the economy.

She also links inequality to the affordability crisis, saying that wealthy households increasingly shape markets for housing, healthcare, and cars through their spending patterns.

To illustrate the point, she notes that San Francisco home prices have surged, reflecting the concentration of AI wealth in that region. In her view, once a narrow group of buyers dominates demand, companies design products and services around that group's preferences, pushing essentials further out of reach for everyone else.

Fiscal and political effects

Another theme is the strain AI could place on government finances. Because investment income is taxed less heavily than wages, Harris argues that shifting value from labour to capital will shrink the tax base. She says that means fewer public revenues for safety-net programs, workforce training, and other services that support lower- and middle-income families. That fiscal weakening, she warns, would reinforce the same inequality that caused the problem in the first place.

The article also links wealth concentration to politics. Harris says billionaire influence has grown sharply in recent election cycles and that AI companies are spending more to shape political races. She warns that as economic power turns into political power, democracy itself becomes more fragile.

Possible responses

Harris does not present the situation as inevitable. She says AI could, in principle, be designed to complement workers, expand human expertise, or create new tasks that need people rather than replacing them. But she says current trends point in the opposite direction.

She outlines several policy responses, including taxing investment income more like wages, creating public equity stakes in AI companies, and en-

couraging worker-owned cooperatives.

The broader message is that AI should not be treated as a neutral technological wave. Harris argues that without deliberate policy changes, it will deepen wealth concentration, worsen affordability, shrink tax revenues, and weaken democratic accountability. Her closing warning is that preserving “democratic capitalism” will require action soon, before AI locks in a more aristocratic economic order.

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Wildlife and the Chernobyl Disaster

Adapted from: “Surviving in a poisoned land: Chernobyl’s wildlife is different, but not in the ways you might think.” BBC.

<https://www.bbc.com/future/article/20260424-chernobyl-wildlife-forty-years-on>.

Author: Chris Baraniuk

This webpage argues that Chernobyl’s wildlife has changed in complex ways, but not in the simple “radiation made everything mutate” sense. It emphasizes that the biggest ecological shift may be the absence of humans, while radiation, heavy metals, habitat change, and long-term evolutionary pressures all likely interact.

The article opens with evolutionary biologist Pablo Burraco catching a small tree frog in the Chernobyl exclusion zone and noticing that it was darker than frogs from elsewhere. That observation becomes the starting

point for a wider question: whether radiation from the 1986 disaster has altered the animals and plants living near the ruined reactor.



Figure 9: Feral dogs, descendants of pets abandoned around Chernobyl, now roam close to the containment shield where the devastated reactor is now housed (Credit: Timothy Mousseau/ CFF)

A major theme is uncertainty. The article explains that researchers have seen unusual traits in Chernobyl wildlife, such as dark frogs, twisted trees, tumours in swallows, feral dogs with genetic patterns of interest, and black fungi living in the reactor ruins, but it is often difficult to prove that radiation alone caused these changes. Other contaminants, especially heavy metals, may also be involved, and some effects may simply reflect a changed environment after people left.

The tree frog research is presented as suggestive but not conclusive. Bur-raco and colleagues sampled more than 250 frogs and reported that frogs

inside the exclusion zone were on average darker, which they hypothesize could be linked to higher melanin levels offering some radiation protection. However, the article also notes criticism from Timothy Mousseau, who says the sampling was not comprehensive enough and that the results do not prove a direct relationship with current radiation levels.

Another section explains that the landscape itself changed dramatically after the disaster. Many pine trees died, birch trees took over in some places, and the forest around Prip'yat became a very different ecosystem from the one that existed before 1986. This matters because animals respond to habitat changes as well as contamination, so some wildlife differences may reflect the new forest structure rather than radiation effects alone.

The article also stresses the role of human absence. Once the area was evacuated, species such as wolves, bears, bison, deer, wild boar, elk, and Eurasian lynx either expanded or returned, benefiting from the lack of human disturbance and the abundance of prey. Wolves in particular are described as being much more numerous in the exclusion zone than in nearby reserves, and abandoned dogs still roam the area as descendants of pets left behind.

The piece then turns to the question of adaptation. Some scientists think certain organisms may have evolved traits that help them cope with radiation, but the article presents this as controversial and still unproven in most cases. It mentions evidence that soybeans grown in the area may have adapted to radioactivity and heavy metals, that bank voles show greater DNA-damage resistance, and that melanin-rich black fungus may benefit from darker pigmentation in radioactive ruins.

At the same time, the article is careful not to overstate these findings. It says there is still no answer on whether the darker frogs truly evolved

their colour as a protective adaptation, and it rejects claims that the fungus “eats” radiation as unsupported. It also highlights the importance of distinguishing newly acquired mutations from changes passed down over generations, noting one study in which chromosome abnormalities persisted in bank voles even after they were removed from Chernobyl conditions.



Figure 10: Elk have been documented in growing numbers following the evacuation of humans from the exclusion zone around Chernobyl (Credit: Getty Images)

The conclusion is that Chernobyl's ecosystem is neither dead nor simply "mutated." Instead, it is a living system shaped by radiation, abandonment, habitat shifts, and time, producing a mix of resilience, adaptation, and damage. The article ends by showing that the disaster's effects extend beyond the exclusion zone, with trace radionuclides found in items such as mushrooms, blueberries, and firewood far from Ukraine.

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Adapted from: "Decades after Chernobyl disaster, this radioactive landscape has become one of wildlife's most unlikely strongholds." ScienceX online magazine

<https://sciencex.com/news/2026-05-decades-chernobyl-disaster-radioactive-landscape.html>

Author: Sanjukta Mondal & Robert Egan

The article explains how the Chernobyl Exclusion Zone (CEZ for short), despite remaining highly radioactive, has become an important refuge for wildlife after nearly four decades of low human presence. It focuses on a recent study showing that the least disturbed protected areas in northern Ukraine, especially the Ukrainian CEZ and the adjacent Drevlianskyi Reserve, support the highest mammal diversity and occupancy rates.



Figure 11: The Ukrainian CEZ and adjacent Drevlianskyi Reserve had the highest mammal diversity, including rare species like lynx and Przewalski's horse. Credit: Anton Zelenov, Wikimedia Commons.

The piece begins with the 26 April 1986 Chernobyl disaster, when a reactor explosion forced people to flee and turned Pripyat and the surrounding area into a deserted landscape. It then contrasts that human catastrophe with the ecological outcome: over time, the abandoned zone has developed into a space where many animals can live with limited interference.

The study discussed in the article examined about 60,000 km² in northern Ukraine and compared several land categories, including the CEZ, nature reserves, national parks, and areas with no legal protection. Researchers used 174 motion-sensing camera traps to record wildlife and applied Bayesian occupancy models to estimate which species were present and how often they were detected.

According to the article, the CEZ and Drevlianskyi Reserve stood out for having the greatest concentration of rare or sensitive mammals. Species highlighted include the Eurasian lynx, gray wolf, red deer, and the endangered Przewalski's horse, along with moose, roe deer, wild boar, brown hare, red fox, raccoon dog, and European badger. The article says these findings support the idea that low human activity can allow wildlife populations to recover, even in a landscape still marked by radiation.

The article also places the Chernobyl case in a broader conservation context. It notes that biodiversity loss is accelerating worldwide because of habitat destruction, agriculture, urbanization, and industrialization, and that protected areas are one of the best tools for reducing disturbance and helping ecosystems rebound. In that sense, the Chernobyl zone is presented as an unusual but powerful example of how wildlife can flourish when human pressure is removed.

A further point made in the article is that small protected areas may not always be enough for large mammals. Some species do well in larger, strictly protected landscapes, while smaller parks may offer too little space or protection to make a major difference. The article closes by emphasizing that the Chernobyl Exclusion Zone shows nature's capacity to reclaim land once people are gone, even under extreme conditions.

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Fires in Europe Are Breaking Records: Here's What the Future Holds

Adapted from: "Fires in Europe are breaking records: here's what the future holds." *Nature*

<https://www.nature.com/articles/d41586-026-02361-3>.
Nature

Author: James Dinneen

The article examines the record-breaking wildfires affecting France and Spain during Europe's 2026 fire season and explains why scientists believe such events may become increasingly common. Fueled by an unprecedented heatwave, fires have spread rapidly through exceptionally hot, dry vegetation, breaking national records even though the season is only partly

complete. By 29 July, approximately 91,000 hectares had burned in France — far exceeding the previous full-season record of just over 66,000 hectares. Spain had recorded more than 200,000 hectares of burned land, while a fire near Madrid had become the largest single wildfire in the country's history.



Figure 12: Firefighters watch a blaze west of Madrid

The immediate effects have been severe. Hundreds of thousands of people have been evacuated, and major infrastructure has been threatened. The Madrid fire nearly destroyed a NASA facility used to communicate with spacecraft operating in deep space. In France, firefighters confronted a huge “fire cloud” powerful enough to create its own unpredictable weather, an event described as the first of its kind observed in the country. With another heatwave developing, authorities and scientists remain concerned that the season could worsen before it ends in late October.

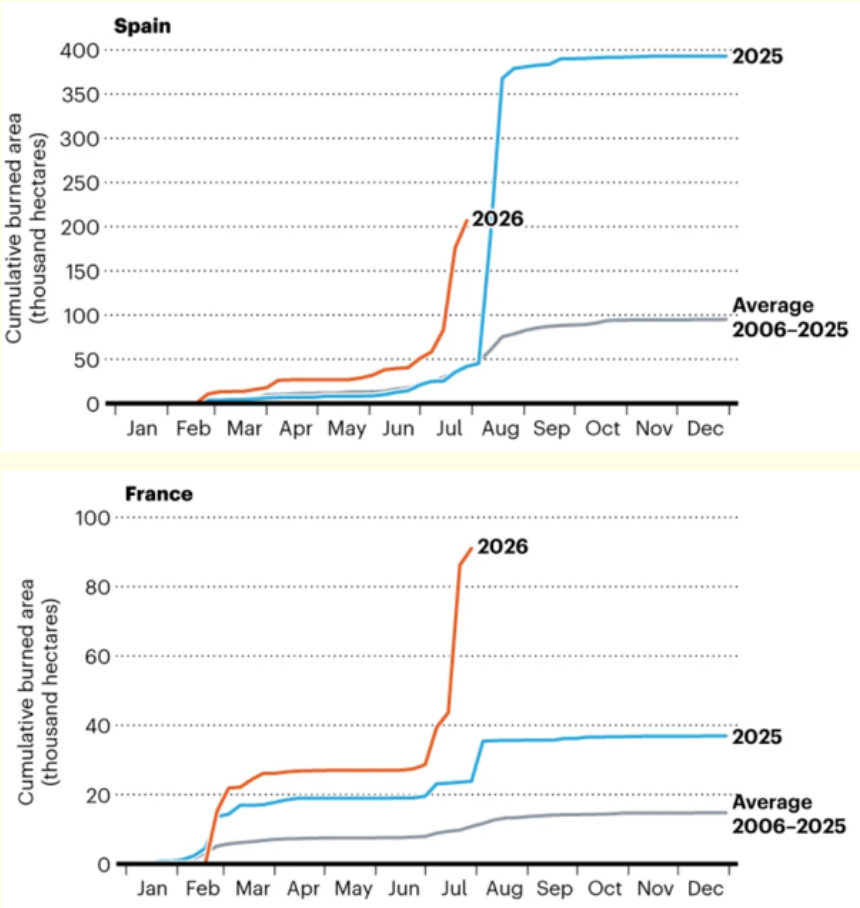


Figure 13: Europe ablaze. Wildfires in France have burnt a record area only halfway through the fire season, which ends in late October. Spain’s fires could also set a record, just one year after a devastating fire season in 2025. Credit: *Nature*

The article presents these fires as evidence of a possible transformation in southern Europe's fire regime. For decades, countries such as Spain, Portugal, France, Italy and Greece experienced comparatively manageable wildfire activity, despite having naturally fire-prone Mediterranean climates. Researchers now say the region may be moving toward conditions more familiar in places such as California, where extreme fires are a recurring hazard. A recent study found that the intensity of hot, dry weather conducive to fire in parts of France and Spain was up to 50% greater during the past decade than during the 1981–2010 period. Areas of Spain and Portugal that previously experienced fewer than ten highly fire-promoting days annually now record as many as 25.

However, the article highlights an important complication: wildfire activity actually declined across much of southern Europe between 2016 and 2025. Raúl Cordero, a meteorologist at the University of Groningen, and colleagues examined records dating back to the 1980s. They found that the number of fires fell by at least 40% in every country studied, while the total area burned also decreased in most places. Portugal was the main exception because of its exceptionally destructive fires in 2017. This earlier decline occurred even as temperatures and fire-weather risk were increasing.

Researchers suggest that improved prevention, firefighting, land management and other factors may have helped suppress many fires during that period. Yet suppression does not eliminate the underlying danger. When extreme weather eventually overwhelms these protections, accumulated dry vegetation can provide abundant fuel, allowing fires to become unusually intense and difficult to control. The record fires of 2025 and 2026 therefore suggest that the earlier period of relatively low activity may

have been temporary rather than evidence that climate-related risks were diminishing.

Climate change is identified as the central long-term driver. Human-caused warming raises temperatures and makes heatwaves more likely; heatwaves, in turn, produce the hot, dry and sometimes windy conditions that dry vegetation and accelerate combustion. Multiple heatwaves since May 2026 have left landscapes especially parched, creating a pattern similar to Spain’s intense heat followed by devastating fires in 2025. Previous projections indicate that continued warming this century will substantially increase Europe’s fire risk. The article’s overall warning is that larger, more dangerous wildfires are no longer merely a future possibility: they are already occurring, and adaptation efforts will need to keep pace with a worsening climate.

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Ants That Navigate Using the Moon

Adapted from: "These ants navigate with a compass tuned to the moon." *Scientific American*.

<https://www.scientificamerican.com/article/these-ants-navigate-with-a-compass-tuned-to-the-moon/>.

Author: K. R. Callaway (edited by Sarah Lewin Frasier)

This article from *Scientific American* reports that researchers have discovered a previously unknown nocturnal navigation system in a species of bull ant: these ants use the moon as a compass and compensate for its movement over time to travel between nest and food at night. The finding overturns the assumption that bull ants must rely on daytime cues or scent trails to orient after dark, showing instead that the ants combine a time-compensated lunar compass with terrestrial and solar cues to maintain accurate routes across changing night conditions. Experimental

manipulations — holding ants in darkness so they could not track time, then releasing them in new locations — produced predictable, systematic errors that reveal the ants' dependence on the moon's position and an internal timekeeping mechanism. The study's authors and outside experts emphasize the surprise and elegance of the mechanism, note that other nocturnal animals sometimes use lunar cues, and argue that this discovery both expands understanding of ant navigation and suggests new lines of inquiry into the sensory and neural systems that support orientation in low light.

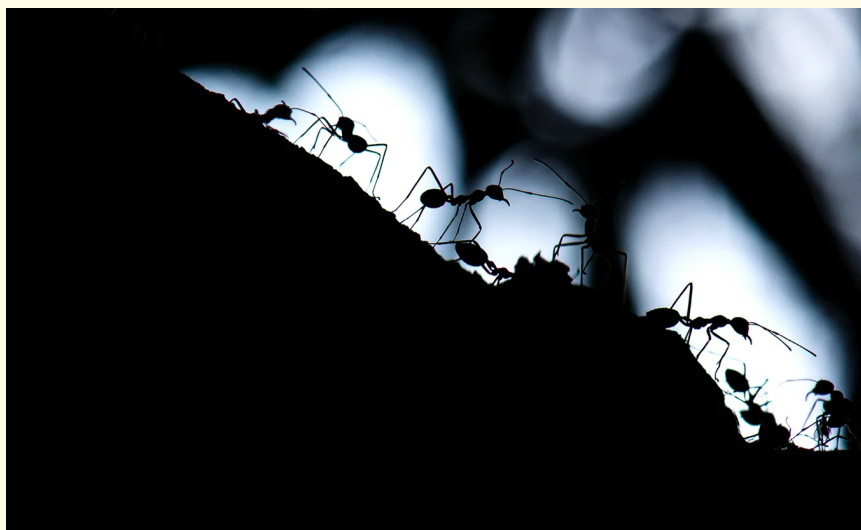


Figure 14: Ants show newfound lunar navigation skills

Background and previous assumptions

Before this work, entomologists understood that different ant species use a mix of cues — scent trails, landmarks, the sun's position and polarized light patterns — to navigate. Many nocturnal foragers use chemical trails or follow daytime-established routes, and bull ants had been thought to rely primarily on non-olfactory cues available near dusk or dawn rather than an active night-time celestial system. The new study challenges that picture by showing that at least one bull ant species continues to navigate through the night using information from the moon, rather than simply reusing day-established visual references or following scent alone.

Key experiment and methods

Researchers captured foraging bull ants that were en route to food and divided them into treatment groups: some ants were kept in transparent containers that preserved environmental light and temporal cues, while others were placed in darkened boxes that removed time-of-night information. After several hours, ants were released at unfamiliar locations and their homing behaviour observed. Ants deprived of time cues exhibited consistent deviations in heading that corresponded with where the moon had moved while they were held in darkness — evidence that the animals normally track the moon's position and compensate for its orbital motion by using an internal clock to update the moon-based heading.

Interpretation and mechanism

The behaviour is described as “time compensation”: ants estimate how much time has passed since leaving the nest so they can predict where the

moon should be, then steer accordingly — a strategy comparable to how some early human navigators used stellar or solar references with temporal knowledge to fix direction. The authors argue that this is an *innate* lunar compass combined with other cues; ants do not rely solely on the moon but integrate lunar position with terrestrial landmarks and occasional solar information at dawn and dusk to produce robust navigation even when the moon is dim or absent during parts of the lunar

Context and significance

Entomologists called the finding surprising and impressive because compensating for the moon's trajectory requires integrating timekeeping with celestial observation — a computationally non-trivial task for a small insect brain. Although other nocturnal animals (for example, some moths and sand hoppers) have been suspected to use the moon for orientation, these bull ants are the first documented case of a time-compensated lunar compass of this sophistication in ants. Because there are more than 12,000 ant species, the discovery suggests that other nocturnal ants might use related or different mechanisms; studying this species may therefore reveal broader principles about sensory integration and navigation in low-light ecological niches.

Implications and next steps

The researchers and outside experts propose follow-up work to identify the sensory and neural circuits that let ants detect lunar position and keep time, to test how the lunar compass interacts with other cues across different phases of the moon, and to examine whether related species use similar strategies. The result also prompts re-evaluation of assumptions about noc-

turnal insect orientation and encourages field and laboratory studies that probe how animals maintain accurate bearings when primary cues become unreliable. Overall, the study both reveals an elegant natural solution to a navigation problem and opens multiple avenues for comparative and mechanistic research.

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Right-Handed or Left-Handed?

Adapted from: “Why is almost everyone right-handed? The answer may lie in how we learned to walk.” University of Oxford.

<https://phys.org/news/2026-05-why-is-almost-everyone-right.html>.

Author: Lisa Lock & Lisa Lock

New research led by the University of Oxford argues that humans’ near-universal right-handedness can be explained by two linked features of our evolution — upright, bipedal walking and the dramatic expansion of the human brain — rather than by factors like tool use or social structure alone. The research (by Thomas A. Püschel et al.) appears in PLOS Biology (2026) under the title “Bipedalism and brain expansion explain human handedness.”

Study overview and methods

The study pooled data on 2,025 individuals across 41 monkey and ape species and used Bayesian models that account for evolutionary relationships to test major hypotheses about the origins of handedness, including tool use, diet, habitat, body mass, social organization, brain size and locomotion. When analyzed against those factors, humans initially appeared as an outlier; however, adding brain size and a limb-proportion indicator of bipedalism (arm-to-leg length ratio) removed that exceptional status, showing that upright walking and larger brains together explain the human pattern.

Key findings about hominin history

Applying the same models to extinct hominins produced a gradient of hand preference across our ancestors: early hominins such as *Ardipithecus* and *Australopithecus* likely had only mild rightward biases similar to great apes, while members of the genus *Homo* (including *H. ergaster*, *H. erectus* and Neanderthals) showed progressively stronger rightward biases, culminating in the extreme modern pattern in *H. sapiens*. An important exception is *Homo floresiensis*, which the models predict had a much weaker right-hand preference; the authors link this to that species' small brain and partly arboreal body plan rather than full bipedalism.

Proposed two-stage evolutionary story

The authors propose a two-stage explanation: first, the adoption of bipedalism freed the hands from locomotor duties and created selective pressure for more specialized, lateralized manual tasks; second, later brain expansion

sion and reorganization strengthened and stabilized a rightward manual bias, producing the near-universal right-handedness seen today.

Implications and open questions

The study is framed as the first to test multiple leading hypotheses for handedness within a single comparative framework, allowing researchers to separate ancient, shared traits from uniquely human developments. Remaining questions include the role of cumulative culture in stabilizing right-handedness, reasons why left-handedness persists at lower frequencies, and whether similar lateralization patterns in other animals (for example parrots or kangaroos) reflect convergent evolutionary processes.

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David Attenborough at 100

Adapted from: “At 100, David Attenborough’s Voice Is a Lesson in Wonder and Planetary Stewardship.”

<https://time.com/article/2026/05/07/david-attenborough-100-birthday-prince-harry/>.

Author: Prince Harry, Duke of Sussex

This article documents a deeply felt appreciation of the life and work of Sir David Attenborough on the occasion of his 100th birthday, arguing that his greatest achievement is not just his career as a broadcaster, but the way his voice has taught audiences to see nature with wonder and to treat the planet with care. It presents him as a trusted cultural figure whose calm, non-preachy style has helped turn environmental concern into something emotionally urgent and personally meaningful.

Core message

The author says Attenborough matters because he makes people care before he makes them worry. His narration of shows like *Planet Earth* and *Blue Planet* is described as a “Voice of Nature” that invites viewers into the natural world rather than lecturing them about it.

The essay’s central idea is that **wonder** is the starting point for protection. By helping audiences feel admiration for glaciers, forests, oceans, and wildlife, Attenborough makes the case that people are more likely to defend what they truly value.

Attenborough’s influence

The piece emphasizes that Attenborough’s authority comes from decades of consistency, humility, and close observation of the natural world. The author notes that when he warns about climate instability, it carries unusual weight because he is speaking from long experience rather than from ideology or performance.

It also argues that he helped dismantle the idea that environmental damage happens “somewhere else.” His work has connected distant natural places to everyday human life, showing that ecosystems, consumption, and climate are all linked to the stability of communities everywhere.

Human moments

A memorable part of the article focuses on the more human, unscripted moments in his work. The author highlights times when Attenborough seemed amused by accidents or delighted when animals interrupted his

narration, suggesting those moments revealed real reverence and curiosity rather than just technical expertise.

These scenes are used to show that his connection to nature is not detached or academic. Instead, the article portrays him as someone who is genuinely at home in the natural world and who finds joy in its unpredictability.

Reaching younger audiences

The article says Attenborough's message has become even more relevant for younger generations who grew up with his documentaries and now encounter his work through streaming and social media. Rather than sounding didactic, he comes across as credible and steady in a noisy and uncertain age.

For young viewers, the author suggests, he offers more than facts about wildlife. He provides a model of lifelong environmental stewardship rooted in patience, attention, and the willingness to stay engaged even when the scientific outlook is grim.

Closing argument

In the end, the piece frames Attenborough as both a witness to environmental decline and a guide toward action. The author argues that his legacy is a blend of clear-eyed realism and childlike wonder, and that this combination is exactly what decision-makers and citizens need if they are to respond meaningfully to ecological crisis.

The final note is sobering: Attenborough has spent a century showing humanity the beauty and fragility of the world, and the remaining question is whether people in power will act before more of that world is lost.

Adapted from: “As David Attenborough turns 100, four experts explore his legacy, from science to storytelling.”

<https://theconversation.com/as-david-attenborough-turns-100-four-experts-explore-his-legacy-from-science-to-storytelling-279275>.

Author: Saffron O'Neill

The article presents David Attenborough's 100th birthday as a moment to assess how he transformed science communication, natural history television, and public thinking about climate change. It is organized into four expert perspectives, and the summary below preserves that structure while condensing the article's main arguments.

Introduction

The piece opens by arguing that Attenborough is not just a famous broadcaster but one of the most trusted and effective storytellers ever to appear on television; one whose career has connected generations of viewers to the natural world, while also tracking the growing environmental crisis over the same period. It frames his 100th birthday as an occasion to examine the full scope of his influence, from scientific education to conservation and climate communication. The central claim is that Attenborough's legacy lies in combining wonder, accuracy, and narrative skill in a way that made complex ecological and scientific ideas accessible to large audiences. Rather than simply celebrating his fame, the article asks what his methods changed in broadcasting and what they have meant for public understanding of nature and climate.

Scientific insight

Ben Garrod's section emphasizes Attenborough's role as a bridge between science and the public. Garrod describes a field experience with Attenborough that captures his childlike excitement about nature, showing that his curiosity has remained vivid even after decades of work. The article uses this anecdote to stress that Attenborough's communication style is rooted in genuine wonder, not performance alone.

Garrod argues that Attenborough helped make evolutionary biology understandable to millions through series such as *Life on Earth*. Concepts like natural selection, adaptation, ecology, and behaviour are presented in his work not as abstract theory but as living processes visible in real animals and ecosystems. The article says that this helped normalize evolutionary thinking in popular culture while keeping the science accurate and current through collaboration with researchers.

This section also notes that Attenborough's tone evolved as the evidence changed. Early programs emphasized abundance and discovery; later ones increasingly focused on habitat loss, biodiversity decline, and extinction risk. Garrod presents this as a sign of credibility: Attenborough adapted his message as science developed, rather than freezing it in nostalgia.

The section ends by arguing that Attenborough's greatest contribution to conservation was helping people care, not just know. By fostering empathy, curiosity, and emotional attachment to nature, he created the conditions in which conservation action becomes possible. In this view, his legacy is inseparable from the environmental crisis of the present, because his work taught audiences why protecting life matters.

Natural history film-making

Jean-Baptiste Gouyon's section focuses on Attenborough's impact on the form of natural history television itself. It explains that in the early 1950s British television was still developing its visual language, with many early programs resembling radio with pictures. Into that environment came Attenborough, who helped redefine how wildlife stories could be told on screen.

The article identifies *The Coelacanth* and *Zoo Quest* as early examples of this transformation. *The Coelacanth* avoided sensationalism and instead connected a newly discovered fish to Darwinian evolution, while *Zoo Quest* introduced a more controlled, film-based documentary style with Attenborough as both on-screen presenter and narrative centre. These programs established a format in which wildlife films became not just records of animals, but guided scientific stories.

Gouyon stresses that Attenborough valued detailed scripting and controlled recording over live television. This preference helped shape a more polished and repeatable model for documentary production, which in turn made wildlife programming more sustainable and more widely distributable. The article suggests that his methods helped shift natural history television from a niche interest for amateur naturalists into a mainstream genre.

The section also highlights Attenborough's BBC management role, where he pushed for natural history TV to become serious science communication rather than simple visual spectacle. That vision later culminated in *Life on Earth* and continues in later series such as *Planet Earth II*, *Blue Planet II*, and *Our Planet*. The conclusion is that Attenborough helped create the

modern model of natural history film-making — cinematic, informative, and emotionally compelling.

Communicating research

Saffron O'Neill's section shifts to climate communication and the question of why Attenborough remains so persuasive. The article says he is one of the few climate voices that broad audiences across the political spectrum are willing to trust. His reach is tied to both his long career and his ability to adapt to changing media habits and platforms.

The section points to evidence that his name recognition is extraordinarily high and his audience unusually diverse. It argues that this matters because climate communication is not only about factual accuracy but also about delivery, trust, and storytelling. Attenborough's broad appeal allows him to communicate urgency without alienating audiences in the way more partisan voices sometimes do.

O'Neill also notes that Attenborough has kept pace with new media, including Instagram and streaming platforms. The article presents this as part of his strength: he meets audiences where they are rather than insisting they come to old formats. His documentaries do not necessarily produce immediate behaviour change on their own, but they can generate attention, concern, and policy interest.

A key point is that effective engagement with climate change requires more than information. People must care, feel motivated, and believe action is possible. The article argues that Attenborough's work succeeds because it combines emotional storytelling, striking visuals, and scientific credibility, making him a model for climate communication in the digital age.

Speaking up on climate

Chloe Brimicombe's section examines whether Attenborough's climate warnings came too late, and how his messaging changed over time. It notes that early documentaries emphasized beauty and wonder, while later work increasingly addressed human harm to the environment. The article says some of his earlier framing was influenced by ideas linking climate change to overpopulation, which the author contrasts with evidence that the wealthiest groups produce the most pollution.

The section argues that Attenborough's views evolved with the science. In more recent years he has spoken more directly about climate change in films such as *Climate Change: The Facts* and *A Perfect Planet*. This shift is treated as a significant move from implicit concern to explicit advocacy.

Brimicombe also reflects personally on how Attenborough shaped her own path into climate science. She describes his films as part of UK culture and as a source of inspiration for students and researchers alike. The article uses these examples to show that his influence extends beyond television viewing into education, scientific careers, and public imagination.

The section ends by arguing that documentaries remain a powerful way to communicate climate change to audiences who might otherwise stay disengaged. It acknowledges criticism that documentaries can emphasize the film-maker's perspective over scientific context, but says this has led to better collaboration between scientists and producers. The article concludes that Attenborough was late in addressing climate change directly, but that his recent work has made him a strong and influential advocate.

Overall message

Taken together, the four sections argue that Attenborough's importance comes from more than longevity or popularity. He helped shape the language of natural history television, made scientific ideas understandable to broad audiences, and later became a powerful voice on climate and conservation. The article treats him as a rare communicator whose authority comes from trust, craftsmanship, and the ability to evolve with the evidence.

The final message is hopeful but urgent: Attenborough's legacy matters because it has helped people care about the natural world, and that care now needs to become action. The article presents his 100th birthday not as an ending, but as a reminder that the stories he told are still unfinished.



Figure 15: Attenborough looks at an armadillo for his show *Attenborough and Animals* in 1963. Credit: BBC Public Service



Figure 16: Attenborough, his wife Jane and his children Robert and Susan at home in Richmond with their pet tortoise in 1955. Credit: Smith Archive



Figure 17: Attenborough standing within a Creosote Bush in the Mojave Desert, which he previously visited 40 years ago. The bush had grown an inch since he last saw it. Credit: Paul Williams

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Acknowledgment: This article has been prepared from the published article whose source is given above. The summary was made using the *Gemini 3.1 Pro/Perplexity LLM*.

Understanding One of the Greatest Mysteries of the Sun

Adapted from: “Unprecedented images may explain one of greatest mysteries of the sun.” *The Guardian*.

<https://www.theguardian.com/science/2026/aug/05/unprecedented-images-may-explain-one-of-greatest-mysteries-of-the-sun>.

Author: Ian Sample (Science editor)

New Images May Help Solve a Solar Mystery

Astronomers have captured the most detailed images ever obtained of the Sun’s turbulent surface, revealing tiny golden bands and swirling structures that may help explain one of solar physics’ greatest mysteries: why the Sun’s outer atmosphere, or corona, is vastly hotter than its visible surface.

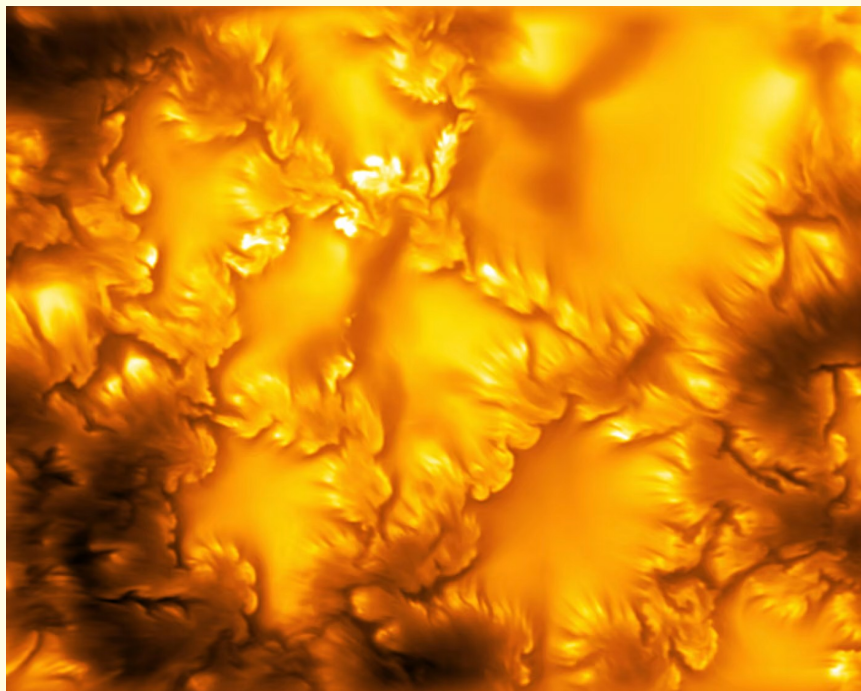


Figure 18: The most high-res image of the sun. Credit: NSO/Aura/NSF

The observations were made with the Daniel K. Inouye Solar Telescope in Hawaii, the most powerful solar telescope in the world. Researchers examined a magnetically active region near a sunspot and identified features smaller than 20 kilometres across. The images reach the telescope's current limits and show details that previous observations could not resolve.

Earlier pictures revealed the Sun's surface as a patchwork of bright, grain-like structures, each roughly comparable in size to France. Those obser-

ventions could distinguish features measuring about 30 kilometres across. The new images provide a much sharper view, allowing scientists to study previously unseen activity on a much smaller scale.

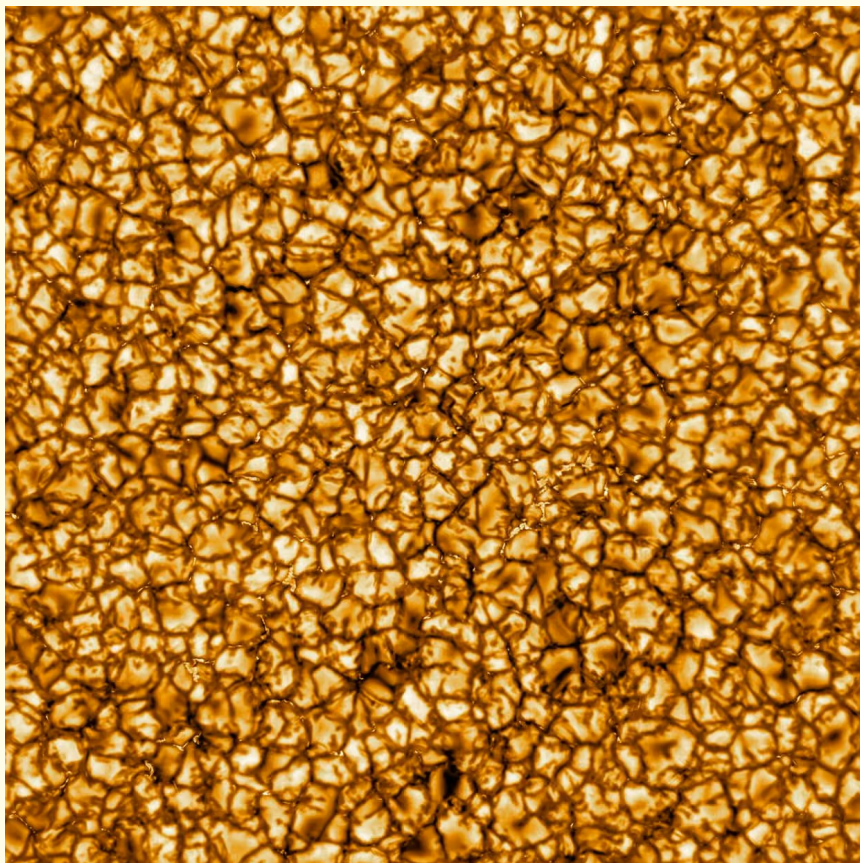


Figure 19: Image of the sun's surface taken in early 2020. Photograph: NSO/Aura/NSF

Researchers were especially interested in small, whirlpool-like patterns visible in the solar plasma. They believe these structures are signs of *Kelvin–Helmholtz instabilities* — physical disturbances that develop when a fast-moving fluid or plasma flows alongside a slower-moving one. The difference in speed creates shear at the boundary, causing small waves to grow into spiralling vortices that resemble breaking waves.

Kelvin–Helmholtz instabilities have been observed in Earth’s oceans, lakes and clouds, as well as in the atmospheres of Jupiter and Saturn. However, scientists had not previously confirmed them on the Sun. The new observations, reported in *Nature*, could therefore provide an important advance in understanding how energy moves through the solar atmosphere.

The discovery may also shed light on the Sun’s powerful magnetic activity. Solar flares, jets and coronal mass ejections are driven by rapidly changing magnetic fields and can disrupt satellites, GPS systems, communications networks and power grids on Earth.

Scientists understand that the Sun’s magnetic fields are generated by the movement of hot, electrically charged plasma, but many details remain unresolved. Magnetic field lines can twist and become tangled, storing energy before suddenly releasing it. What initiates this process, however, has remained uncertain.

The newly observed vortices may be part of the answer. By stirring the plasma, they could twist and braid magnetic field lines, allowing energy to move from large structures to increasingly smaller ones. Eventually, that energy may dissipate as heat. This process could help explain why the corona reaches temperatures of several million degrees, even though the Sun’s visible surface is approximately 6,000°C.

The findings do not settle the coronal-heating problem entirely, but they offer scientists a promising explanation. Further observations will be needed to determine how common these instabilities are and how much energy they transfer. If confirmed, the process could also explain the hot outer atmospheres of other stars.

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Acknowledgment: This article has been prepared from the published article whose source is given above. The summary was made using the *Gemini 3.1 Pro/Perplexity* LLM.

Part VI

Contributed Articles – 2

The Bucket Game: Strategy, Proof, and Generalization

Article contributed by Aarav Lad, Akshobhyan R., Hari Annamalai, Nidhi Nayak, and Shaunak Okhde

1 The Problem

Alice and Bob are playing a game. There are 7 white counters and 3 black counters in a bucket. On each turn, a player must do exactly one of the following:

- Either remove between 1 and 3 white counters; or
- Remove 1 or 2 black counters.

Players alternate turns. **Whoever removes the last counter from the bucket wins.** Alice moves first.

Question. What should Alice's first move be, so that she is guaranteed to win no matter how Bob plays?

A. 1 white B. 2 white C. 3 white D. 1 black E. 2 black

This looks like a self-contained puzzle. But the idea behind the solution scales up to a game with any number of counters and any removal limits. We will build up the solution in stages.

2 What Happens With a Single Pile of Counters?

Before mixing two colours together, it helps to understand how a single pile behaves in this kind of game.

One colour at a time: Suppose there is only one pile of counters, and each turn a player may remove between 1 and k of them (for some fixed k). As before, whoever takes the last counter wins.

Claim 1. *If it is your opponent's turn and the pile size is a multiple of $k + 1$, then you are guaranteed to win.*

Why this works: Suppose the pile has $(k + 1)$ counters and it is your opponent's turn. Whatever number r (where $1 \leq r \leq k$) your opponent removes, you respond by removing $(k + 1) - r$ counters. This is possible since $1 \leq (k + 1) - r \leq k$. Together, the two of you have removed $k + 1$ counters in that round, and it is your opponent's turn again.

Now suppose instead the pile has $x(k + 1)$ counters for some positive integer x . By repeating the above strategy, you can ensure that after each round, $k + 1$ counters vanish from the pile. After x such rounds, all $x(k + 1)$ counters are gone, and as you took the last counter, you win. ■

Applying this to our two colours separately:

- **White counters** (up to 3 removable per turn): a multiple of $3 + 1 = 4$ white counters, on your opponent's turn, guarantees your win.
- **Black counters** (up to 2 removable per turn): a multiple of $2 + 1 = 3$ black counters, on your opponent's turn, guarantees your win.

Combining both colours: If both piles are present, the two ideas above combine very naturally.

Claim 2. *If, at the start of your opponent's turn, the number of white counters is a multiple of 4 and the number of black counters is a multiple of 3, then you are guaranteed to win.*

Why this works: Whichever colour your opponent removes counters from, you respond with the pairing move for that colour. This keeps the other colour's count fixed (still a multiple of its magic number) while bringing the chosen colour down, 4 or 3 at a time, with your opponent moving first in each round and you moving second.

Eventually one colour runs out. At that point, whichever colour remains is a multiple of its magic number, and it is your opponent's turn – exactly the single-pile situation from before. From there you finish the job using the single-colour strategy. 

We call a position where the white count is a multiple of 4 *and* the black count is a multiple of 3 a **loss case**: whoever is forced to move from a loss case is guaranteed to lose, if the opponent plays correctly.

3 Solving the Original Problem

We now return to the starting position: $w = 7$ white counters and $b = 3$ black counters, with Alice to move.

Note that $b = 3$ is already a multiple of 3, but $w = 7$ is not a multiple of 4. Rather, 7 leaves a remainder of 3 when divided by 4.

So, if Alice removes 3 white counters she leaves behind

$$w = 7 - 3 = 4, \quad b = 3,$$

which is a loss case, and it is now Bob's turn. Alice is guaranteed to win from here. Therefore the answer is: **C (3 white counters)**.

None of the other options reach a loss case in one move: removing 1 or 2 white counters leaves $w = 6$ or $w = 5$ (neither is a multiple of 4), and removing 1 or 2 black counters leaves $b = 2$ or $b = 1$ (neither is a multiple of 3, and $w = 7$ isn't a multiple of 4). So C is the unique correct first move.

4 Partial Generalization: Any Number of Counters

What strategy should we use if the bucket starts with w white counters and b black counters (arbitrary w and b) and the removal limits are the same (up to 3 white or up to 2 black per turn)?

Partially generalized problem. A and B take turns removing counters from a bucket with w white and b black counters. The removal limits are up to 3 white or up to 2 black per turn. A moves first. How should A play to guarantee a win?

From Section 2, we know that a loss case is any position with $w \equiv 0 \pmod{4}$ and $b \equiv 0 \pmod{3}$. The strategy is therefore to steer the game towards the nearest loss case. We split into four cases based on the remainders of w and b .

Case 1: $w \equiv 0 \pmod{4}$ and $b \equiv 0 \pmod{3}$. The starting position is itself a loss case, and it is A 's turn. So A is guaranteed to lose!

Case 2: $w \not\equiv 0 \pmod{4}$ and $b \equiv 0 \pmod{3}$. Let k be the remainder when w is divided by 4 (so $1 \leq k \leq 3$). If A removes k white counters, the position becomes $w - k \equiv 0 \pmod{4}$ and $b \equiv 0 \pmod{3}$: a loss case, with B to move. So A wins by taking k whites.

Case 3: $w \equiv 0 \pmod{4}$ and $b \not\equiv 0 \pmod{3}$. This is symmetric to Case 2. Let ℓ be the remainder when b is divided by 3. Removing ℓ black counters leaves a loss case for B . So A wins by taking ℓ blacks.

Case 4: $w \not\equiv 0 \pmod{4}$ and $b \not\equiv 0 \pmod{3}$. This is the interesting case, as neither remainder is 0. Let

$$i = w \bmod 4 \quad (1 \leq i \leq 3), \quad j = b \bmod 3 \quad (1 \leq j \leq 2).$$

Definition 1 (Equalization). *If a player's move brings the white and black remainders to the same value (i.e., makes $i = j$), we say that player has **equalized**.*

We now show that equal, non-zero remainders behave just like the all-zero loss case: whoever must move from a position with $i = j \neq 0$ is doomed to lose. The argument is the same as earlier; we use pairing, centred on i rather than 0 as the target remainder.

Claim 3. *If $i = j$, the player to move loses. If $i \neq j$, the player to move wins by equalizing in a single move.*

Why this works: If $i \neq j$, say $j > i$ (the other case is symmetric). Since $1 \leq j \leq 2$, the difference $j - i$ is at most 2, so it is a legal number of black counters to remove. Taking $j - i$ black counters brings the black remainder down to $j - (j - i) = i$, matching the white remainder. The position now has equal remainders, and it is the opponent's turn.

If $i = j$, the mover cannot avoid handing over an unequal (and hence losing) position. For, if the mover removes some white counters, changing the white remainder from i to a new value i' ; since the amount removed is between 1 and 3, we get $i' \neq i$ within the cycle of remainders mod 4 (removing a non-zero amount always changes the remainder). So now $i' \neq j$, since $j = i \neq i'$. The position is unequal, and it is the opponent's turn – who can equalize by the argument above. The same reasoning applies if the mover instead removes black counters. Either way, the mover is forced to hand over an unequal position for the opponent to fix.

So whenever the remainders are equal, the mover is forced to break the equality, and the opponent is always able to restore it. Since the total number of counters decreases every turn, eventually the piles run out entirely, so both remainders hit 0 (the original loss case!), and it will be the mover's turn.

So the player who is stuck starting from $i = j$ ultimately loses. ■

Summarizing the winning strategy: This lets us describe the winning strategy for any starting position (w, b) : compute $i = w \bmod 4$ and $j = b \bmod 3$. If $i = j$ (including $i = j = 0$), there is no winning first move.

Otherwise, remove $|i - j|$ counters from whichever colour has the larger remainder. You are then guaranteed to win.

Describing every outcome of Case 4 explicitly

We can now describe every outcome of Case 4 explicitly, in terms of i and j , where $i = w \bmod 4$ and $j = b \bmod 3$:

Remainders	Winner	Winning move (if A wins)
$i = 1, j = 1$	B	– (A has no winning move)
$i = 1, j = 2$	A	remove 1 black
$i = 2, j = 1$	A	remove 1 white
$i = 2, j = 2$	B	– (A has no winning move)
$i = 3, j = 1$	A	remove 2 whites
$i = 3, j = 2$	A	remove 1 white

Each of these follows directly from the claim that equal remainders ($i = j$) are losses for the mover, and unequal remainders are wins, achieved by removing the difference from the larger side.

5 Full Generalization: Any Removal Limits

Why stop at “up to 3 white or up to 2 black”? The same reasoning goes through unchanged if we allow up to m white counters or up to n black counters per turn, for any positive integers m and n .

Fully generalized problem. A and B take turns removing counters from a bucket with w white and b black counters.

Each turn, a player removes between 1 and m white counters, or between 1 and n black counters. A moves first. How should A play to guarantee a win?

Every argument above goes through with 4 replaced by $m + 1$ and 3 replaced by $n + 1$.

Definition 2 (Generalized loss case). *A position is a **generalized loss case** if $w \equiv 0 \pmod{m+1}$ and $b \equiv 0 \pmod{n+1}$. Whoever must move from such a position is guaranteed to lose.*

Let $i = w \bmod (m + 1)$ and $j = b \bmod (n + 1)$. As before, there are four cases:

Case 1 ($i = 0, j = 0$): The starting position is already a loss case, so A loses.

Case 2 ($i = 0, j \neq 0$): A removes j black counters, reaching a loss case for B .

Case 3 ($i \neq 0, j = 0$): A removes i white counters, reaching a loss case for B .

Case 4 ($i \neq 0, j \neq 0$): This splits further, as before:

- If $i = j$, the mover loses – there's no way to avoid breaking the equality, and the opponent will always restore it.
- If $i \neq j$, say (without loss of generality) $j > i$. Since $j \leq n$, the amount $j - i$ is a legal number of black counters to remove. Taking $j - i$ blacks equalizes the remainders and the mover wins.

The proofs of these facts are the same as in Section 4, with 4 and 3 replaced by $m + 1$ and $n + 1$.

This completes the solution to the fully general version of the bucket game: for any choice of starting counts w, b and removal limits m, n , the first player can immediately tell whether they are destined to win or lose, and exactly which counters to remove if they can win, just by comparing $w \bmod (m + 1)$ and $b \bmod (n + 1)$.

Comment: The key idea throughout was to find a “safe” combination of remainders that the opponent can never escape, and steer the game there as quickly as possible. Starting from single-colour pairing strategies, we built up to a complete solution for any starting numbers, and for any removal limits.

Acknowledgment

The authors would like to thank Shri Vinay Nair for posing the problem and urging us to work on it. The original problem is drawn from *Computational and Algorithmic Thinking 2011–2015, Book 2* by David I. Clark.

About the authors

AARAV LAD is a 13-year-old 8th grader from Bangalore. He is passionate about mathematics, playing the piano and cubing. His favourite topics include number theory and algebra. He has attended math camps such as Square Roots and Proofs. He can be reached at ladaarav@gmail.com.

AKSHOBHYAN R. is a 12-year-old 7th grader from Chennai. He is passionate about mathematics and likes to play chess. He has attended math

camps such as Epsilon India 2025. He can be reached at akshobhyan.r.p@gmail.com.

HARI ANNAMALAI, a 14-year-old 10th grader from Bangalore. He is passionate about mathematics who primarily enjoys working on problems from number theory and mathematical games. He has attended math camps such as Epsilon India 2024, Epsilon India 2025 and PACT in the USA. He also loves playing chess. He can be reached at hari.k.annamalai@gmail.com.

NIDHI NAYAK is a 14-year-old homeschooler from Bangalore. She is passionate about mathematics and has attended math programs such as All Girls Math Nurture Camp, Raising A Mathematician Training Program and Math Path in the USA. She also enjoys swimming and crocheting. She can be reached at nidhijnayak@gmail.com.

SHAUNAK OKHDE is a 13-year-old homeschooler from Bangalore. He is passionate about mathematics, physics, piano and badminton and has attended math camps like Vedic Mathrix, Epsilon India 2024, Epsilon India 2025 and Cube Roots. He can be reached at shaunakokhde@gmail.com.

Semi-Magic Squares With a Prime Pattern

Article contributed by Anand Prakash

In this article, I explore the construction of semi-magic squares using numbers with a prime pattern. To start with, let me explain what is a semi-magic square, and what we mean by “numbers with a prime pattern.”

Semi-magic square

As is well known, a magic square is a square array of distinct positive integers where the sums of the numbers in each row, each column, and both main diagonals are the same. If we insist only that the row sums and the column sums are all the same, with no condition on the diagonal sums, then the resulting array is called a *semi-magic square*.

Numbers with a prime pattern

Consider stretches of consecutive positive integers whose highest prime divisors occur in increasing order.

For example, consider the consecutive integers 24, 25, 26. Their prime factorizations are

$$24 = 2^3 \times 3, \quad 25 = 5^2, \quad 26 = 2 \times 13,$$

so their highest prime divisors are (respectively) 3, 5, 13, which are in increasing order.

These are the sequences which will be of interest to us.

Semi-magic squares using numbers with a prime pattern

What we wish to construct here are semi-magic squares of small order using stretches of consecutive integers with the above property.

To start with, we construct lists of such consecutive integers. The tables below give a few such lists. We will construct a few semi-magic squares using these lists. In each case, we have just the right amount of “raw material” for the task.

Three consecutive integers

Consecutive integers	Prime divisors
24, 25, 26	3, 5, 13
27, 28, 29	3, 7, 29
32, 33, 34	2, 11, 17

Below, we construct a semi-magic square using these nine numbers.

A semi-magic square of order three using the above data

26	27	33
28	34	24
32	25	29

The magic sum of this semi-magic square is 86.

Four consecutive integers

Consecutive integers	Prime divisors
8, 9, 10, 11	2, 3, 5, 11
20, 21, 22, 23	5, 7, 11, 23
56, 57, 58, 59	7, 19, 29, 59
220, 221, 222, 223	11, 17, 37, 223

A magic square of order four using the above data

221	11	20	58
22	56	223	9
59	21	10	220
8	222	57	23

Remarkably, this is a magic square and not just a semi-magic square. The magic sum is 310.

Indeed, it is a **pan-magic square**. Please check!

Five consecutive integers

Consecutive integers	Prime divisors
90, 91, 92, 93, 94	5, 13, 23, 31, 47
175, 176, 177, 178, 179	7, 11, 59, 89, 179
450, 451, 452, 453, 454	5, 41, 113, 151, 227
495, 496, 497, 498, 499	11, 31, 71, 83, 499
960, 961, 962, 963, 964	5, 31, 37, 107, 241

A semi-magic square of order five using the above data

177	93	450	963	497
961	496	179	91	453
90	451	962	499	178
498	176	94	452	960
454	964	495	175	92

The magic sum of the semi-magic square is 2180.

Curiously, one of the diagonals has the sum 2180 while the other diagonal has the sum 2179, which is 1 less than the magic sum.

Six consecutive integers?

To continue, we now seek sequences of consecutive integers of length 6, having the number pattern noted earlier. Here are two such sequences:

Consecutive integers	Prime divisors
168, 169, 170, 171, 172, 173	7, 13, 17, 19, 43, 173
2058, 2059, 2060, 2061, 2062, 2063	7, 71, 103, 229, 1031, 2063

However, I have not yet been able to construct a semi-magic square of order six using such lists.

I have also not yet been able to construct a semi-magic square of any higher order using such lists.

I leave these questions open for the reader.

About the author

ANAND PRAKASH runs a small garment shop at Kesariya village in the state of Bihar. He has a keen interest in number theory and recreational mathematics and has published many papers in international journals in these fields. He also has a deep interest in classical Indian music, poetry, and cooking. He has written a large number of poems in Hindi. He may be contacted at prakashanand805@gmail.com.

Representing Similar Triangles in 2D and 3D

Article by Dr. J Chattopadhyay and Mr. S S Chattopadhyay

Introduction: What Does “Similar” Mean?

Two triangles $\mathcal{T}$ and $\mathcal{T}'$ are said to be **similar** (written as $\mathcal{T} \sim \mathcal{T}'$) if one is a scaled-up (or scaled-down) copy of the other. More precisely, if one triangle has side lengths a, b, c and another has side lengths a', b', c' , then they are similar whenever

$$a : b : c = a' : b' : c'$$

(or some permutation of a', b', c' on the right-hand side). We denote the relation of similarity by the symbol $\sim$.

In everyday language, similar triangles have the same shape but not necessarily the same size. We note here that similarity is an **equivalence relation**. The class of all triangles having a given shape forms an equivalence class.

(An equivalence class is a class of objects that are all “equivalent” to each other under some rule (here, similarity).) This means that the following three properties hold:

1. Every triangle is similar to itself: $\mathcal{T} \sim \mathcal{T}$.
2. If triangle 1 is similar to triangle 2, then triangle 2 is similar to triangle 1: if $\mathcal{T}_1 \sim \mathcal{T}_2$, then $\mathcal{T}_2 \sim \mathcal{T}_1$.
3. If triangle 1 is similar to triangle 2, and triangle 2 is similar to triangle 3, then triangle 1 is similar to triangle 3: if $\mathcal{T}_1 \sim \mathcal{T}_2$ and $\mathcal{T}_2 \sim \mathcal{T}_3$, then $\mathcal{T}_1 \sim \mathcal{T}_3$.

Because of these properties, we can sort all triangles into equivalence classes where every triangle in a class is similar to every other triangle in the same class. Each class represents one “shape” of triangle.

Our goal is to find a way to *picture* all these different triangle-shapes as points in space.

Representation in 2D (Dhar and Sinha, 1994)

Setting Up Coordinates

Take any triangle with sides a, b, c . We can always label the sides so that $a \leq b \leq c$ (shortest to longest). Now define:

$$x = \frac{a}{c}, \quad y = \frac{b}{c}.$$

Since $a \leq b \leq c$, we get $0 < x \leq y \leq 1$.

There is one more constraint: the **triangle inequality**. So we have $x + y > 1$.

Clearly, two triangles that are similar to each other give the same values of x and y . So each “shape” of triangle corresponds to exactly one point (x, y) .

The Region of All Triangle-Shapes

The conditions $0 < x \leq 1$, $0 < y \leq 1$, $x \leq y$, and $x + y > 1$ carve out a triangular region in the plane. It has the appearance shown below.

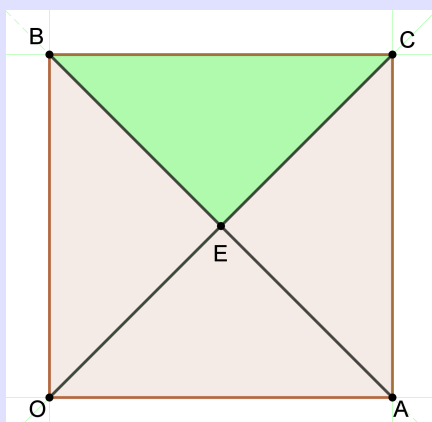


Figure 20: The region is bounded by the line $x + y = 1$ (but points on the line are excluded, since $x + y > 1$), the line $y = x$, and the lines $x = 1$ and $y = 1$. The points of interest are $O = (0,0)$, $A = (1,0)$, $B = (0,1)$, $C = (1,1)$ and $E = (1/2, 1/2)$.

Where Do Special Triangles Live?

Equilateral triangles. If all three sides are equal ($a = b = c$), then $x = y = 1$. So the single point $C = (1,1)$ represents equilateral triangles.

Right-angled triangles. By the Pythagorean theorem, $a^2 + b^2 = c^2$, which means $x^2 + y^2 = 1$. This is the equation of a circular arc from $B = (0,1)$ to $D = (1/\sqrt{2}, 1/\sqrt{2})$ inside our region.

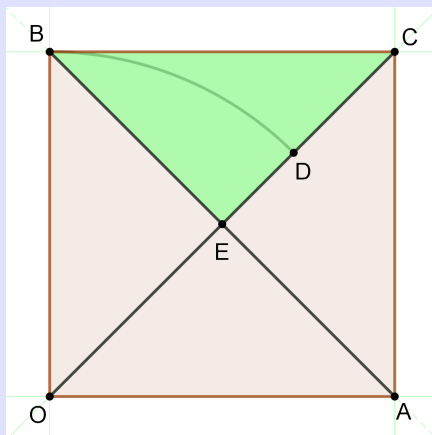


Figure 21

Obtuse-angled triangles. These correspond to the region *below* the arc (between the arc and the line $x + y = 1$).

Acute-angled triangles. These correspond to the region *above* the arc (between the arc and the corner C).

Isosceles triangles. These appear along certain boundary line segments where two of the sides are equal.

Representation in 3D (The New Approach)

The Idea: Use Angles Instead of Sides

The 2D method above uses side-length ratios. Can we do something similar using the angles of a triangle instead?

Let the three angles of a triangle be x , y , and z (measured in radians). We know that:

$$x + y + z = \pi \quad (\text{i.e., } 180^\circ),$$

with each angle satisfying $0 < x < \pi$, $0 < y < \pi$, $0 < z < \pi$.

Since similar triangles have the same angles, we can represent each triangle-shape by the point (x, y, z) in three-dimensional space.

The Surface of All Triangle-Shapes

The equation $x + y + z = \pi$ describes a flat plane in 3D space. The extra conditions (x, y, z all positive and less than π) cut out a triangular region on this plane. The three corners of this triangle are:

$$A = (0, 0, \pi), \quad B = (\pi, 0, 0), \quad C = (0, \pi, 0).$$

Points on the edges AB , BC , and CA are excluded as they would require one angle to be zero. (That would correspond to a degenerate triangle.)

Where Do Special Triangles Live in 3D?

Equilateral triangles. All three angles are equal, so the equilateral class corresponds to the single point $(\pi/3, \pi/3, \pi/3)$, which is the centroid of triangle ABC .

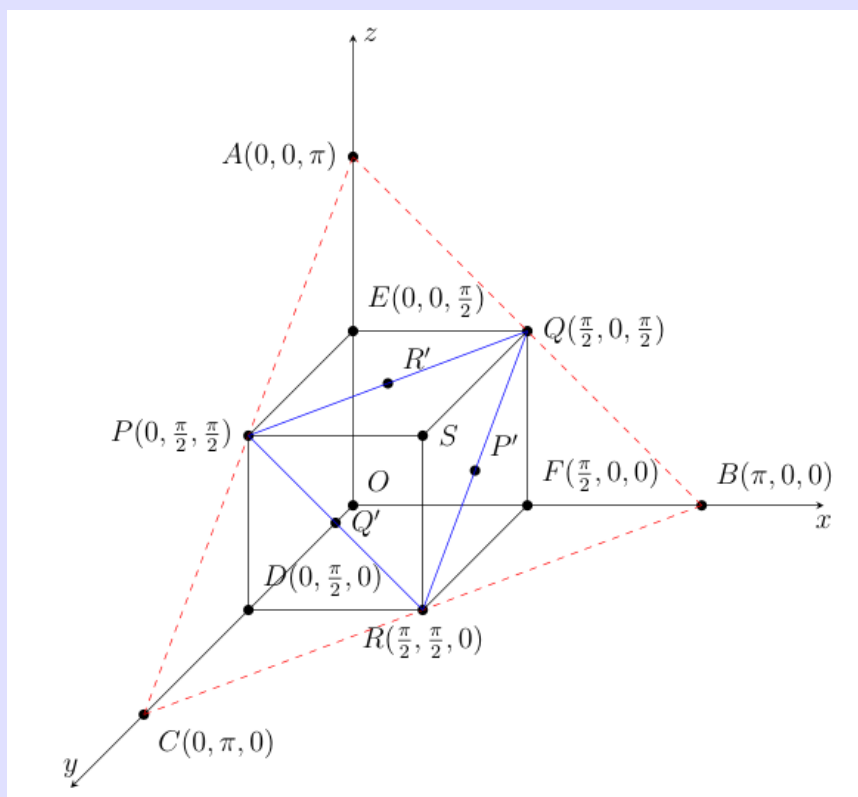


Figure 22

Right-angled triangles. One angle equals $90^\circ = \pi/2$. Consider the case $z = \pi/2$; then $x + y = \pi/2$, which is a straight line segment connecting:

$$P = \left(0, \frac{\pi}{2}, \frac{\pi}{2}\right) \quad \text{and} \quad Q = \left(\frac{\pi}{2}, 0, \frac{\pi}{2}\right).$$

Similarly, setting $y = \pi/2$ or $x = \pi/2$ gives two more line segments. To-

gether, these three segments form a smaller triangle PQR inside the larger triangle ABC . All points on the edges of this smaller triangle (excluding the vertices P, Q, R) represent right-angled triangles.

Acute-angled triangles. All three angles are less than 90° . These correspond to the points strictly inside the smaller triangle PQR .

Obtuse-angled triangles. One angle is greater than 90° . These correspond to the points outside the smaller triangle PQR but still inside the larger triangle ABC .

Isosceles triangles. Two angles are equal. For example, if $x = y$, the point lies on the line from $A = (0, 0, \pi)$ to the midpoint $R = (\pi/2, \pi/2, 0)$. There are three such lines (one for each pair of equal angles), and they are the medians of triangle ABC . All three medians meet at the centroid $(\pi/3, \pi/3, \pi/3)$, which is the equilateral triangle.

Right-angled isosceles triangles. The angles are $\pi/4, \pi/4, \pi/2$. There are three such points; they are the midpoints of the sides of the smaller triangle PQR .

How Common Are Different Types of Triangles?

Since every triangle-shape corresponds to a point on the triangular surface ABC in 3D, we can measure how “common” a type is by looking at the area it occupies on that surface.

Obtuse-angled triangles occupy the three corner regions outside the inner triangle PQR . These three smaller triangles together take up $3/4$ of the total area.

Acute-angled triangles occupy the inside of the triangle PQR , which is $1/4$ of the total area.

Right-angled triangles lie on the boundary of PQR , which is a set of line segments. Lines have zero area, so the proportion is 0.

This leads to the following conclusion:

In the sense described above, 75% of all triangle-shapes are obtuse, 25% are acute, and right-angled triangles, while infinite in number, are vanishingly rare.

This is similar to the fact that prime numbers, though infinite in number, make up a proportion of zero among all positive integers. Likewise for perfect square numbers.

What About the 2D Approach?

If we compute areas in the Dhar–Sinha picture, we find similar proportions. In that setting we get,

$$\text{Area of obtuse region} = \frac{\pi - 2}{8},$$

$$\text{Area of acute region} = \frac{4 - \pi}{8}.$$

The sum of these two areas is, of course $1/4$. Since $\pi - 2 \approx 1.14$ and $4 - \pi \approx 0.86$, we again find that obtuse triangles are more common than acute ones.

About the authors

SIDDHARTHA SANKAR CHATTOPADHYAY retired as a mathematics teacher from Bidhannagar Government High School, Kolkata in 2019. He has been actively associated with the organization *Association for Improvement of Mathematics Teaching* (AIMT) over more than three decades. He regularly writes articles on school mathematics in various magazines, both in Bangla and English. He may be contacted at 1959ssc@gmail.com.

JAITRA CHATTOPADHYAY did his PhD in mathematics (number theory) from Harish-Chandra Research Institute. After spending two years at IIT Guwahati, he joined Visva-Bharati, Santiniketan as an assistant professor in the department of mathematics. Apart from research, he is keenly interested in mathematics competition problems at the high school level; he loves to interact with enthusiastic students. He may be contacted at jaitra.chattopadhyay@visva-bharati.ac.in.

Further reading

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The Puzzle of the Bakhshali Manuscript Square Root Formula

Article contributed by S. C. Dutta Roy

Introduction: This note describes a way of deriving a very old formula for approximating square roots. It was used by ancient Indian mathematicians roughly 1,500 years ago.

The formula, from a document called the *Bakhshali manuscript*, is sophisticated and very accurate. How might the original authors of the manuscript have come upon it? The answer to this question may forever remain a mystery. There are no written records that may offer an explanation for the puzzle. In this note we describe a possible way of arriving at the formula.

1. Why do we need a way to approximate square roots?

Some numbers have square roots that come out exactly, like $\sqrt{9} = 3$ or $\sqrt{25} = 5$. But most numbers do not; e.g., $\sqrt{10}$ is an endless, never-repeating decimal (3.16227766...). Much before calculators existed, mathematicians still needed good approximations to such numbers, e.g., for astronomy.

The Bakhshali manuscript, an ancient Indian mathematical text, contains a clever formula for doing exactly this. But no one knows for certain how the ancients hit upon it, since they left no explanation. We thus have only the formula to go by. Mathematicians have proposed several different ways one *could* have arrived at it.

This article presents one such way.

2. Setting up the problem

Suppose we want to find a good approximation for $\sqrt{N}$ for some positive integer N that is not a perfect square. The most natural first step would be to identify the nearest perfect square to N and then make use of that information in some clever way. So let us write

$$N = A^2 + b, \tag{6}$$

where A is a whole number chosen so that A^2 is as close to N as possible, and b is whatever is left over.

We opt to approximate from below, so that $A^2 < N$ and $b > 0$. Since A^2 is close to N , the leftover part b is small compared to A^2 .

Running example. Let's find $\sqrt{10}$. The nearest perfect square below 10 is $9 = 3^2$, so we take $A = 3$ and

$$N = 10 = 3^2 + 1 \implies b = 1.$$

We will use this example at each step below.

3. The first approximation

We want a number close to $\sqrt{N} = \sqrt{A^2 + b}$. Since b is small, $\sqrt{N}$ should be only slightly larger than A . So let's guess that

$$\sqrt{N} = A + d,$$

where d is a small correction we need to find. Squaring both sides, we get

$$N = (A + d)^2 = A^2 + 2Ad + d^2,$$

$$\therefore A^2 + b = A^2 + 2Ad + d^2 \quad (\text{since } N = A^2 + b),$$

$$\therefore b = 2Ad + d^2.$$

Since d is small, d^2 is smaller still, and can thus be dropped as a good first approximation. This leads to:

$$b \approx 2Ad, \quad \therefore d \approx \frac{b}{2A}.$$

So our first, rough approximation to $\sqrt{N}$ is

$$x_1 = A + \frac{b}{2A}. \tag{7}$$

With $A = 3$, $b = 1$:

$$x_1 = 3 + \frac{1}{2 \times 3} = 3 + \frac{1}{6} = 3.1\bar{6}.$$

Compared with the true value $\sqrt{10} = 3.16228\dots$, our first, rough approximation is already correct to two decimal places.

4. How good is the first approximation?

How far is x_1 from the true $\sqrt{N}$? We cannot compare x_1 with $\sqrt{N}$ directly (as we don't know $\sqrt{N}$ exactly), but we can compare x_1^2 with N , which we do know exactly.

Squaring $x_1 = A + \frac{b}{2A}$, we get:

$$x_1^2 = A^2 + 2 \cdot A \cdot \frac{b}{2A} + \left(\frac{b}{2A}\right)^2 = A^2 + b + \left(\frac{b}{2A}\right)^2,$$

$$\therefore x_1^2 - N = \left(\frac{b}{2A}\right)^2 \quad (\text{since } N = A^2 + b). \quad (8)$$

This quantity, being a square, is always positive, so x_1^2 is always a little larger than N . Thus, x_1 slightly overestimates $\sqrt{N}$. (If you think about it for a moment, you will see that it makes sense.)

$$x_1^2 = (3.1\bar{6})^2 = 10.02\bar{7}, \text{ while } N = 10.$$

The difference is $0.02\bar{7} = \left(\frac{1}{6}\right)^2$, as the formula predicts.

5. The second approximation

Since x_1^2 is too big by $\left(\frac{b}{2A}\right)^2$, we look for a slightly smaller number $x_2 = x_1 - e$ (with e small and positive) whose square is closer to N .

Write $c = \frac{b}{2A}$, so $x_1 = A + c$ and

$$x_1^2 - N = c^2, \quad \text{i.e.} \quad x_1^2 - c^2 = N.$$

We want x_2 such that

$$x_2^2 = x_1^2 - c^2.$$

To find x_2 we proceed as in Section 3: let $x_2 = x_1 - e$. Squaring, we get

$$x_2^2 = x_1^2 - 2x_1e + e^2 \approx x_1^2 - 2x_1e \quad (\text{since } e^2 \text{ is very small}),$$

$$\therefore x_1^2 - c^2 \approx x_1^2 - 2x_1e, \quad (\text{since } x_2^2 = x_1^2 - c^2),$$

$$\therefore 2x_1e \approx c^2, \quad \therefore e \approx \frac{c^2}{2x_1}.$$

Hence

$$x_2 = x_1 - \frac{c^2}{2x_1} = \left(A + \frac{b}{2A}\right) - \frac{\left(\frac{b}{2A}\right)^2}{2\left(A + \frac{b}{2A}\right)}. \quad (9)$$

This is exactly the Bakhshali square root formula!

Note the flow: the idea of guessing and then correcting for the square of the leftover piece is applied twice in a row.

This makes the reasoning easy to understand and appreciate.

Continuing the example, $c = \frac{1}{6} = 0.1\bar{6}$ and $x_1 = 3.1\bar{6}$, so

$$x_2 = 3.1\bar{6} - \frac{(0.1\bar{6})^2}{2 \times 3.1\bar{6}} = 3.1\bar{6} - \frac{0.02\bar{7}}{6.3\bar{3}} \approx 3.1\bar{6} - 0.00439 \approx 3.16228.$$

This matches the true value $\sqrt{10} = 3.16227766\dots$ to five decimal places, from just two rounds.

6. Going further: a third approximation

If still more accuracy is needed, the same idea can be applied a third time. We can check how far x_2^2 is from N , and then use the part leftover to obtain an even better approximation x_3 , using the same logic.

The algebra now gets tedious to write out in full. But it uses no new ideas, so we shall not repeat it here.

About the author

PROF. DUTTA ROY spent about four decades in the Department of Electrical Engineering, Indian Institute of Technology Delhi, and is now retired. Currently, his interests are in exploring and appreciating the beauty of Classical Music and Mathematics.

He may be contacted at s.c.dutta.roy@gmail.com.

Further reading

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Easy Division of Numbers

Article contributed by Dr. Ramnath Aiyar

Introduction

In this article, we present a method for dividing numbers that mostly uses only multiplication and division by 2, operations that are simpler than traditional long division. The method works for division by any number, no matter how large.

How Numbers Work: Bases

Base 10 (Decimal): You already know how numbers work in everyday life. The number 493 really means:

$$493 = 3 + 9 \times 10 + 4 \times 100$$

In general, any number with digits $a_0, a_1, a_2, \dots$ (where a_0 is the ones digit) has the value:

$$N = a_0 + a_1 \times 10 + a_2 \times 100 + a_3 \times 1000 + \dots$$

Base 2 (Binary): Computers can only recognize two states: OFF and ON, represented by 0 and 1. So computers use **base 2** (also called **binary**). In binary, each digit is multiplied by a power of 2:

$$N = a_0 + a_1 \times 2 + a_2 \times 4 + a_3 \times 8 + \dots$$

Each digit a_k can only be 0 or 1. These digits are better known as **bits**.

Converting from Base 10 to Base 2

To convert a decimal number to binary, repeatedly divide by 2 and write down the remainders. The remainders (read in reverse) give you the binary number.

Example: Convert 785 to binary.

Number	Divide by 2	Remainder
785	392	1
392	196	0
196	98	0
98	49	0
49	24	1
24	12	0
12	6	0
6	3	0
3	1	1
1	0	1

Reading the remainders from bottom to top: $785 = (1100010001)_2$.

Note: It helps to recall that in binary, dividing by 2 is the same as dropping the last digit, and multiplying by 2 is the same as appending a 0 at the end. For example:

$$(1100010001)_2 \div 2 = (110001000)_2$$

$$(1100010001)_2 \times 2 = (11000100010)_2$$

Checking If a Number Is Divisible (when no quotient is needed)

Method Using Base 10

An odd number that is not a multiple of 5 ends in 1, 3, 7, or 9. For any such number p , either p or $3p$ can be written in the form $10m + 1$ or $10m - 1$ for some integer m . Let m be defined in this manner.

The rule: To check whether a given number N is divisible by p (where $p = 10m + 1$):

1. Let a_0 be the last digit of N .
2. Let A be the number you get by dropping the last digit.
3. Compute $A - m \times a_0$.
4. Repeat until the result is small enough to check by inspection.

If the final result is divisible by p , then so is the original number.

Example: Is 19129 divisible by 37?

First, note that $37 \times 3 = 111 = 10 \times 11 + 1$, so $m = 11$.

- Step 1: Drop last digit 9 from 19129, giving $A = 1912$.

Compute $1912 - 11 \times 9 = 1912 - 99 = 1813$.

- Step 2: Drop last digit 3 from 1813, giving $A = 181$.

Compute $181 - 11 \times 3 = 181 - 33 = 148$.

- Step 3: Drop last digit 8 from 148, giving $A = 14$.

Compute $14 - 11 \times 8 = 14 - 88 = -74$.

Since $-74 = -2 \times 37$ is divisible by 37, we conclude that 19129 is divisible by 37.

A Better Method Using Binary Ideas

We can make this even simpler by working with base 2. Any odd number p can be written as $p = 2m + 1$ for some integer m . The task is to check whether a given number N is divisible by p .

The algorithm:

1. Compute $m = (p - 1)/2$.
2. Check if N is odd. If it is, divide N by 2 (rounding down) and subtract m . If N is even, just divide by 2.
3. Repeat until N is less than or equal to p .
4. If N equals p or 0, the original number is divisible by p . Otherwise, it is not.

Example: Is 61013 divisible by 17?

Here $17 = 2 \times 8 + 1$, so $m = 8$. At each step: divide by 2, and subtract 8 if the number was odd.

$$61013 \xrightarrow{\div 2, -8} 30498 \xrightarrow{\div 2} 15249 \xrightarrow{\div 2, -8} 7616 \xrightarrow{\div 2} 3808 \xrightarrow{\div 2} 1904 \xrightarrow{\div 2} 952 \\ 952 \xrightarrow{\div 2} 476 \xrightarrow{\div 2} 238 \xrightarrow{\div 2} 119 \xrightarrow{\div 2, -8} 51 \xrightarrow{\div 2, -8} 17$$

The final result is 17, which equals p . So 61013 is **divisible by 17**.

Quick Tricks for Multiplying and Dividing by 2

Multiplying by 2

Work from left to right through each digit:

- If a digit is 4 or less, double it.
- If a digit is 5 or more, subtract 5, double the result, and carry 1 (add 1 to the previous result digit).
- If the leftmost digit is 5 or more, the answer gets an extra digit at the front.

Example: $9157356 \times 2 = 18314712$.

Dividing by 2

Work from left to right:

- Divide each digit by 2 (ignore any remainder).
- If a digit is odd, add 5 to the *next* digit before processing it.
- If the last digit is odd, the number has a remainder of 1 (i.e., it does not divide evenly).

Example: $95938024 \div 2 = 47969012$.

Finding the Quotient and Remainder

The previous section only told us whether or not a number divides evenly. But we can go further and find the exact quotient and remainder.

The Idea

Suppose we want to divide N by an odd number $p = 2m + 1$. We use the same halving process as before, but this time we record all the steps. Then we work *backwards* to build up the quotient.

Forward Pass: Record the Steps

At each step:

1. Record whether N is odd or even (call this d_i , which is 0 or 1).
2. Replace N with $\lfloor N/2 \rfloor - m \times d_i$ (divide by 2, subtract m if odd).
3. Stop when $N < p$. This final value is related to the remainder.

Backward Pass: Compute the Quotient

Starting from the last step (where the quotient is 0), work backwards using:

$$q_i = 2 \times q_{i+1} + d_i - \text{flag}$$

where:

$$\text{flag} = \begin{cases} -1 & \text{if } 2 \times r_{i+1} \geq p \\ 0 & \text{otherwise} \end{cases}$$

and the remainders are updated by:

$$r_i = (2 \times r_{i+1}) \mod p$$

The final value q_0 is the quotient, and r_0 is the remainder.

Worked Example

Divide 1057 by 37. Here $37 = 2 \times 18 + 1$, so $m = 18$.

Forward pass:

Step	N	d_i	Calculation
0	1057	1	$1057 = 2 \times 528 + 1$
1	510	0	$528 - 18 = 510$; $510 = 2 \times 255 + 0$
2	255	1	$255 - 0 = 255$; $255 = 2 \times 127 + 1$
3	109	1	$127 - 18 = 109$; $109 = 2 \times 54 + 1$
4	36	0	$54 - 18 = 36$; $36 < 37$, so stop.

We stop because $36 < 37$. So the last remainder is $r_4 = 36$ and $q_4 = 0$.

Backward pass:

Step	r_i	$2r_{i+1} \geq 37?$	flag	q_i
4	36	—	—	0
3	$2 \times 36 \mod 37 = 35$	Yes	-1	$1 + 2(0) - (-1) = 2$
2	$2 \times 35 \mod 37 = 33$	Yes	-1	$1 + 2(2) - (-1) = 6$
1	$2 \times 33 \mod 37 = 29$	Yes	-1	$0 + 2(6) - (-1) = 13$
0	$2 \times 29 \mod 37 = 21$	Yes	-1	$1 + 2(13) - (-1) = 28$

Result: $1057 \div 37 = 28$ remainder 21.

Check: $37 \times 28 + 21 = 1036 + 21 = 1057$.

Dividing by Even Numbers

Any even number can be written as $2^k \times p$ where p is odd. For example, $416 = 2^5 \times 13$.

Strategy:

1. First, divide N by the odd part p using the method above. Get quotient q_0 and remainder r_0 .
2. Then divide q_0 by 2^k by halving it k times. Each time q_0 is odd before halving, record that bit.
3. The final quotient is q_0 after k repetitions of the halving step.
4. The final remainder is: (the recorded bits, interpreted as a binary number) $\times p + r_0$.

Example: Divide 1575 by $416 = 13 \times 32$.

First, dividing 1575 by 13 gives quotient = 121 and remainder = 2.

Now divide 121 by $32 = 2^5$ by halving 5 times:

Iteration	q	Odd?	Running sum
0	121 (odd)	Yes	$0 + 1 = 1$
1	60 (even)	No	$1 + 0 = 1$
2	30 (even)	No	$1 + 0 = 1$
3	15 (odd)	Yes	$1 + 8 = 9$
4	7 (odd)	Yes	$9 + 16 = 25$
—	3		

Final quotient = 3.

Final remainder = $25 \times 13 + 2 = 325 + 2 = 327$.

Check: $416 \times 3 + 327 = 1248 + 327 = 1575$.

About the author

RAMNATH P R C AIYAR worked as Principal Research Scientist in CRNTS, IIT Bombay. He has 40 years research experience in the field of magnetism and magnetic materials, microwave engineering including computational electromagnetics. Currently, he is interested in Mathematics related to number theory. He may be contacted on aiyar123@gmail.com

Further reading

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Meeting Fibonacci at the Vertices of an Octahedron

Article contributed by Jyotirmoy Sarkar

1. Fibonacci sequence

The Fibonacci sequence is likely a familiar concept to many high school students today. Perhaps a teacher introduced it to them; perhaps they have read about it in a book or on the internet. Nonetheless, it is good to begin with the definition of the Fibonacci sequence, enumerated as A000045 in the OEIS [1]:

$$F_0 = 0, F_1 = 1, \text{ and } F_k = F_{k-1} + F_{k-2}, \text{ for all } k \geq 2. \quad (10)$$

The first few Fibonacci numbers F_0 to F_9 are 0, 1, 1, 2, 3, 5, 8, 13, 22, 34,

Fibonacci (1202) proposed the recurrence relation (10) to model the growth of an idealized rabbit population. Later, the sequence appeared in many

areas of mathematics, physics, biology, computer science, social networks, and in nature.

In fact, the sequence was known before Fibonacci! Sanskrit poets (as early as 200 BC) enumerated all patterns of 2-unit-long syllables, interlaced with one-unit syllables. The number of (possibly empty) sequences of 1s and 2s whose sum is n is F_{n+1} . See [2] and Figure 1.

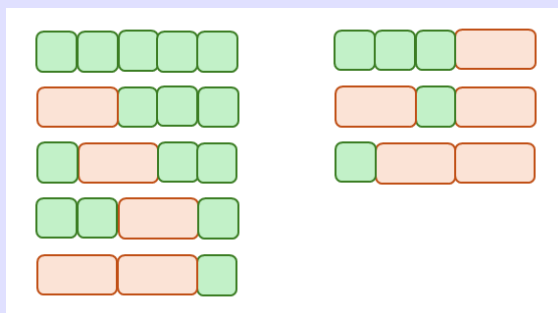


Figure 23: A five-unit-long cadence consists of short and long syllables in $F_6 = 8$ ways: Of these, $F_5 = 5$ end with a short syllable and $F_4 = 3$ with a long syllable. Discover the pattern and exhibit all $F_7 = 13$ six-unit long cadences.

Avid readers of *The Compass* magazine may have read [3], where Professor Sury defined a T-map, from natural (or counting) numbers to natural numbers, as $T(n) = n/2$ if n is even, and $T(n) = n + 1$ if n is odd. Starting from any natural number, one (if n is even) or two (if n is odd) applications of the T-map produce a smaller number. So, if you iterate the T-map, in a finite number of steps, you will surely reach 1. After that, the T-map images will alternate as $2, 1, 2, 1, 2, 1, \dots$. How many natural numbers $k \geq 1$

hit 1 (not necessarily for the first time) after n iterations of the T-map? The unexpected answer is F_{n+1} . Read the beautiful proof in [3].

I studied random walks on the vertices of an octahedron a decade ago (see [4]). However, I met the Fibonacci numbers in the walk only recently when I asked a new question. Please join me in exploring that random walk.

A random walk on an octahedron

A regular octahedron is one of the five Platonic solids. (Why exactly five? Check out [5].) It has $F = 8$ equilateral triangular faces; hence, the name octahedron (eight faces). It has $V = 6$ vertices, at each of which four edges meet; hence, its graph is regular. It has $E = (6 \times 4)/2 = 12$ edges. (Divide by 2 because pairs of edges on adjacent faces are glued together to become one). (Indeed, for any 3D convex polyhedron, Euler (1758) proved that $V - E + F = 2$. See [6].)

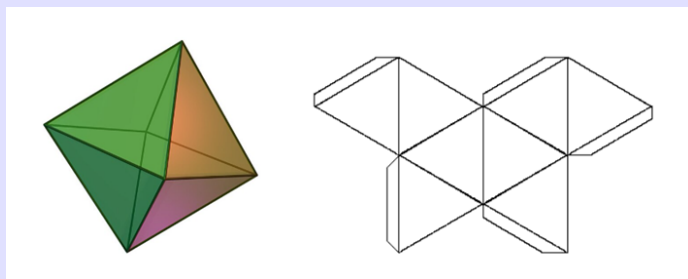


Figure 24: A regular octahedron has 8 faces, six vertices, and 12 edges. To efficiently make a regular octahedron with cardboard, use the outline given.

What is a symmetric random walk on the vertices of an octahedron?

At time $t = 0$, a particle appears at a vertex, which we shall call the origin, and dwells there for one unit of time. At time $t = 1$, it moves to one of the adjacent vertices *with equal probability* $1/4$, and dwells there for the next unit of time. Thereafter, at each successive epoch, the particle moves from its current vertex to one of its adjacent vertices with probability $1/4$ each. We are interested in the epoch T_R when the particle returns to the origin for the first time.

Note that T_R is a random variable that takes values $2, 3, 4, \dots$ (without any upper bound). Also, note that $P\{T_R = 1\} = 0$ and $P\{T_R = 2\} = 1/4$ (because in one step the particle leaves the origin, and in the second step, it returns to the origin with probability $1/4$). What are the probabilities associated with $T_R = 3, 4, \dots$? Do all these probabilities add to one, as they ought to?

To answer, we simplify the graph on which the random walk takes place.

Return to origin on a simpler graph

If T_R is the only random variable you wish to study for a random walk on the vertices of a regular octahedron, then you are better off studying a related random walk on a much simpler graph. Here's how to obtain that simpler graph.

The graph of an octahedron can be thought of as follows: Begin with a square whose four (blue) vertices, labelled counter-clockwise as 1, 2, 4, 3, are cyclically connected. Next, connect each of them to a fifth (red) vertex, labelled 0, above the square, and to a sixth (green) vertex, labelled 5, below

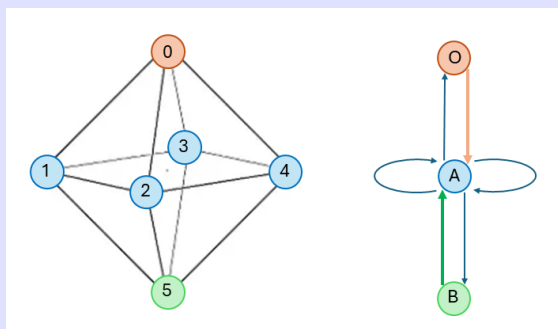


Figure 25: The return time to the origin, for a random walk on the vertices of an octahedron, is equivalent to that on a simpler graph with vertex set $\{O, A, B\}$.

the square. (The graph is not complete because diametrically opposite vertices, $(0,5), (1,4), (2,3)$, are not connected by an edge.) Classify the six vertices of the octahedron into three types: (O) the singleton Vertex 0, called the origin; (A) four Vertices 1, 2, 4, 3, adjacent to the origin (one step from the origin); and (B) the singleton Vertex 5, diametrically opposite the origin (two steps from the origin).

The equivalent random walk (Figure 3, right panel) starts at Vertex O. At time $t = 1$, the particle leaves the origin O and goes to Vertex A. Therefore,

$$T_R = 1 + T_A, \quad (11)$$

where T_A denotes the random number of steps needed for the particle to return to the origin O from Vertex A. After entering Vertex A at time $t = 1$, the particle can return to the origin O at time $t = 2$ with probability $1/4$, or remain in Vertex A with probability $1/2$, or go to Vertex B with probability

1/4. Whenever it goes to B, in the next step, it surely (with probability one) returns to Vertex A.

The waiting time (or the number of steps) T_A is stochastically related to itself as follows:

$$T_A = \begin{cases} 1 & \text{with probability } 1/4 \\ 1 + \bar{T}_A & \text{with probability } 1/2 \\ 2 + \tilde{T}_A & \text{with probability } 1/4 \end{cases} \quad (12)$$

where the new random variables $\bar{T}_A$ and $\tilde{T}_A$ have the same distribution as T_A . The relationship (12) is called a renewal property of T_A .

Probabilities associated with T_A and T_R

Writing $p_k = P\{T_A = k\}$, we note that $p_0 = 0, p_1 = 1/4$, and thereafter (partitioning according as the first step is towards A or towards B) the renewal property (12) implies the following recursive relation:

$$p_k = \frac{1}{2} p_{k-1} + \frac{1}{4} p_{k-2}, \text{ for any } k \geq 2. \quad (13)$$

Using the starting values $p_0 = 0, p_1 = 1/4$ and the recursive relation (13), we can compute all $p_k, k \geq 2$. The answers involve the Fibonacci sequence!

Theorem 1: The probabilities associated with different values of T_A are

$$P\{T_A = k\} = p_k = \frac{F_k}{2^{k+1}}, \text{ for all } k \geq 0. \quad (14)$$

Proof: The proof is by mathematical induction on k .

For $k = 0$, we have $P\{T_A = 0\} = 0 = F_0/2^1$, and for $k = 1$, we have $P\{T_A = 1\} = 1/4 = F_1/2^2$. So, the theorem holds for $k = 0, 1$.

Next, assume the theorem holds for $0, 1, \dots, (k-1)$ for some $k \geq 2$. We must show that the theorem holds for k . Using the recursive relation (13) and the strong induction hypothesis, we have

$$\begin{aligned} p_k &= \frac{1}{2} p_{k-1} + \frac{1}{4} p_{k-2} \\ &= \frac{1}{2} \frac{F_{k-1}}{2^k} + \frac{1}{4} \frac{F_{k-2}}{2^{k-1}} \\ &= \frac{F_{k-1} + F_{k-2}}{2^{k+1}} = \frac{F_k}{2^{k+1}}. \end{aligned} \tag{15}$$

The last equality in (15) uses the definition (10). Hence, the theorem holds for k . Therefore, the theorem holds for all $k \geq 0$. This completes the proof.



Corollary 1.1: Because of (11), the probabilities associated with T_R are

$$P\{T_R = k + 1\} = P\{T_A = k\} = p_k = \frac{F_k}{2^{k+1}}, \text{ for all } k \geq 0. \tag{16}$$

Total probability

To describe any discrete random variable, we list its values and the associated (non-negative) probabilities, ensuring that the total probability is one.

Proposition 1: The probabilities associated with different possible values of T_A (or T_R) add to one. That is,

$$\sum_{k=0}^{\infty} p_k = \sum_{k=0}^{\infty} \frac{F_k}{2^{k+1}} = 1.$$

Proof: Denote the sum of all p_k 's by S . Then using (10) and changing variables of the summation

$$\begin{aligned}
 S &= \sum_{k=0}^{\infty} \frac{F_k}{2^{k+1}} = \left(0 + \frac{1}{4}\right) + \sum_{k=2}^{\infty} \frac{F_k}{2^{k+1}} \\
 &= \left(0 + \frac{1}{4}\right) + \frac{1}{4} \sum_{k=2}^{\infty} \frac{F_{k-1} + F_{k-2}}{2^{k-1}} \\
 &= \left(0 + \frac{1}{4}\right) + \frac{1}{4} \left\{0 + \sum_{i=1}^{\infty} \frac{F_i}{2^i}\right\} + \frac{1}{4} \sum_{j=0}^{\infty} \frac{F_j}{2^{j+1}} \\
 &= \frac{1}{4} + \left\{\frac{1}{2}S\right\} + \frac{1}{4}S = \frac{1}{4} + \frac{3}{4}S,
 \end{aligned}$$

solving which, we get $S = 1$. This completes the proof. ■

Remark 1: By (10), the ratios F_{k+1}/F_k of successive terms in the Fibonacci sequence converge to a limit x which satisfies $x > 0, x^2 = x + 1$. The solution $x = (\sqrt{5} + 1)/2 = 1.61803398 \dots$ is called the golden ratio, ϕ . See details in [2]. Thus, the Fibonacci sequence grows indefinitely at a limiting geometric rate of $\phi < 2$. Hence, $\lim_{k \rightarrow \infty} F_k/2^k = 0$, which also follows from $\sum_{k=0}^{\infty} F_k/2^k = 2$.

The mean and the variance of T_A and T_R

Some important properties of a random variable are its mean (location centre), mean-square, and standard deviation (spread about the mean). For T_A , these are defined as $\mu_A = E[T_A] = \sum_{k=0}^{\infty} k p_k$, $\mu'_{A2} = E[T_A^2] = \sum_{k=0}^{\infty} k^2 p_k$, and $\sigma_A = \sqrt{\mu'_{A2} - \mu_A^2}$, respectively. Likewise, for T_R , one can define μ_R and σ_R . In fact, by (11), we know that $\mu_R = \mu_A + 1$ and $\sigma_R = \sigma_A$. The

reader may follow these definitions to compute the mean and the standard deviation.

However, unlike most textbooks, we shall derive these expressions using the renewal property (12).

Proposition 2: The mean of T_A is 5, and the mean of T_R is 6.

Proof: Because $E[T_A] = E[\tilde{T}_A] = E[\tilde{\tilde{T}}_A]$, using (12), we have

$$E[T_A] = \frac{1}{4} \times 1 + \frac{1}{2} \times (1 + E[T_A]) + \frac{1}{4} \times (2 + E[T_A]) = \frac{5}{4} + \frac{3}{4}E[T_A],$$

solving which, we get $E[T_A] = 5$. Next, since $T_R = 1 + T_A$ as explained in (11), we have $E[T_R] = 1 + E[T_A] = 6$, completing the proof. ■

Remark 2: It is somewhat surprising that $E[T_R] = 6$, the same as the number of vertices in an octahedron! But it is no coincidence: Such a result holds for a symmetric random walk on any regular graph. See [7].

Proposition 3: The standard deviation of T_A (and T_R) is $\sqrt{22}$.

Proof: In Lemma 2, we proved that $E[T_A] = E[\tilde{T}_A] = E[\tilde{\tilde{T}}_A] = 5$. Because $\mu'_{A2} = E[T_A^2] = E[\tilde{T}_A^2] = E[\tilde{\tilde{T}}_A^2]$, using (12), we have

$$\begin{aligned} \mu'_{A2} &= \frac{1}{4} \times 1^2 + \frac{1}{2} \times E[(1 + \tilde{T}_A)^2] + \frac{1}{4} \times E[(2 + \tilde{\tilde{T}}_A)^2] \\ &= \frac{1}{4} + \frac{1}{2} \times (1 + 2E[\tilde{T}_A] + \mu'_{A2}) + \frac{1}{4} \times (4 + 4E[\tilde{\tilde{T}}_A] + \mu'_{A2}) \\ &= \frac{1}{4} + \left(\frac{1+10}{2} + \frac{1}{2}\mu'_{A2} \right) + \left(\frac{4+20}{4} + \frac{1}{4}\mu'_{A2} \right) = \frac{47}{4} + \frac{3}{4}\mu'_{A2}, \end{aligned}$$

solving which, we get $\mu'_{A2} = E[T_A^2] = 47$. Hence, the variance of T_A is $47 - 5^2 = 22$, and the standard deviation of T_A is $\sqrt{22} = 4.6904$. Also, since $T_R = 1 + T_A$, the standard deviation of T_R is also $\sqrt{22}$, completing the proof. ■

Take-home lessons and open problems

Our objectives in writing this paper are fourfold:

1. Reinforce the notion of the Fibonacci sequence.
2. Simplify a random walk on an elaborate graph to a much simpler graph.
3. Illustrate a proof by mathematical induction.
4. Utilize the renewal property of a random variable to evaluate the total probability, the mean, and the standard deviation.

If the reader grasps these pedagogical intentions, our effort has been worthwhile.

Observant students will realize that T_R , the return time to origin, for a symmetric random walk on the vertices of a tetrahedron, satisfies $P\{T_R = 2\} = 1/3$, and $P\{T_R = m + 1\} = (2/3)P\{T_R = m\}$ for all $m \geq 2$.

Motivated students may discover that the return time to origin T_R , for a symmetric random walk on the vertices of a regular hexahedron (cube), is an *even* natural number with $P\{T_R = 2\} = 1/3$, $P\{T_R = 4\} = 4/27$, and

$$P\{T_R = 2m + 2\} = \frac{7}{9} P\{T_R = 2m\} \text{ for all } m \geq 2.$$

More adventurous students may study the return time to origin T_R for a symmetric random walk on the vertices of the remaining two Platonic solids: a regular icosahedron (a 20-faced solid with 12 vertices, each of degree 5), or a regular dodecahedron (a 12-faced solid with 20 vertices, each of degree 3).

Dedication

I wrote this paper to inspire the younger generation enjoy mathematics and delve into problem-solving. Therefore, I dedicate this paper to the one who inspired me into this noble pursuit. I offer this paper as a token of gratitude to my teacher, guide, and beloved Mamababu, Professor Bikas Kumar Sinha.

About the author

JYOTIRMOY SARKAR is a professor at the School of Science (Dept... of Statistics & Biostatistics), Indiana University Indianapolis (USA). His main research areas include enumeration, probability, statistics, and reliability theory. He enjoys reading, 'riting, 'rithmetic and R-coding. He may be contacted at jsarkar@iu.edu.

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Part VII

Culture & History of Mathematics

How Stamps and Postcards Helped India Count Its People

Adapted from: “How stamps and postcards helped India count its people.” <https://www.bbc.com/>

<https://www.bbc.com/news/articles/c4g4yyxxwp2o>

Author: Soutik Biswas

India’s census history is a story about communication, trust, and nation-building. It shows how stamps, postmarks, postcards, and the postal network helped the Indian state persuade people to take part in the count, from the first post-independence census to the digital era.

A census told through mail

The article opens with a striking image from the 2001 census: a commemorative stamp, a first day cover, and a cancellation mark that together



Figure 26: An envelope mailed from Nandikotkur in January 1951 and delivered in Madras (now Chennai) days later bears one of the earliest known bilingual postmarks promoting India's first post-independence census. Credit: Vikas Kumar & BBC

presented India as a family of diverse people counting itself. That visual idea sets the tone for the entire piece. Rather than treating the census as a dry administrative exercise, the article shows it as a national project that needed public belief as much as bureaucratic machinery. The postal system became the medium through which the state spoke to its citizens across language, geography, and literacy barriers.

Why the census mattered

After independence, India urgently needed reliable demographic data to support universal adult franchise, elections, and planned economic development. The Census Act was passed in 1948, even before the constitution was finalised, which underlines how central the exercise was to the new republic. But the state faced a practical problem: India was vast, largely rural, and unequal in access to communication, so it needed a system that could reach ordinary people quickly and repeatedly. The post office filled that gap because it was the largest unified communications network available to the government.

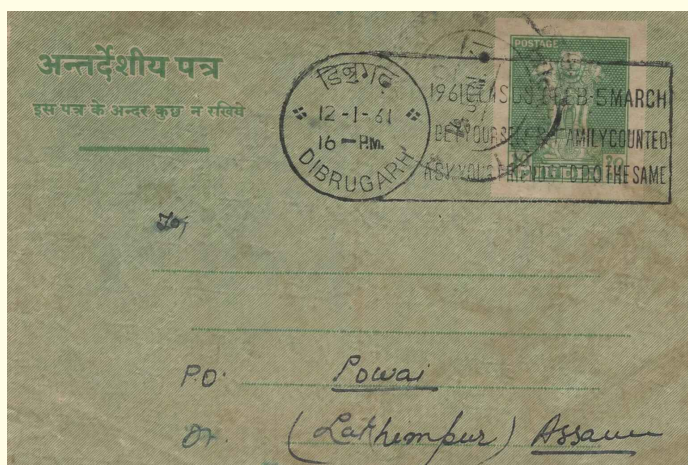


Figure 27: This inland letter card, mailed within Assam in January 1961, carried a census postmark urging Indians to “Get yourself & family counted” – and to persuade friends to do the same. Credit: Vikas Kumar & BBC

Early campaigns

The article traces how census promotion appeared on actual mail. One of the earliest post-independence examples is a bilingual pictorial postmark from 1951, showing a family of three and the words “Census of India” in Hindi and English. In 1961, inland letters carried messages such as “Get yourself & family counted” and urged people to persuade friends as well. These campaigns were not just about announcing dates; they were designed to create participation and normalise the idea that every household had a role in the national count.

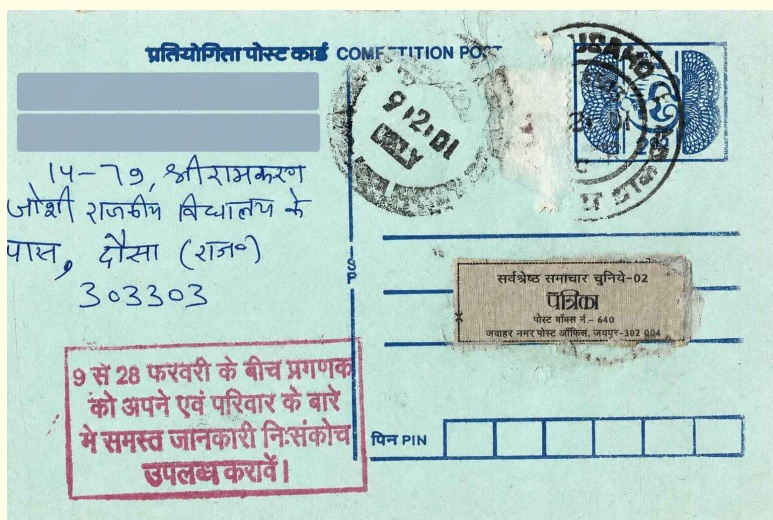


Figure 28: Mailed from Dausa to Jaipur in February 2001, this postcard carried a Hindi census postmark urging Indians to share details about themselves and their families “without any hesitation.” Credit: Vikas Kumar & BBC

The postal network was especially effective because it worked in low-literacy settings. Postmen often served as readers, scribes, and informal intermediaries in villages, so they were more than mail carriers. They helped translate the state's message into everyday life, which made census outreach feel more personal and practical. That human layer mattered because the census depended on trust, not coercion.



Figure 29: Three million commemorative stamps were issued in 1971 to mark the centenary of the census, celebrating the country's diversity through faces embedded within the number 100. Credit: Vikas Kumar & BBC

Trust and memory

Trust was a recurring concern because earlier censuses had left a difficult legacy. The article notes that colonial censuses in 1931 and 1941 faced boy-



Figure 30: Issued for the 2011 Census, this commemorative stamp showed families holding hands alongside an enumerator and the census emblem. Credit: Vikas Kumar & BBC

cotts in some areas, and the 1941 count in Punjab and Bengal was criticised for alleged communal manipulation. Independent India therefore had to prove that the census was legitimate, neutral, and worth participating in. Public messaging through stamps and postcards became part of that effort, turning the act of counting into a shared civic ritual.

As the decades passed, the imagery evolved with the country's own changing ambitions. In 1971, three million commemorative stamps marked the centenary of the census and celebrated India's diversity through faces embedded in the number 100. Those stamps also described the census as one of the world's largest administrative operations and highlighted the use of electronic computers to process population data. The tone had shifted from simple participation toward modernity, scale, and administrative pride.

A changing national image

The article pays close attention to how the census was framed in public material. In the 2000 advertisements for the 2001 census, the exercise was described as the “Mirror of the nation” and a “Group Photograph of the nation”. That language is important because it presents the census as a collective self-portrait, not merely a government inventory. Multilingual postcards carried these slogans in 11 or 13 languages, helping the message travel across India’s linguistic landscape. Later imagery also reflected demographic anxiety, including references to the two-child norm and population control.

फार्म नं. 5

भारत की जनगणना 1971 राजस्थान
सुपरवाइजर की साप्ताहिक रिपोर्ट रिपोर्ट नं. 1.024

जिला मिर्जापुर तहसील/शहर बस गहर स्थ. चार्ज 2239/गणनाखंड नं. 9...

नाम सुपरवाइजर स्कूल नंबर 14...

सफिल में कुल ब्लॉकों की सं. 4 (80, 81, 82, 83)

तथा ग्रॉफिट में उनके नम्बर

कितने ब्लॉकों के नज 1 नवशों बन चुके हैं	कितने ब्लॉकों में भवनों पर डॉकिल किये जा चुके हैं	कितने ब्लॉकों में गंजरी नवशों बनाने का कार्य चल रहा है	कितने ब्लॉकों में भवनों पर नम्बर डॉकिल करने का कार्य चल रहा है
4	2	X	2

1. सप्ताह में कितने ब्लॉकों में गये 3 2. कितने भवनों के नम्बर स्रोतों पर चैक किये 40

3. कितने भवनों के बावत म. नु. व प्रति. अनुसूची के इम्प्राज चैक किये X

4. उन ब्लॉकों के नं. जिसमें काम पिछड़ा हुआ है 8/1

दिनांक 23.2.70 हस्ताक्षर सुपरवाइजर G. B. Malik
23.2.70

Figure 31: A supervisor in Rajasthan’s Bhilwara district mailed this pre-printed postcard on 23 February 1970 to track mapping, house numbering and enumeration work for the 1971 Census. Credit: Vikas Kumar & BBC

These materials reveal a deeper shift in how the Indian state imagined

itself and its citizens. In the early years, the emphasis was on getting people counted and building basic legitimacy. Later, the census became linked with development, diversity, and population management. The stamps and postcards therefore serve as miniature records of changing policy priorities and political moods.

The digital present

The article ends by connecting that history to the present-day census preparation. The upcoming count is described as crucial for policy planning, welfare delivery, and political representation, and it will collect caste data for the first time in decades. It will also be enormous in scale, covering 36 states and federally administered territories, more than 7,000 sub-districts, over 9,700 towns, and nearly 640,000 villages. Yet one major change stands out: for the first time, the census will be conducted digitally, with enumerators using mobile apps to collect and upload data in real time.

That shift gives the article its closing tension. Technology may make data collection faster, but it does not automatically create confidence. The core challenge, the article suggests, remains what it has always been: convincing a massive and diverse population to trust the state enough to be counted. From family-shaped postmarks to smartphone uploads, the tools have changed, but the democratic question behind the census has not.

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Acknowledgment: This article has been prepared from the published article whose source is given above. The summary was made using the *Gemini 3.1 Pro/Perplexity LLM*.

Part VIII

Miscellaneous Articles – 2

Tales about Triangular Numbers – 2

We continue our study of the triangular numbers (“T-numbers”). As noted in the April issue, there are many relationships connecting the T-numbers with one another and to numbers in other families, e.g., the square numbers. We examine one more such property in this article.

Triangular numbers that exceed a square by 1

We observe that the triangular number $T_4 = 10$ exceeds the square $3^2 = 9$ by 1. Are there other such instances? How do we find all triangular numbers that share such a property? This question requires us to solve the following equation for m and n over the positive integers:

$$\frac{m(m+1)}{2} = n^2 + 1. \tag{17}$$

We shall show that this equation has infinitely many solutions and describe how to generate all the solutions.

Multiplying the above equation by 8, we obtain:

$$\begin{aligned}4m(m + 1) &= 8n^2 + 8, \\ \therefore 4m^2 + 4m + 1 &= 8n^2 + 9, \\ \therefore (2m + 1)^2 &= 2(2n)^2 + 9, \\ \therefore X^2 &= 2Y^2 + 9,\end{aligned}$$

with $X = 2m + 1$ and $Y = 2n$.

What are the positive integer solutions of the equation $X^2 = 2Y^2 + 9$, and therefore of $\frac{1}{2}m(m + 1) = n^2 + 1$? It is easy to generate solutions using a computer. We find the following. Keep in mind that $m = (X - 1)/2$ and $n = Y/2$. So from the first two rows (giving X and Y) we can generate the rest of the data (m and n , and then T_m and n^2).

X	9	51	297	1731	10089	58803	...
Y	6	36	210	1224	7134	41580	...
m	4	25	148	865	5044	29401	...
n	3	18	105	612	3567	20790	...
T _m	10	325	11026	374545	12723490	432224101	...
n ²	9	324	11025	374544	12723489	432224100	...

Patterns

Can we uncover some nice patterns in the above data? Patterns that yield the law of formation of each row? Indeed we can.

The second item below describes the law of formation of the n -row. From this we deduce that there are infinitely many solutions to the given problem.

1. Observe that all numbers in the X and Y rows are multiples of 3. Could this have been anticipated? Yes, and here's how. Let us look closely at the equation

$$X^2 = 2Y^2 + 9.$$

Suppose that Y is **not** a multiple of 3. Then we have

$$Y^2 \equiv 1 \pmod{3},$$

$$\therefore 2Y^2 + 9 \equiv 2 \pmod{3},$$

$$\therefore X^2 \equiv 2 \pmod{3}.$$

But this congruence is not possible; all squares are either $0 \pmod{3}$ or $1 \pmod{3}$. Hence it cannot be that Y is not a multiple of 3. In other words, Y is a multiple of 3.

But it follows easily from this that X^2 is a multiple of 9, and hence that X too is a multiple of 3. Thus, both X and Y are multiples of 3. Writing $x_1 = X/3$ and $y_1 = Y/3$, we now get

$$x_1^2 = 2y_1^2 + 1,$$

which is a much more familiar looking equation. But we will not say anything further about this for now.

2. Much more interesting is the pattern we find in the n -sequence. The successive terms are

$$3, 18, 105, 612, 3567, 20790, \dots$$

These numbers too are all multiples of 3. Dividing them by 3 we get:

$$1, 6, 35, 204, 1189, 6930, \dots$$

Do you see the pattern? Observe that from the third term onwards,

$$35 = 6 \times 6 - 1,$$

$$204 = 6 \times 35 - 6,$$

$$1189 = 6 \times 204 - 35,$$

$$6930 = 6 \times 1189 - 204,$$

and so on.

So if we denote the n -th term of the sequence $1, 6, 35, 204, \dots$ by u_n , with $u_1 = 1, u_2 = 6, \dots$ then our observation is that

$$u_n = 6u_{n-1} - u_{n-2} \quad \text{for all } n > 2.$$

Proofs

What we have noted above are all in the nature of observations. Now it is up to us to show that these patterns are genuine. So we shall set ourselves the following task.

Claim

Define the u_n sequence recursively as follows.

$$u_0 = 0, \quad u_1 = 1, \quad u_2 = 6, \quad u_n = 6u_{n-1} - u_{n-2} \quad \text{for all } n \geq 2. \quad (1)$$

Then the following is true for all positive integers n :

$$(3u_n)^2 + 1 \quad \text{is a triangular number.}$$

Proof of the claim

We shall make use of the following criterion:

A criterion for triangular numbers: A positive integer x is a triangular number if and only if $8x + 1$ is a perfect square.

So it suffices to show that

$$8\left((3u_n)^2 + 1\right) + 1$$

is a perfect square for every $n \geq 1$.

We shall base the proof on the following identity.

$$u_n^2 - 6u_n u_{n-1} + u_{n-1}^2 = 1 \quad \text{for all } n \geq 1. \quad (2)$$

Proof of identity: Let $D_n := u_n^2 - 6u_n u_{n-1} + u_{n-1}^2$ for $n \geq 1$. Then

$$D_1 = u_1^2 - 6u_1 u_0 + u_0^2 = 1 - 0 + 0 = 1.$$

Now fix $n \geq 2$. Using (1) in the form $u_{n-2} = 6u_{n-1} - u_n$, we compute:

$$\begin{aligned} D_n - D_{n-1} &= (u_n^2 - 6u_n u_{n-1} + u_{n-1}^2) - (u_{n-1}^2 - 6u_{n-1} u_{n-2} + u_{n-2}^2) \\ &= u_n^2 - 6u_n u_{n-1} + u_{n-2}(6u_{n-1} - u_{n-2}). \end{aligned}$$

Since $6u_{n-1} - u_{n-2} = u_n$ by (1), the last term equals $u_{n-2}u_n$, so

$$\begin{aligned} D_n - D_{n-1} &= u_n^2 - 6u_n u_{n-1} + u_n u_{n-2} \\ &= u_n(u_n - 6u_{n-1} + u_{n-2}) \\ &= u_n \cdot 0 = 0, \end{aligned}$$

again by (1). Hence $D_n = D_{n-1}$ for all $n \geq 2$, so

$$D_n = D_{n-1} = \cdots = D_1 = 1$$

for every $n \geq 1$. This proves the identity. ■

Now we are in a position to complete the proof of the main result. For $n \geq 1$ define the integer w_n by

$$w_n := 3u_n - u_{n-1}.$$

Squaring and using the identity just proved:

$$\begin{aligned} w_n^2 &= 9u_n^2 - 6u_n u_{n-1} + u_{n-1}^2 \\ &= 8u_n^2 + (u_n^2 - 6u_n u_{n-1} + u_{n-1}^2) = 8u_n^2 + 1. \end{aligned}$$

Therefore

$$\begin{aligned} 8((3u_n)^2 + 1) + 1 &= 72u_n^2 + 9 \\ &= 9(8u_n^2 + 1) = 9w_n^2 = (3w_n)^2, \end{aligned}$$

which is a perfect square, since w_n is an integer. It follows that $(3u_n)^2 + 1$ is a triangular number for every positive integer n . ■

We will study more such relationships in subsequent articles. We must mention, in passing, that the sequence 1, 6, 35, 204, ... has a connection with Ramanujan's solution to the "house number problem." (More on this later.)

Article contributed by Shailesh Shirali (with some assistance from Claude AI)

Computational Problems in Geometry – 1

In this series, we discuss a few miscellaneous problems in geometry that are difficult to tackle using synthetic approaches. We are forced to fall back on computational methods.

Square in a quadrant

Problem 1. AOB is a quadrant of a circle with centre O and radius 1 (so $OA = OB = 1$). A square $PQRS$ is inscribed in the quadrant, with P on OA , Q on OB , and R and S on arc AB . Find the area of the square.

Solution. We opt to use complex numbers. Place the quadrant on the complex plane with its centre O at the origin, A on the positive imaginary axis, and B on the positive real axis. Let the affixes of P , Q be ip , q respectively. Here p, q are positive real numbers. The arc AB is the portion of the circle $|z| = 1$ in the first quadrant. Let the affixes of R and S be r and s respectively.

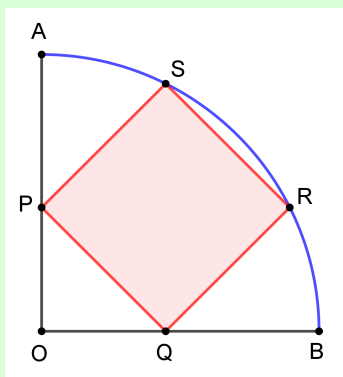


Figure 32

The square $PQRS$ is traversed counter-clockwise ($P \rightarrow Q \rightarrow R \rightarrow S$). Hence,

$$ip - q = i(r - q), \quad \therefore r = (p + q) + iq,$$

$$s - ip = i(q - ip), \quad \therefore s = p + i(p + q).$$

Since both R, S lie on the arc $|z| = 1$,

$$(p + q)^2 + q^2 = 1 = p^2 + (p + q)^2,$$

giving $p^2 = q^2$. Since $p, q > 0$, we get $p = q$. Hence $(2p)^2 + p^2 = 1$, giving $5p^2 = 1$. Therefore,

$$p^2 = \frac{1}{5},$$

and

$$\text{Area of square} = |PQ|^2 = p^2 + p^2 = \frac{2}{5}.$$

Comment: The reader should try proving by vectors that $p = q$.

Hypotenuse of a right angled triangle

Problem 2. Triangle ABC is right-angled at A and AD is its altitude (with D on BC). The areas of triangles ABD and ACD are u^2 and v^2 respectively. Find the length of the hypotenuse BC in terms of u and v .

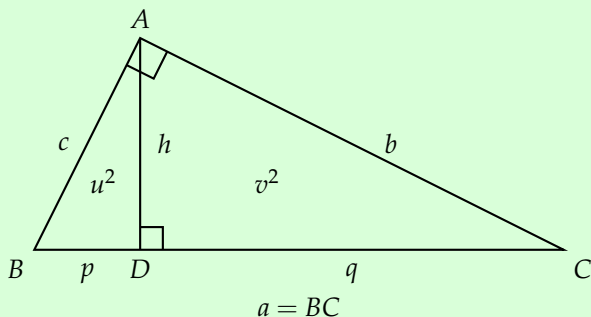


Figure 33

Solution. Let $BD = p$, $DC = q$, and the altitude $AD = h$.

From the given data,

$$\text{Area of } \triangle ABD = \frac{1}{2} \cdot p \cdot h = u^2, \quad \therefore ph = 2u^2,$$

$$\text{Area of } \triangle ACD = \frac{1}{2} \cdot q \cdot h = v^2, \quad \therefore qh = 2v^2.$$

Since $\triangle ABC$ is right-angled at A , we have

$$h^2 = p \cdot q.$$

Hence,

$$h^2 = \frac{2u^2}{h} \cdot \frac{2v^2}{h} = \frac{4u^2v^2}{h^2},$$

$$\therefore h^4 = 4u^2v^2, \quad \therefore h^2 = 2uv, \quad \therefore h = \sqrt{2uv}.$$

Finally,

$$BC = p + q = \frac{2u^2}{h} + \frac{2v^2}{h} = \frac{2(u^2 + v^2)}{h} = \frac{2(u^2 + v^2)}{\sqrt{2uv}}.$$

So we have our answer:

$$BC = \frac{\sqrt{2}(u^2 + v^2)}{\sqrt{uv}}.$$

Triangle with a double angle

Problem 3. Triangle ABD has sides $AD = 10$ units and $BD = 11$ units. Point C lies on side BD with $BC = 6$ units and $CD = 5$ units. Given that $\angle BAC = 2\angle CAD$, show that $AB = AC$

Solution. Let $\angle CAD = \alpha$, so that $\angle BAC = 2\alpha$ and $\angle BAD = 3\alpha$.

This problem is going to require several applications of the sine rule. So get ready!

Applying the sine rule in triangle ACD , we get:

$$\frac{AD}{\sin(\angle ACD)} = \frac{CD}{\sin(\angle CAD)},$$

$$\therefore \frac{10}{\sin(\angle ACD)} = \frac{5}{\sin \alpha},$$

giving $\sin(\angle ACD) = 2\sin\alpha$ and $\sin(\angle ACB) = 2\sin\alpha$, since angles $\angle ACB$ and $\angle ACD$ are supplementary.

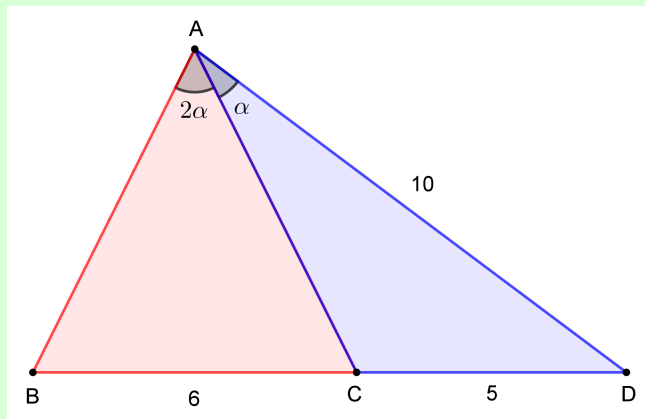


Figure 34

Next, we apply the sine rule in triangle ABC :

$$\begin{aligned}\frac{AB}{\sin(\angle ACB)} &= \frac{BC}{\sin(\angle BAC)}, \\ \therefore \frac{AB}{2\sin\alpha} &= \frac{6}{\sin 2\alpha} = \frac{6}{2\sin\alpha \cos\alpha}, \\ \therefore AB &= \frac{6}{\cos\alpha}.\end{aligned}$$

Applying the sine rule in triangle ABD , we get

$$\frac{AD}{\sin(\angle ABD)} = \frac{BD}{\sin(\angle BAD)},$$

$$\therefore \sin(\angle ABD) = \frac{10 \cdot \sin 3\alpha}{11}.$$

Applying the sine rule in triangle ABC again, we get:

$$\frac{AC}{\sin(\angle ABC)} = \frac{BC}{\sin(\angle BAC)} = \frac{6}{2\sin\alpha\cos\alpha},$$

$$\therefore AC = \frac{6}{11} \cdot \frac{10\sin\alpha(4\cos^2\alpha - 1)}{2\sin\alpha\cos\alpha} = \frac{30(4\cos^2\alpha - 1)}{11\cos\alpha}.$$

Here we have used the identity $\sin 3\alpha = \sin\alpha(4\cos^2\alpha - 1)$.

Now we invoke the cosine rule. In triangle ACD , the cosine rule gives:

$$CD^2 = AC^2 + AD^2 - 2 \cdot AC \cdot AD \cdot \cos(\angle CAD),$$

$$\therefore 25 = AC^2 + 100 - 20 \cdot AC \cdot \cos\alpha.$$

Let $u = 4\cos^2\alpha - 1$, so that

$$AC = \frac{30u}{11\cos\alpha}.$$

Substituting, we get:

$$25 = \frac{900u^2}{121\cos^2\alpha} + 100 - 20\cos\alpha \cdot \frac{30u}{11\cos\alpha}$$

$$= \frac{3600u^2}{121(u+1)} + 100 - \frac{600u}{11},$$

$$\therefore -75 = \frac{3600u^2}{121(u+1)} - \frac{600u}{11},$$

$$\therefore 121u + 121 = -48u^2 + 88u^2 + 88u.$$

Thus we obtain the following quadratic equation:

$$40u^2 - 33u - 121 = 0.$$

The discriminant is $33^2 + 4(40)(121) = 1089 + 19360 = 20449 = 143^2$. Thus:

$$u = \frac{33 + 143}{80} = \frac{176}{80} = \frac{11}{5}.$$

(The negative root $u = \frac{33-143}{80} = -\frac{11}{8}$ can be discarded since we must have $u = 4\cos^2\alpha - 1 \geq -1$.)

Therefore $4\cos^2\alpha - 1 = \frac{11}{5}$, giving:

$$\cos^2\alpha = \frac{4}{5}, \quad \cos\alpha = \frac{2}{\sqrt{5}}.$$

Hence,

$$AB = \frac{6}{\cos\alpha} = \frac{6}{2/\sqrt{5}} = 3\sqrt{5},$$

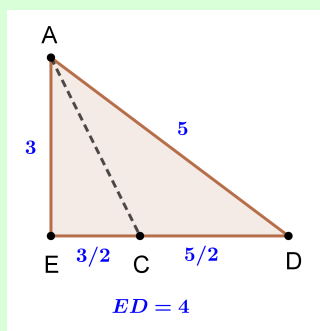
$$AC = \frac{30 \cdot \frac{11}{5}}{11 \cdot \frac{2}{\sqrt{5}}} = \frac{66}{\frac{22}{\sqrt{5}}} = \frac{66\sqrt{5}}{22} = 3\sqrt{5},$$

so $AB = AC$, as required. ■

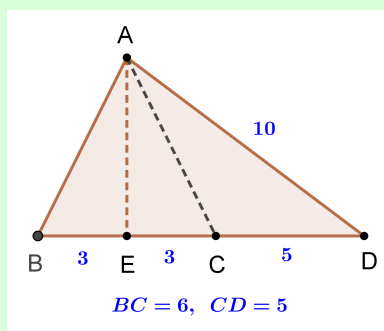
Remark

The problem features the numbers 6, 5, 10. The solution made heavy use of these numbers. Evidently, the conclusion (the fact that $AB = AC$) depends on this choice of numbers. We are led to ask whether there is some other choice of integers for which $AB = AC$. This question leads to an interesting number-theoretic exploration.

Let us examine the data in the problem more closely. It is easy to work out how the problem must have been composed. We can anticipate that the composer started with the PPT (3,4,5) and the right triangle (AED) shown in Figure (a), with $AB = 3$, $ED = 4$, $AD = 5$. The bisector (AC) of $\angle EAD$ divides ED into two parts in the ratio 3 : 5, with lengths $3/2$ and $5/2$. Scaling up by a factor of 2 to clear the fractions, and reflecting C in AE to give B , we obtain the given configuration in Figure (b).



(a)



(b)

Figure 35: Starting with the 'seed' 3,4,5, we have replicated the given data

It should be clear that infinitely many such problems can be devised. Starting with the PPT (5,12,13), we get a new problem in which 6,5,10 are replaced by 20,26,39 respectively. And so on.

Article contributed by the editors

We will study more such problems in future issues of the magazine.

Triangle with a Double Angle and Sides in GP

Problem

Determine the angles of $\triangle ABC$ given the following data about the triangle:

$$\angle C = 2\angle A, \quad b^2 = ac.$$

Is the triangle constructible using compass and straightedge?

Solution

The given conditions are $C = 2A$ and $b^2 = ac$. Since $A + B + C = 180^\circ$, we get $B = 180^\circ - 3A$. It follows that $0 < A < 60^\circ$.

The sine rule yields

$$\begin{aligned}\frac{a}{\sin A} &= \frac{b}{\sin B} = \frac{c}{\sin C}, \\ \therefore \frac{a}{\sin A} &= \frac{b}{\sin 3A} = \frac{c}{\sin 2A}, \\ \therefore \frac{a}{\sin A} &= \frac{b}{\sin A(4\cos^2 A - 1)} = \frac{c}{2\sin A \cos A}, \\ \therefore a &= \frac{b}{4\cos^2 A - 1} = \frac{c}{2\cos A}.\end{aligned}$$

It follows that

$$\begin{aligned}\cos A &= \frac{c}{2a}, \quad 4\cos^2 A - 1 = \frac{b}{a}, \\ \therefore \frac{b}{a} &= \frac{c^2}{a^2} - 1, \quad \therefore ab = c^2 - a^2, \\ \therefore c^2 &= a(a + b).\end{aligned}$$

We also know that $b^2 = ac$.

Let us solve this combined equation system parametrically in terms of b .

Put $b = 1$. Since $b^2 = ac$, we get

$$c = \frac{1}{a}.$$

Substituting into the equation $c^2 = a(a + b)$ we get

$$c^2 = \frac{1}{c^2}(c + 1), \quad \therefore c^4 - c - 1 = 0.$$

So, to solve the given problem we must solve the polynomial equation

$$x^4 - x - 1 = 0.$$

The solution will tell us the sides of the triangle and hence the angles of the triangle.

We shall prove that the polynomial $x^4 - x - 1$ cannot be factorized over the rational numbers. This means that it does not have a rational root, nor a root that is the root of a quadratic equation with rational coefficients.

Proof of irreducibility of the polynomial

The strategy we shall use to prove irreducibility is by arguing modulo 2. Recall that in the mod 2 world we have the arithmetic rules

$$1 + 0 = 0 + 1 = 1, \quad 1 + 1 = 0, \quad 1 \cdot 1 = 1.$$

In particular, $-1 \equiv 1$ and $2 \equiv 0$.

There are just two 1st-degree polynomials in this reduced world: x and $x + 1$.

There are four 2nd degree polynomials: x^2 , $x^2 + 1$, $x^2 + x$ and $x^2 + x + 1$. Of these, the first three are factorizable:

$$x^2 = x \cdot x, \quad x^2 + 1 = (x + 1) \cdot (x + 1), \quad x^2 + x = x \cdot (x + 1).$$

An easy check for factorizability is via the factor theorem. The polynomial $x^2 + 1$ takes the value 0 at $x = 1$ and hence must be factorizable. (It yields the factor $x + 1$.)

On the other hand, $x^2 + x + 1$ is irreducible. (Check: It takes the value 1 at both $x = 0$ and $x = 1$.) It is the only irreducible 2nd degree polynomial in this world.

In the present context, the technical term for the “mod 2 world” is $\mathbb{F}_2[x]$. We shall use this term from now on.

What relevance does this have for checking a polynomial for irreducibility over $\mathbb{Q}$? Here is the connection.

Let $g(x)$ be any polynomial in x with integer coefficients. Let $\bar{g}(x)$ be the corresponding polynomial reduced mod 2. Let us check for factorizability of $\bar{g}(x)$ in $\mathbb{F}_2[x]$.

Suppose we find that $\bar{g}(x)$ is irreducible in $\mathbb{F}_2[x]$. Then we can definitely conclude that $g(x)$ is irreducible over the rationals. That is,

$$\bar{g}(x) \text{ irreducible over } \mathbb{F}_2[x] \implies g(x) \text{ irreducible over } \mathbb{Q}.$$

Typically, checking whether $\bar{g}(x)$ is irreducible over $\mathbb{F}_2[x]$ is much easier than checking irreducibility of $g(x)$ over $\mathbb{Q}$. This is what makes the above finding so useful.

Important note: The converse of the above statement is **not** true. In other words, if $\bar{g}(x)$ is factorizable over $\mathbb{F}_2[x]$, we cannot conclude anything about the factorizability of $g(x)$ over $\mathbb{Q}$. (One of the examples presented above highlights this fact. Do you notice which example?)

Example of an irreducibility check: Consider the polynomial $h(x) = x^3 + 3x + 5$. Is this factorizable over $\mathbb{Q}$? Let us examine the reduced version of $h(x)$ mod 2 over $\mathbb{F}_2[x]$. We have,

$$\bar{h}(x) = x^3 + x + 1.$$

It is easy to check that the reduced polynomial is irreducible over $\mathbb{F}_2[x]$. (It does not vanish at either $x = 0$ or $x = 1$.) Hence $h(x)$ is irreducible over $\mathbb{Q}$.

Now it is time to return to our original problem! We prove the following.

Theorem. *The polynomial $f(x) = x^4 - x - 1$ is irreducible over $\mathbb{Q}$.*

Proof: We shall follow verbatim the strategy described earlier. We first reduce $f(x) = x^4 - x - 1$ modulo 2 to obtain

$$\bar{f}(x) = x^4 + x + 1 \in \mathbb{F}_2[x].$$

We claim that $\bar{f}(x)$ is irreducible over $\mathbb{F}_2$. Since the degree of $\bar{f}$ is 4, if $\bar{f}$ were reducible, it would have a factor of degree 1, 2, or 3. We consider each case in turn.

Linear factors: A linear factor exists if and only if $\bar{f}$ has a root in $\mathbb{F}_2$. We have,

$$\bar{f}(0) = 0 + 0 + 1 = 1 \neq 0, \quad \bar{f}(1) = 1 + 1 + 1 = 1 \neq 0.$$

Hence $\bar{f}$ has no roots in $\mathbb{F}_2$ and therefore no linear factors.

Obviously, this rules out cubic factors as well.

Quadratic factors: If $\bar{f}$ factored as a product of two quadratics, each must be irreducible (since we have already excluded linear factors). The only irreducible quadratic polynomial over $\mathbb{F}_2$ is $x^2 + x + 1$. Therefore, if $\bar{f}$ factors into two quadratics, it would mean that

$$\bar{f}(x) = (x^2 + x + 1)^2.$$

But in $\mathbb{F}_2[x]$,

$$(x^2 + x + 1)^2 = x^4 + x^2 + 1 \neq x^4 + x + 1 = \bar{f}(x).$$

Thus $\bar{f}$ does not factor as a product of two quadratics.

Conclusion: Since $\bar{f}(x) = x^4 + x + 1$ has no factors of degree 1, 2, or 3 over $\mathbb{F}_2$, it is irreducible over $\mathbb{F}_2$.

Since $f(x)$ is monic and $\deg f = \deg \bar{f} = 4$, it follows that $f(x) = x^4 - x - 1$ is irreducible over $\mathbb{Q}$. ■

Alternative Proof: A Direct Argument

Rather than consider the problem modulo 2, it is possible to argue directly. Some may prefer this direct approach. Here's how this works.

Since $f(x) = x^4 - x - 1$ is monic with integer coefficients, if f is reducible over $\mathbb{Q}$, then it factors over $\mathbb{Z}$ (this follows from Gauss's Lemma). The possible factorizations are:

- $f(x) = \text{linear} \times \text{cubic}$, or
- $f(x) = \text{quadratic} \times \text{quadratic}$.

We consider both possibilities in turn.

The case of linear factors: As the leading coefficient of $f(x)$ is 1 and the constant term is -1 , the only possible rational roots of $f(x) = x^4 - x - 1$ are $x = \pm 1$. But,

$$f(1) = 1 - 1 - 1 = -1 \neq 0, \quad f(-1) = 1 + 1 - 1 = 1 \neq 0.$$

Hence f has no rational roots and thus no linear factors over $\mathbb{Z}$.

The case of quadratic factors: Suppose to the contrary that

$$x^4 - x - 1 = (x^2 + ax + b)(x^2 + cx + d)$$

for some $a, b, c, d \in \mathbb{Z}$.

Expanding the right-hand side and comparing coefficients of x^3, x^2, x^1, x^0 , we obtain:

$$a + c = 0, \quad \therefore c = -a.$$

$$b + d + ac = 0, \quad \therefore b + d = a^2.$$

$$ad + bc = -1, \quad \therefore a(d - b) = -1.$$

$$bd = -1. \quad \therefore (b, d) = (1, -1) \text{ or } (-1, 1).$$

If $(b, d) = (1, -1)$, then $b + d = 0 = a^2$, so $a = 0$. But then $a(d - b) = 0 \neq -1$, a contradiction.

If $(b, d) = (-1, 1)$, then $b + d = 0 = a^2$, so $a = 0$. But then $a(d - b) = 0 \neq -1$, again a contradiction.

In both cases we reach a contradiction. Therefore no factorization into two quadratics over $\mathbb{Z}$ exists, and $f(x) = x^4 - x - 1$ is irreducible over $\mathbb{Q}$. ■

This tells us that to solve the original construction problem, we must solve a 4th degree equation whose roots are not the roots of any linear or quadratic equation with rational coefficients.

This strongly suggests that the required triangle is **not** constructible using Euclidean methods, i.e., using straight edge and compass.

This is so. Such a triangle cannot be constructed following the usual rules. But we will not give the complete proof of this here as it involves ideas from higher mathematics.

Numerical approach: Let $x = 2 \cos A$. Using the sine rule and plenty of algebra, we arrive at the equation $(x^2 - 1)^2 = x$, i.e.,

$$h(x) = x^4 - 2x^2 - x + 1 = 0.$$

The polynomial $h(x)$ is irreducible. This may be proved using the same way as earlier. Rather than obtain a closed-form solution of this equation in terms of radicals (which does not cast fresh light on the problem), we approach the problem numerically.

We first check that $h(x)$ has a unique root in the interval $(1, 2)$. Observe that

$$h(1) = 1 - 2 - 1 + 1 = -1 < 0, \quad h(2) = 16 - 8 - 2 + 1 = 7 > 0.$$

Since h is continuous, there exists at least one root in $(1, 2)$. Examining $h'(x) = 4x^3 - 4x - 1$ on this interval: $h'(1) = -1$ and $h'(2) = 23 > 0$, one can verify that h has exactly one local minimum in $(1, 2)$ with a negative value, followed by a monotone increase. A sign analysis confirms there is exactly one root of h in $(1, 2)$. This root can be approximated by Newton's method. We obtain:

$$x_0 = 2 \cos A \approx 1.49042,$$

giving:

$$A = \arccos\left(\frac{1}{2}x_0\right) \approx 41^\circ 51',$$

$$C = 2A \approx 83^\circ 41',$$

$$B = 180^\circ - 3A \approx 54^\circ 28'.$$

Article contributed by Shailesh Shirali (with lots of help from Claude AI)

Grandma's Identities – II

In the January 2026 issue of *The Compass*, we described a trigonometric identity playfully called *Grandma's Identity*. Not only was the identity very beautiful, but the method used to prove it was equally striking and noteworthy. We describe another identity of this genre in this article.

Recap: In the January 2026 article, we used the following algebraic identity which is true for all odd positive integers n ,

$$x^n - 1 = (x - 1) \prod_{k=1}^{(n-1)/2} \left(x^2 - 2x \cos \frac{2\pi k}{n} + 1 \right),$$

to prove that

$$\prod_{k=1}^{(n-1)/2} 2 \cos \frac{\pi k}{n} = 1.$$

This is the 'Grandma identity.'

A twin sister

In [2], Sury mentions that ‘Grandma’s identity’ has a ‘twin sister,’ namely:

$$\prod_{s=1}^{2k} 2 \cos \frac{2\pi s}{4k+1} = (-1)^k = \prod_{s=1}^{2k-1} 2 \cos \frac{2\pi s}{4k-1}.$$

He writes, “We call this the twin sister because it is seen to be equivalent to the Grandma identity on using the formula $\cos(\pi - \theta) = -\cos \theta$.”

Here are some instances of the twin sister identity:

$$2 \cos \frac{2\pi}{3} = -1,$$

$$2 \cos \frac{2\pi}{5} \cdot 2 \cos \frac{4\pi}{5} = -1,$$

$$2 \cos \frac{2\pi}{7} \cdot 2 \cos \frac{4\pi}{7} \cdot 2 \cos \frac{6\pi}{7} = 1,$$

$$2 \cos \frac{2\pi}{9} \cdot 2 \cos \frac{4\pi}{9} \cdot 2 \cos \frac{6\pi}{9} \cdot 2 \cos \frac{8\pi}{9} = 1,$$

etc.

We leave the task of explaining Sury’s remark to the reader. In what way does the identity $\cos(\pi - \theta) = -\cos \theta$ explain the ‘twin sister’ identity?

A distinguished Italian cousin

Sury then points out that there is a closely related identity, which he calls an “Italian cousin.”

This is the following identity which is true for all odd positive integers n :

$$\prod_{r=1}^{(n-1)/2} \left(3 + 2 \cos \frac{2\pi r}{n} \right) = F_n.$$

Here F_n is the n -th number in the Fibonacci sequence

$$1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, \dots$$

This relation certainly puts the Grandma identity into very distinguished company!

Older sister of the Italian cousin

There is a slightly more general result than the statement quoted above:

Theorem. *Let $n > 1$ be an integer. Let*

$$f(n) = \prod_{r=1}^{\lfloor n/2 \rfloor} \left(3 + 2 \cos \frac{2\pi r}{n} \right).$$

Then $f(n) = F_n$, the n -th Fibonacci number.

Let us verify the theorem for a few small values of n . For $n = 1, 2$, there is nothing to verify.

n = 3: We have,

$$f(3) = 3 + 2 \cos \frac{2\pi}{3} = 3 - 1 = 2,$$

so $f(3) = F_3$ (verified).

n = 4: We have,

$$f(4) = \left(3 + 2 \cos \frac{2\pi}{4} \right) \cdot \left(3 + 2 \cos \frac{4\pi}{4} \right) = 3 \cdot 1 = 3,$$

so $f(4) = F_4$ (verified).

n = 5: We have,

$$\begin{aligned} f(5) &= \left(3 + 2\cos\frac{2\pi}{5}\right) \cdot \left(3 + 2\cos\frac{4\pi}{5}\right) \\ &= \frac{5 + \sqrt{5}}{2} \cdot \frac{5 - \sqrt{5}}{2} \\ &= \frac{20}{4} = 5, \end{aligned}$$

so $f(5) = F_5$ (verified).

n = 6: We have,

$$\begin{aligned} f(6) &= \left(3 + 2\cos\frac{2\pi}{8}\right) \cdot \left(3 + 2\cos\frac{4\pi}{6}\right) \cdot \left(3 + 2\cos\frac{6\pi}{6}\right) \\ &= 4 \cdot 2 \cdot 1 = 8, \end{aligned}$$

so $f(6) = F_6$ (verified).

The proof of this remarkable identity is more involved than one may expect. We will postpone it to the next article of this series. But we list the main steps of the proof here. The reader could try filling in the missing details.

Step 1. Invoke the “Binet formula” for the n -th Fibonacci number:

$$F_n = \frac{u^n - v^n}{u - v},$$

where u, v are the roots of the quadratic equation $z^2 = z + 1$ or $z^2 - z - 1 = 0$:

$$u = \frac{1 + \sqrt{5}}{2}, \quad v = \frac{1 - \sqrt{5}}{2}.$$

Step 2. Recall that $u + v = 1$ and $uv = -1$ which imply that $u^2 + v^2 = 3$ and $u^2v^2 = 1$. So u^2, v^2 are the roots of the quadratic equation $z^2 - 3z + 1 = 0$, and we have,

$$z^2 - 3z + 1 = (z - u^2)(z - v^2), \quad z^2 + 3z + 1 = (z + u^2)(z + v^2).$$

Step 3. Recall that the numbers L_n defined by

$$L_n = u^n + v^n$$

yield the *Lucas sequence*: 1, 3, 4, 7, 11, 18, 29, 47, The Lucas sequence is intricately connected with the Fibonacci sequence.

Step 4. Write down the complete factorization of $3 + 2\cos\theta$ in terms of the n -th roots of unity.

Step 5. Pair up the factors suitably and simplify to the extent possible.

After resolutely following each step to its logical conclusion, we arrive at the stated identity.

in [2], Sury follows a different approach, starting with an algebraic identity. We will describe both these approaches in the next issue. ■

Further reading

[1] Steve Humble, "Grandma's Identity" from *The Mathematical Gazette*, Nov., 2004, Vol. 88, No. 513 (Nov., 2004), pp. 524-525,
<http://www.jstor.com/stable/3620735>

[2] B Sury, "Of grand-aunts and Fibonacci" from *The Mathematical Gazette*, Vol. 92, No. 523 (March 2008), pp. 63-64,
<https://www.jstor.org/stable/27821717>

A Fermat-Type Primality Problem

The following statement is well known and easy to prove:

If $n \in \mathbb{N}$ is such that $2^n + 1$ is prime, then n is a power of 2.

Here, 1 is considered to be a power of 2 (with $1 = 2^0$).

The proof of the above statement rests on the observation that $x + 1$ is a divisor of $x^m + 1$ for all odd values of m . (This in turn follows from the factor theorem.)

Primes of the form $2^n + 1$ are known as *Fermat primes*, because at one point Fermat appears to have had the impression that all numbers of the form $2^{2^n} + 1$ ($n = 0, 1, 2, 3, \dots$) are prime.

But this statement is not true. Euler showed that the number $2^{32} + 1$ is composite. (The story of how Euler performed this extraordinary feat can wait for another telling.) So, a necessary condition for $2^n + 1$ to be prime is that n must be a power of 2. But it is not a sufficient condition.

In this article, we shall explore another such family. We ask:

For $n \in \mathbb{N}$, if $4^n + 2^n + 1$ is prime, what can be said about n ?

Our objective is to prove the following striking result which will surely remind you of the above result about the Fermat numbers:

Theorem. *If $n \in \mathbb{N}$ is such that $4^n + 2^n + 1$ is prime, then n is a power of 3.*

Thus, a necessary condition for $4^n + 2^n + 1$ to be prime is that n must be a power of 3. But is this a sufficient condition? We shall explore this question later.

Proof of theorem

We shall prove the contrapositive of the stated proposition. That is, we shall show that if $n \geq 1$ is *not* a power of 3, then $4^n + 2^n + 1$ is composite.

We write

$$n = 3^k \cdot m,$$

where $k \geq 0$ and $\text{GCD}(m, 3) = 1$. Since n is not a power of 3, we must have $m > 1$. Let

$$a = m, \quad b = 3^k,$$

so $n = ab$ with $a > 1$ and $3 \nmid a$. Let $y = 2^b$. Then

$$4^n + 2^n + 1 = 2^{2ab} + 2^{ab} + 1 = y^{2a} + y^a + 1.$$

Claim:

$$y^2 + y + 1 \mid y^{2a} + y^a + 1,$$

whenever $3 \nmid a$.

Proof of the claim: Let the non-real cube roots of unity be ω and ω^2 . Write

$$f(y) = y^{2a} + y^a + 1.$$

If we can show that the following two equalities hold,

$$f(\omega) = 0, \quad f(\omega^2) = 0,$$

then the stated conclusion will follow immediately, for then it would be true that the expressions $y - \omega$ and $y - \omega^2$ are both factors of $y^{2a} + y^a + 1$ (implying that $y^2 + y + 1$ is a factor of $y^{2a} + y^a + 1$).

But this verification is immediate. The statement is true precisely because a is not a multiple of 3, and therefore $\{\omega^{2a}, \omega^a\} = \{\omega^2, \omega\}$. The reader should be able to verify this statement. The claim follows.

Back to the proof of the main theorem: Substituting $y = 2^b$, we see that the integer divisibility

$$4^b + 2^b + 1 \mid 4^n + 2^n + 1$$

follows immediately.

We have thus found a divisor of $4^n + 2^n + 1$. It remains to check that the divisor is proper; that is, $4^b + 2^b + 1$ is neither 1 nor equal to $4^n + 2^n + 1$. The first part is obvious since $b \geq 1$. The second part follows from the fact that $a > 1$ (and hence $b < n$). We ask the reader to fill in the details here.

It follows that $4^n + 2^n + 1$ has a proper divisor and hence is composite. This completes the proof of the theorem. ■

Illustration with actual data

Some small cases illustrate the claim:

n	Power of 3?	$4^n + 2^n + 1$	Status
1	Yes	7	prime
2		21	$= 3 \cdot 7$
3	Yes	73	prime
4		273	$= 3 \cdot 7 \cdot 13$
5		1057	$= 7 \cdot 151$
9	Yes	262657	prime

What about the converse? The converse of the claim is *false*: there exist powers of 3, such as $n = 27$, for which $4^n + 2^n + 1$ is composite:

$$4^{27} + 2^{27} + 1 = 18014398643699713$$

$$= 2593 \cdot 71119 \cdot 97685839.$$

It follows that the condition n is a power of 3 is *necessary but not sufficient* for primality of $4^n + 2^n + 1$.

The conclusion we reach is thus exactly like that for the Fermat numbers. The necessary condition is not sufficient.

Do there exist values of $n \neq 1, 3, 9$ for which $4^n + 2^n + 1$ is prime? The answer is not clear.

Article contributed by Shailesh Shirali (with help from Claude AI)

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We invite mathematics teachers and practitioners to contribute short articles to The Compass.

Articles could deal with mathematical content or the teaching of mathematics. They could feature real-life classroom experiences with projects and explorations.

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They could feature reflections on the teaching of mathematics, or on the nature of mathematics.

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